

SHARPNESS AND UNIQUENESS FOR POSITIVE SOLUTIONS OF THE EXPLICIT FORMULA FOR $\zeta(s)$

NIC JOHNS

ABSTRACT. Weil’s explicit formula for $\zeta(s)$ equates a functional $\mathcal{A}(F)$, which depends only on the archimedean data of ζ , with sums over the zeros and over the prime powers. We construct an explicit even function $F = \Xi^2 H$, with H entire and positive on $\mathbb{R}$, such that $F \geq 0$ and $\widehat{F} \geq 0$ on $\mathbb{R}$, $\widehat{F}$ vanishes at $\log n/2\pi$ for every integer $n \geq 2$, and $\mathcal{A}(F) = 0$. Since Ξ vanishes at every non-trivial zero, on or off the critical line, no information about the zeros is used. F is an exact magic function, in the sense of the sphere-packing bounds of Cohn–Elkies and Viazovska, for the linear programme of Odlyzko, Poitou and Serre: for the data of ζ the programme has no positive slack, so the method is sharp, and F is a minimiser under RH. With a support theorem of Part I, every pair of positive measures satisfying the formula, on the zero side and on the prime side beyond $\log 2/2\pi$, is then ζ ’s own. This resolves Conjectures S and U of Part I. H comes from an eigenvalue- $\frac{1}{4}$ eigenfunction of the hyperbolic Laplacian through a contour argument of Viazovska’s type, with Green’s identity in place of Cauchy’s theorem; its positivity is partly computer-assisted. We do not prove RH; by these results it is equivalent to the existence of such a pair.

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1. INTRODUCTION

1.1. Positive solutions of the explicit formula. Weil's explicit formula [52] (in Guinand's form [26] for ζ) is a linear identity between a measure on the zeros of ζ and a measure on the prime powers. For even test functions F in a natural class $\mathcal{T}$ (Definition I.2.1), with $\widehat{F}(\xi) = \int F(t) e^{-2\pi i \xi t} dt$, it reads

$$(1.1) \quad \mathcal{A}(F) = \sum_{\rho} F\left(\frac{\rho - \frac{1}{2}}{i}\right) + \frac{1}{\pi} \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \widehat{F}\left(\frac{\log n}{2\pi}\right),$$

where the left side $\mathcal{A}(F)$ depends only on the archimedean data of ζ : its Gamma factor and its pole at $s = 1$. Under the Riemann hypothesis (RH) the right side is $\int F d\mu + \frac{1}{\pi} \int \widehat{F} d\nu$ for two positive measures, μ on the zero ordinates and ν on the points $\log n/2\pi$. We ask how much of this structure is forced by positivity and the archimedean data alone. Call a pair of positive measures, μ even on $\mathbb{R}$ and ν on $[\log 2/2\pi, \infty)$, that represents $\mathcal{A}$ on $\mathcal{T}$ an *admissible pair* (Definition I.2.4); nothing is assumed about where the measures live or about integrality. Let $\mathcal{K}$ be the set of admissible pairs and p_{ζ} the pair of ζ , which is admissible exactly under RH. Two questions arise.

Sharpness. Testing (1.1) against functions with $F \geq 0$ and $\widehat{F} \geq 0$ is the explicit-formula method of Stark, Odlyzko, Poitou and Serre [46, 39, 44, 41, 40] for lower bounds on conductors and discriminants. For the archimedean data of ζ it leads to the linear programme

$$\kappa^* = \inf \left\{ \mathcal{A}(F) : F \in \mathcal{T}, F \geq 0 \text{ on } \mathbb{R}, \widehat{F} \geq 0 \text{ on } [\frac{\log 2}{2\pi}, \infty), \int F = 1 \right\} :$$

a test function with $\mathcal{A}(F) = \kappa \int F$ proves the conductor bound $e^{-2\pi\kappa}$, and $q_{\min} = e^{-2\pi\kappa^*}$ is the best bound the method can give (Part I, Proposition 2.10). Is the method sharp for ζ itself, that is, does it reach the conductor 1 of ζ ? Odlyzko's numerical optimisation gave 0.997 [40, Table 4]. By linear-programming duality, admissible pairs exist if and only if $\kappa^* \geq 0$ (Theorem I.2.7(a)), a statement that involves neither zeros nor primes.

Rigidity. Do positivity and the archimedean data determine the solution, that is, is p_{ζ} the only possible admissible pair?

Part I [32] set up this framework and posed the two questions as Conjecture S, $\kappa^* \leq 0$, and Conjecture U, $\mathcal{K} \subseteq \{p_{\zeta}\}$ (Section 2).

1.2. Main results. We answer both questions. The tool is an *exact magic function*: a test function F in the cone of the linear programme with $\int F > 0$ and $\mathcal{A}(F) = 0$, which shows that the value 0 is reached by an explicit function. (The term comes from the linear-programming bounds for sphere packing of Cohn and Elkies and of Viazovska [12, 50], where such functions show that a bound is sharp.) We construct such a function.

There is no circularity in building F from ζ itself. The function Ξ vanishes at every non-trivial zero of ζ , on or off the critical line, so the factor Ξ^2 removes the zero side of (1.1) without any information about where the zeros lie, and RH is assumed nowhere. The explicit formula enters only as an unconditional identity; in Theorem 1 it evaluates $\mathcal{A}(F)$, which involves only the archimedean data. The difficulty is to keep F and $\widehat{F}$ non-negative; by Section 3, this is a problem about the Gamma function and the integers alone.

Recall $\Xi(t) = \xi(\frac{1}{2} + it)$ with $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$, $\xi_n = \log n/2\pi$, and the classical cone $\mathcal{C}_{\text{OPS}}$ of functions with $F \geq 0$ and $\widehat{F} \geq 0$ on all of $\mathbb{R}$, with its slack $\kappa_{\text{OPS}}^* \geq \kappa^*$.

Theorem 1. *There is an even entire function H , real on $\mathbb{R}$, with $H(t) > 0$ for every real t and $H(0) = 1$, such that for every $\delta \in (0, 1)$*

$$(1.2) \quad |H(z)| \leq C_{\delta} (1 + |z|)^{-9+\delta} e^{\pi |\operatorname{Re} z|/2} \quad (|\operatorname{Im} z| \leq \tfrac{1}{2} + \delta),$$

and such that $F := \Xi^2 H$ satisfies:

- (a) $F \in \mathcal{C}_{\text{OPS}}$: $F \geq 0$ and $\widehat{F} \geq 0$ on $\mathbb{R}$;
- (b) $\widehat{F}(\xi_n) = \widehat{F}'(\xi_n) = 0$ for every integer $n \geq 2$;
- (c) $\int F = \widehat{F}(0) > 0$ and $\mathcal{A}(F) = 0$.

Consequently $\kappa^* \leq \kappa_{\text{OPS}}^* \leq 0$. In particular Conjecture S holds, and $q_{\min} \geq 1$, already for the classical cone.

So $F/\int F$ is an exact magic function: it is admissible in the definition of κ^* with value 0. In the language of conductors, Theorem 1 says that the method of Odlyzko, Poitou and Serre is sharp for the data of ζ , already in the classical cone: the explicit function F proves the bound 1, the conductor of ζ , where numerical optimisation had given 0.997; under RH no test function does better (Corollary 3). This answers the degree-one case, that of $\mathbb{Q}$ and so of ζ itself, of Odlyzko's Open Problem 2.2 [40, p. 123]: “What functions $F(x)$ give the best GRH bounds for discriminants?” Under RH, F gives the best possible bound in degree one. We do not address the unconditional Open Problem 2.1 of [40], or higher degrees. We do not claim that κ^* is attained, or that $\kappa^* = 0$: by duality $\kappa^* = 0$ holds if and only if admissible pairs exist (Corollary 3). What holds unconditionally is $\kappa^* \leq 0 = \mathcal{A}(F)/\int F$; and among the cone members of the form $\Xi^2 H$ with H analytic on a strip $|\text{Im } t| \leq \frac{1}{2} + \delta$, which all have $\mathcal{A} \geq 0$ by Corollary I.4.4, F realises the least value 0 of $\mathcal{A}/\int$.

Theorem 2. *Every admissible pair is ζ 's pair: $\mathcal{K} \subseteq \{p_\zeta\}$. That is, Conjecture U holds.*

This is a rigidity statement: positivity and the archimedean data force any positive solution of (1.1) to be carried by the zeros of ζ and by the prime powers, with their usual weights. It follows from Theorem 1 by complementary slackness: every admissible pair is carried by the zeros of F and of $\widehat{F}$ (Lemma I.2.13), and, because $H > 0$, the real zeros of F are those of ζ , so the zero-side support theorem of Part I (Theorem I.3.6) applies. The pattern is loosely that of the linear-programming approach to sphere packing, where a certificate vanishing on the expected support constrains every optimal configuration [12, 50].

Corollary 3. *The following are equivalent: (i) RH; (ii) admissible pairs exist; (iii) $\kappa^* \geq 0$; (iv) $\kappa^* = 0$. If they hold, then $\mathcal{K} = \{p_\zeta\}$, $\kappa_{\text{OPS}}^* = 0$, and $F/\int F$, with F as in Theorem 1, is a minimiser for both κ^* and κ_{OPS}^* . If RH fails, then $\kappa^* < 0$ and $\mathcal{K} = \emptyset$.*

We do not prove RH: by Corollary 3 it is equivalent to the existence of an admissible pair, and to $\kappa^* = 0$. We do not claim that $\kappa_{\text{OPS}}^* = 0$ implies RH: the duality theorem of Part I concerns the cone $\mathcal{C}$ with the gap $\log 2/2\pi$, not $\mathcal{C}_{\text{OPS}}$. Under RH, Corollary 3 says that the fully optimised explicit-formula conductor bound for the data of ζ , in either cone, is exactly 1.

Theorems 1 and 2 resolve Conjectures S and U of Part I. There, Conjecture U was proved under support conditions, for instance when the zero measure is carried by the zeros of ζ and the origin [32, Theorem 3.6]; a magic-function principle [32, Corollary 3.9] reduced Conjecture U to the construction of a function as in Theorem 1 whose real zeros are those of ζ ; and $\kappa^* \leq 1.43 \cdot 10^{-1060}$, a conductor bound $1 - 8.98 \cdot 10^{-1060}$, was certified numerically.

The method does not prove too much. For the data of the Dedekind zeta functions of $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\sqrt{-3})$, Part I certifies strictly positive slack, so no exact magic function exists for them, and no analogue of our construction can exist there (Section 9). The existence of F is specific to the data of ζ .

The function H is explicit. Let $S(m, n; c)$ be the Kloosterman sum, $j_2 := \partial_\mu J_\mu|_{\mu=2}$, $i_2 := \partial_\mu I_\mu|_{\mu=2}$, and $\gamma_\infty(t) := \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})$ at $s = \frac{1}{2} + it$, the archimedean factor of Ξ .

Theorem 4 (The object). *For $n \geq 1$ put*

$$\mathcal{S}_n := \sum_{c \geq 1} \frac{1}{c} \left[S(1, n; c) j_2\left(\frac{4\pi\sqrt{n}}{c}\right) - S(-1, n; c) i_2\left(\frac{4\pi\sqrt{n}}{c}\right) \right], \quad a_n := n\mathcal{S}_n + \frac{1}{4}\delta_{n1}.$$

(a) *The series converge absolutely, and with $z_n := 4\pi\sqrt{n}$ and $\bar{a}_n := n^{1/4}e^{4\pi\sqrt{n}}/(4\sqrt{2}\pi^2)$,*

$$\left(1 - \frac{2}{z_n}\right)\bar{a}_n \leq a_n \leq \bar{a}_n \quad (n \geq 1);$$

in particular every a_n is positive.

(b) *The function H of Theorem 1 can be taken such that, for $x > 1$,*

$$\int_{\mathbb{R}} \gamma_{\infty}(t)^2 H(t) x^{-it} dt = C \sin^2(\pi x) \sqrt{x} \sum_{n \geq 1} a_n K_0(4\pi\sqrt{nx}), \quad C = 880.6617 \pm 0.0015,$$

and the left side equals $C/(32\pi^2)$ at $x = 1$. Thus $b_n := Ca_n$ satisfy $(1 - 2/z_n) \cdot 15.77370 n^{1/4} e^{4\pi\sqrt{n}} \leq b_n \leq 15.77376 n^{1/4} e^{4\pi\sqrt{n}}$.

(c) *The left side of (b) is, for every $x > 0$, equal to $-C/(16\pi^2)$ times the four-path function (5.6), which is analytic on $\operatorname{Re} x > 0$; and for real t , H is given by the physical-side formula (7.4): an integral of two K_0 -series in the same coefficients a_n against $\cos(t \log Y)$ and against the positive conical kernel $\mathfrak{c}_t$.*

The factor $\sin^2(\pi x)$ produces the double zeros at the integers; the sum $\sqrt{x} \sum_n a_n K_0(4\pi\sqrt{nx})$ converges only for $x > 1$, and has a double pole at $x = 1$, so the continuation to $(0, 1]$ in (c) is part of the content. In the variable x , the transform $\widehat{F}$ is the positive combination $\sum_m d(m) m^{-1/2} \widehat{G}_H(mx)$ of dilates of the function in (b) (Lemma 3.2), so the integer zeros pass from the Γ -only function to $\widehat{F}$.

Table 1 lists certified enclosures of the first coefficients and of C , for comparison with independent computations. Figure 1 illustrates H on $[0, 40]$ and the function in (b) on $[1, 6]$; it is not part of any proof.

1.3. The construction in outline. *Reduction (Section 3).* For $F = \Xi^2 H$ every zero of ζ is a zero of F , so $\mathcal{A}(F)$ is a sum over prime powers (Part I, Corollary 4.4). On a line where the Dirichlet series of ζ^2 converges absolutely, $\widehat{F}$ decouples into the positive combination above. So it suffices to find an even H of growth (1.2) with $H \geq 0$, whose Γ -only transform $\widehat{G}_H(x) = \int \gamma_{\infty}^2 H x^{-it} dt$ vanishes at every integer $x \geq 2$ and is non-negative for $x \geq 2$: a Γ -only integer-critical function. Arithmetic has disappeared from the problem; what remains is a question about the Gamma factor and the lattice $\mathbb{Z}$.

	certified ball	source
a_1	4581.370553 ± 0.000028	Kloosterman terms $c \leq 6 \cdot 10^5$, and a tail bound
a_2	$1025727.23492 \pm 0.00011$	Kloosterman terms $c \leq 6 \cdot 10^5$, and a tail bound
a_3	$62590223.68999 \pm 0.00060$	Kloosterman terms $c \leq 3 \cdot 10^5$, and a tail bound
a_4	$1968035704.4549 \pm 0.0011$	Kloosterman terms $c \leq 3 \cdot 10^5$, and a tail bound
a_5	$40671323483.6182 \pm 0.0017$	Kloosterman terms $c \leq 3 \cdot 10^5$, and a tail bound
C	880.6617 ± 0.0015	Proposition 6.4

TABLE 1. The coefficients $a_1, \dots, a_5$ and the constant C of Theorem 4, as certified balls (Appendix A.1 and Proposition 6.4). Each printed ball is rounded outward, so it contains the certified one.

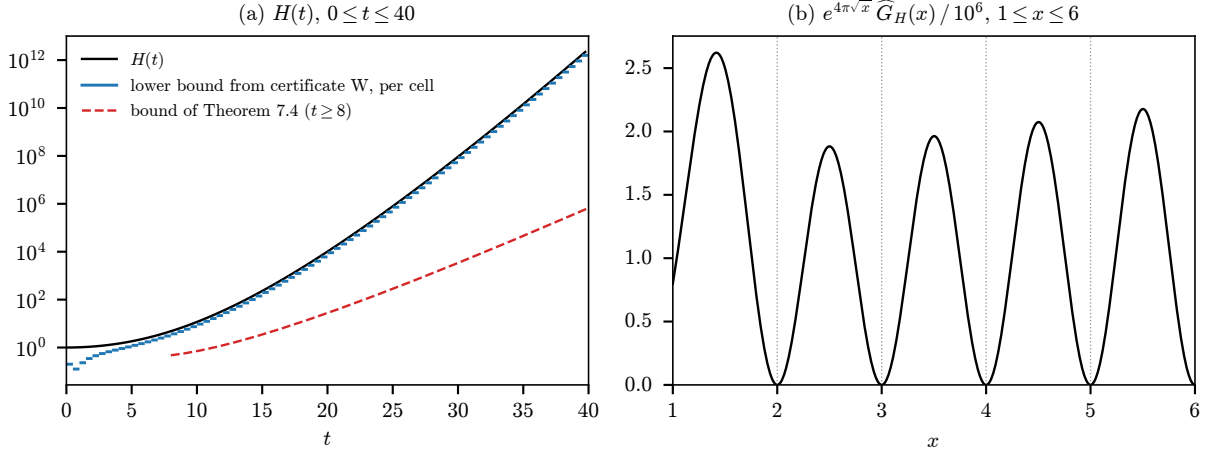


FIGURE 1. An illustration; it is not part of any proof. (a) $H(t)$ for $0 \leq t \leq 40$, on a logarithmic scale. The curve joins $H(0) = 1$ and the values at the 80 cell centres of the window certificate (Theorem 7.10), computed without error terms. The steps are lower bounds for H on each cell, formed from the certified cell bounds and the certified enclosure of $\Pi(0)$. The dashed curve is the bound of Theorem 7.4 for $t \geq 8$. (b) $e^{4\pi\sqrt{x}} \widehat{G}_H(x)/10^6$ for $1 \leq x \leq 6$, where $\widehat{G}_H(x)$ is the left side of Theorem 4(b), computed in floating point from the coefficient midpoints; $\widehat{G}_H$ vanishes doubly at $x = 2, \dots, 6$, and $\widehat{G}_H(1) = C/(32\pi^2)$. Data and script: `figures/` in the ancillary files.

Automorphic input (Section 4). The kernel $K_0(4\pi\sqrt{nx})$ arises by pairing the Whittaker function $\sqrt{v}K_0(2\pi nv)$, of eigenvalue $\frac{1}{4}$ for $\Delta = -v^2(\partial_u^2 + \partial_v^2)$ on the upper half-plane, with the family $\sqrt{xv}K_0(2\pi xv)$ (Lemma 5.5), and $\sin^2(\pi x)$ is, up to the factor -4 , the symbol of the second difference $g(\tau+1) - 2g(\tau) + g(\tau-1)$ acting on that family. So we look for a 1-periodic eigenfunction of Δ with eigenvalue $\frac{1}{4}$, together with a partner whose second difference it is. We start from the odd Niebur–Poincaré series P_ν [38], of eigenvalue $\frac{1}{4} - \nu^2$, whose Fourier coefficients are Kloosterman sums of $J_{2\nu}$ and $I_{2\nu}$ with an explicit constant (Theorem 4.4). The first-order operator $2\partial_v + \nu^2/v$ maps eigenvalue $\frac{1}{4} - \nu^2$ to $\frac{1}{4}$ up to an error $(1 - \nu^2)^2 P_\nu/v$, which has a *double* zero at $\nu = 1$ (Lemma 4.6); so the spectral derivative $\tilde{\Phi} = \partial_\nu[(2\partial_v + \nu^2/v)P_\nu]_{\nu=1}$ is an exact eigenfunction with eigenvalue $\frac{1}{4}$. It is 1-periodic and odd, has no constant term, and decays exponentially at the cusp ∞ , with Fourier coefficients $-16\pi a_n$. It is not invariant under $\mathrm{SL}_2(\mathbb{Z})$, so it is not a Maass cusp form; it is the Y^2 -component of a vector-valued function that is equivariant for the symmetric square of the standard representation, and its X^2 -component $\Phi = \tilde{\Phi} \circ S$ satisfies $\Phi(\tau+1) - 2\Phi(\tau) + \Phi(\tau-1) = 2\tilde{\Phi}(\tau)$.

Why eigenvalue $\frac{1}{4}$, and why the symmetric square. The eigenvalue $\frac{1}{4}$ is that of ζ^2 : up to Gamma factors, $\zeta(s)^2$ is the L -function of the completed Eisenstein series at the centre $s = \frac{1}{2}$, whose non-constant Fourier terms are multiples of $d(|n|)\sqrt{v}K_0(2\pi|n|v)e(nu)$ (cf. [6]), and $d(m)$ is the weight in the decoupling of Section 3. The symmetric square supplies the first-order operator: heuristically (Remark 4.11), tensoring with Sym^2 shifts the spectral parameter, and the eigenvalue $(s-1)(2-s)$ of the shifted constituent is stationary at $s = \frac{3}{2}$, where it equals $\frac{1}{4}$; this stationary point is the double root of Lemma 4.6.

The lift (Section 5). We integrate the closed Green form $\omega(\Phi_j, k_\tau(x))$ of the weights Φ_j and of the kernel $k_\tau(x) = \sqrt{xv}K_0(2\pi xv)\cos(2\pi ux)$ over four paths from the cusps to i . This is

the analogue, for a non-holomorphic input, of the contour integrals of Viazovska and Cohn–Kumar–Miller–Radchenko–Viazovska [50, 14, 15]; Green’s identity replaces Cauchy’s theorem, and every weight vanishes on its own path. The result $\mathcal{G}(x)$ is analytic for $\operatorname{Re} x > 0$, equals $-16\pi^2 \sin^2(\pi x) \sqrt{x} \sum a_n K_0(4\pi \sqrt{nx})$ for $x > 1$, and its Mellin transform, divided by the Gamma factor $\Gamma_{\mathbb{R}}(w + \frac{1}{2})^2$, is even under $w \mapsto -w$ because the kernel intertwines $\tau \mapsto -1/\tau$ with $w \mapsto -w$. The absence of constant terms in the cusp expansions makes $\mathcal{G}$ vanish to order $x^{5/2}$ at 0, and this makes H entire. The growth bound (1.2) follows by integration by parts.

Positivity (Sections 6 and 7). On the Fourier side, every a_n is positive because the term $c = 1$ dominates; the trivial bound $|S| \leq c$ suffices for every n (Theorem 6.1). On the physical side, applying the four-path functional to the Mellin transform of the kernel gives

$$H_{\text{raw}}(t) = \frac{1}{2\pi^2(t^2 + \frac{1}{4})^2} \left[\int_1^\infty A(Y) \cos(t \log Y) dY + \frac{1}{2} \int_{1/2}^\infty B(Y) \mathfrak{c}_t(\theta(Y)) dY \right],$$

where A, B are K_0 -series in the a_n and the kernel $\mathfrak{c}_t$, built from conical Legendre functions, is positive by Mehler–Dirichlet’s formula (Theorems 7.2 and 7.3). An explicit lower bound for the kernel gives $H > 0$ for $|t| \geq 8$ (Theorem 7.4), and a certified computation covers $[0, 40]$ (Theorem 7.10). Section 8 assembles the proofs.

1.4. What is proved, what is computer-assisted, and what is not claimed. *Computer-assisted parts.* Exactly four ingredients rest on computations in rigorous ball arithmetic, each with a script and log in the ancillary files (Appendix A):

- (1) the window: $H > 0$ on $[0, 40]$ (Theorem 7.10), with continuum coverage by Taylor models in t ;
- (2) the numerical inputs of the large- $|t|$ theorem (Proposition 7.9): the sign of B on $[0.870, 4]$, one tail constant $\bar{\rho}(4) < 1$, and three integrals evaluated at $t = 8$;
- (3) the certified enclosures of the coefficients a_n , $n \leq 300$, used by (1), (2) and (4);
- (4) the value of the normalising constant C and its sign (Proposition 6.4); the sign also follows from (1).

Everything else is analytic: the construction and class membership of H , its entireness and the growth bound (1.2), the positivity of every a_n and the two-sided bounds (whose constants are elementary evaluations at the single point $z = 4\pi$), the physical-side formula and the positivity of its kernel, the reduction, and the deduction of Theorems 1–2 and Corollary 3 from Part I. That deduction, from the inputs listed in Remark 8.1, is also checked in Lean 4 (Appendix A.6); the inputs themselves are not formalised.

Growth. One can show that H is entire of order one but *not* of exponential type: along the imaginary axis it grows like $e^{y \log(y/2\pi e)}$. We only sketch this (Remark 5.10), since nothing uses it. What we prove and use is the strip bound (1.2), which allows growth like $e^{\pi |\operatorname{Re} z|/2}$ along horizontal lines in the strip; no bound on the exponential type of H is used or claimed.

Not claimed. We do not prove RH, nor that admissible pairs exist; by Corollary 3 each of these is equivalent to RH. We do not claim that κ^* is attained or that $\kappa^* = 0$ unconditionally, that $\kappa_{\text{OPS}}^* = 0$ implies RH, that the integer-critical function is unique, or anything about the convergence of the exact Hermite family of Part I or of other interpolation schemes. The paper has not yet been checked by a human expert (see the note on the first page and the AI-use statement).

1.5. Related work. *Magic functions and contour integrals.* Viazovska’s solution of the sphere packing problem in dimension 8 [50] and its extensions [14, 15] construct extremal functions for the linear programming bounds of Cohn and Elkies [12] as $\sin^2$ times Laplace transforms of modular forms, through contour integrals; Cohn and Gonçalves [13] gave a sign-uncertainty

analogue, and Feigenbaum, Grabner and Hardin [23] studied the Fourier eigenfunctions of this type with prescribed zeros. In those settings positivity is either visible or certified by interval arithmetic; Lee [34] proves algebraically the positivity, on the imaginary axis, of a family of quasimodular forms that gives sign-uncertainty eigenfunctions of this type. Sections 4–5 follow the architecture of Viazovska’s construction [50]: $\tilde{\Phi}$ and Φ play the role of the depth-2 quasimodular form φ_0 and its S -transform, the four paths and (up to a common factor) their weights are hers, and Green’s identity replaces Cauchy’s theorem. What is new is the input: an exact eigenfunction of eigenvalue $\frac{1}{4}$ that decays at the cusp ∞ and is a component of a Sym^2 -equivariant function, built from the odd Niebur–Poincaré series. There are partial precedents for non-holomorphic input: the closed Green form and the deformation of paths between cusps of Lewis and Zagier [35], and the Maass–Poincaré series that Radchenko and Sun [42] continue in the spectral parameter to build Fourier interpolation bases. The Fourier interpolation basis of Bondarenko, Radchenko and Seip [7] and the interpolation theorem of Radchenko and Viazovska [43] are the analytic input of Theorem I.3.6.

Exact functionals and zero-killing devices elsewhere. Exact extremal functionals with double zeros at the expected spectrum also occur outside modular forms, in the conformal bootstrap: Mazáč [37] constructs them for the one-dimensional bootstrap, and Hartman, Mazáč and Rastelli [27] adapt such functionals to the modular bootstrap, where they reproduce the magic functions of dimensions 8 and 24. Devices that remove the zeros of ζ also appear elsewhere: Connes and Consani [16] note that the map $\mathcal{E}(f)(x) = x^{1/2} \sum_{n \geq 1} f(nx)$, on even Schwartz functions with $f(0) = \hat{f}(0) = 0$, takes values in the radical of the Weil quadratic form, and Benjamin and Chang [3] remove the contributions of the zeros from a crossing equation of the modular bootstrap with functionals whose Mellin symbol carries the factor $\zeta(s)$. Shen [45], in an unrefereed preprint, sets out the dictionary between the Cohn–Elkies bound and Weil’s positivity criterion and shows that no compactly supported test function can play, for ζ , the role of the Cohn–Elkies magic function; the function F of Theorem 1 is not compactly supported. Extremal problems of Fourier analysis attached to the explicit formula, mostly over band-limited test functions, are studied by Carneiro and coauthors, for instance for gaps between primes [11] and for the pair correlation of zeros [10].

Niebur–Poincaré series and spectral derivatives. Niebur [38] and Fay [22] introduced and expanded the series we use; the expansion with the constant 2 is stated by Bringmann, Jorgenson and Smajlović [9, §2.4], and we prove it from scratch (Lemma 4.3). Spectral derivatives of Poincaré and Eisenstein series at degenerate points appear in the theory of polyharmonic and sesquiharmonic Maass forms: Andersen, Lagarias and Rhoades [2, Theorem 1.1] describe the scalar critical point $s = \frac{1}{2}$, Bringmann, Diamantis and Raum [8] and Matsusaka [36] differentiate Poincaré series at harmonic points, and Alfes, Burban and Raum [1] combine symmetric powers with spectral derivatives, in the framework of Bernstein and Gelfand [5], at eigenvalue 0.

Period functions and self-reciprocity. Lewis and Zagier [35] also introduced the Ferrar-type kernels of eigenvalue- $\frac{1}{4}$ forms; Bettin and Conrey [6] treat the period function of the Eisenstein series at $s = \frac{1}{2}$, whose Voronoi kernel is $4K_0 - 2\pi Y_0$. The self-reciprocity of our Γ -only function under the Koshliakov kernel, equivalent to the evenness of H , is the classical Mellin mechanism of Koshliakov, Dixit and others [33, 17, 18, 4, 19].

What is new. We have not found the following in the literature we searched: the function of Theorem 4 and its coefficients; the double-root lift of a spectral derivative of a Niebur–Poincaré series to an exact eigenfunction of eigenvalue $\frac{1}{4}$, with its symmetric-square partner; Viazovska’s contour lift run with Green’s identity on this non-holomorphic input; the positivity of the a_n , which follows from the standard dominance of the term $c = 1$, and the positivity of H ; and the resulting exact magic function for the explicit formula, which gives the bound $\kappa^* \leq \kappa_{\text{OPS}}^* \leq 0$.

Each ingredient has the precedents just listed, and the search covered only literature reachable through standard bibliographic databases.

History. The input was found by analogy with a holomorphic model case. For a model problem whose kernel is K_1 instead of K_0 , Viazovska’s contour argument with the depth-2 quasimodular form $E_4 - E_2^2 = 288 \sum_n n \sigma_1(n) q^n$ gives, up to a constant factor, the function $\sin^2(\pi x) \sum_n \sigma_1(n) \sqrt{nx} K_1(4\pi \sqrt{nx})$, which is positive off the integers and vanishes doubly at the positive integers; it is an object of the type studied by Feigenbaum, Grabner and Hardin, formally the case $d = 0$ of their family [23]. Its Eisenstein analogue, built from the completed Eisenstein series $E^*(\tau, \frac{1}{2})$, is $\sin^2(\pi x) \sum_n n d(n) K_0(4\pi \sqrt{nx})$: it has the right kernel, but the constant term of the Eisenstein series makes the corresponding H singular at $t = \pm \frac{i}{2}$ (compare the remark after Lemma 5.6). Numerical interpolation at the integer nodes, whose canonical interpolants only semi-converge, and a numerical construction of Sym^2 -equivariant eigenvalue- $\frac{1}{4}$ functions without constant terms then singled out one candidate, which was recognised, first numerically and then in closed form, as the spectral derivative of Section 4. The proofs below were written for that function, and no step of them uses these computations.

1.6. Organisation and notation. Section 2 recalls the setting and the results of Part I that are used, with full statements. Section 3 proves the reduction, Section 4 constructs the automorphic input, Section 5 the lift, and Sections 6 and 7 prove positivity. Section 8 proves the main results, and Section 9 discusses other archimedean data and open questions. Appendix A describes the certificates and Appendix B the Bessel-function bounds. We use $x = e^{2\pi\xi}$ throughout, so that the node $\xi_n = \log n/2\pi$ is $x = n$ and the gap $[\xi_2, \infty)$ is $x \geq 2$; $\tau = u + iv$ is the variable on the upper half-plane $\mathbb{H}$; and $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2} \Gamma(s/2)$.

2. THE SETTING, AND THE RESULTS OF PART I THAT ARE USED

This section recalls, with full statements, the framework of Part I [32] and the results of Part I on which the proof rests: the explicit formula (Lemma I.2.3), the a priori bounds (Lemma I.2.6), duality and the logic of the conjectures (Theorems I.2.7(a) and I.2.9), complementary slackness and the zero-side support theorem with its corollary (Lemma I.2.13, Theorem I.3.6, Corollary I.3.9(a)), and the explicit formula for zero-killing functions with the bound it needs (Lemma 2.1, Corollary I.4.4). A result restated from Part I keeps its number there, with the prefix I: Theorem I.3.6 below is Theorem 3.6 of Part I. These numbers, and references of the form “Part I, Theorem 2.7”, are to version 1.3 of [32] (Zenodo, doi:10.5281/zenodo.23217537; the concept DOI 10.5281/zenodo.23197915 resolves to the latest version).

What is reproduced, and what is not. The proofs of Lemmas I.2.2(a), I.2.6 and I.2.13, of Corollaries I.3.9(a) and I.4.4, and of the easy implications of Theorems I.2.7(a) and I.2.9 are given below. The following proofs and inputs are *not* reproduced; for each we give the exact reference and the idea:

- (P1) the explicit formula on $\mathcal{T}$ (Lemma I.2.3) and the bound (2.4) (Lemma 2.1), which are classical;
- (P2) the implication (ii) $\Rightarrow$ (i) of the duality theorem (Theorem I.2.7(a); Part I, Appendix B.1.1);
- (P3) part (b) of the logic theorem (Theorem I.2.9(b); Part I, Appendix B.1.2);
- (P4) the zero-side support theorem (Theorem I.3.6; Part I, Appendix B.1.4), whose proof uses Lemma I.2.6, Theorem I.2.9(b), [7, Theorem 1.1], [43] and (P5);
- (P5) the membership in $\mathcal{T}$ of the basis functions U_m of [7, Theorem 1.1] (more precisely, of their real and imaginary parts), on which the proof of Theorem I.3.6 tests admissibility: Bondarenko, Radchenko and Seip state rapid decay in vertical strips, and that this decay is uniform on every closed strip $|\text{Im } z| \leq \frac{1}{2} + \delta$ is Part I’s reading of [7, §4.3] (Part I, Definition 3.4), which this paper does not re-verify.

Accordingly, Theorem 1 uses from Part I only definitions and the classical facts (P1), and does not depend on (P2)–(P5); Theorem 2 rests in addition on (P3), (P4) and (P5); and Corollary 3 rests on (P1)–(P5). Part I's Propositions 2.8 and 2.10 are mentioned below only for interpretation and, in Appendix A.6, for the Lean formalisation; no proof in this paper depends on them.

2.1. Notation. We write $\widehat{F}(\xi) = \int_{\mathbb{R}} F(t)e^{-2\pi i \xi t} dt$, $\xi_n = \log n/(2\pi)$ for real $n > 0$, and $\xi_2 = \log 2/(2\pi)$ for the *gap*. The multiplicative coordinate is $x = e^{2\pi i \xi}$: the node ξ_n corresponds to $x = n$ and the gap $[\xi_2, \infty)$ to $x \geq 2$. $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$, $\xi(s) = \frac{1}{2}s(s-1)\Gamma_{\mathbb{R}}(s)\zeta(s)$, $\Xi(t) = \xi(\frac{1}{2} + it)$, and

$$(2.1) \quad \gamma_{\infty}(t) := \frac{1}{2}s(s-1)\Gamma_{\mathbb{R}}(s) \Big|_{s=\frac{1}{2}+it}, \quad \text{so that} \quad \Xi(t) = \gamma_{\infty}(t)\zeta(\tfrac{1}{2} + it).$$

Ξ is entire, even and real on $\mathbb{R}$; γ_{∞} is analytic for $\text{Im } t < \frac{5}{2}$ (its poles are at $t = i(2k + \frac{1}{2})$, $k \geq 1$), has no zeros there except $t = -\frac{i}{2}$ ($s = 1$), and $\gamma_{\infty}(-t) = \overline{\gamma_{\infty}(t)}$ for real t . $Z_{\zeta} = \{\gamma \in \mathbb{R} : \zeta(\frac{1}{2} + i\gamma) = 0\}$, and for a non-trivial zero ρ we put $t_{\rho} := -i(\rho - \frac{1}{2})$, so that t_{ρ} is real exactly when ρ is on the critical line. Λ is von Mangoldt's function, $d(m)$ the number of divisors of m , and PP the set of prime powers.

2.2. The test class and the explicit formula. The archimedean data of ζ are the density and the functional

$$(2.2) \quad \Omega_{\infty}(t) := \frac{1}{2\pi} \left(\text{Re } \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \log \pi \right), \quad \mathcal{A}(F) := F\left(\frac{i}{2}\right) + F\left(-\frac{i}{2}\right) + \int_{\mathbb{R}} F(t)\Omega_{\infty}(t) dt,$$

where $\psi = \Gamma'/\Gamma$; Ω_{∞} is even, increasing in $|t|$, and $|\Omega_{\infty}(t)| \leq C \log(2 + |t|)$.

Definition I.2.1 (Test class). For $0 < \delta < \frac{1}{2}$ let $\mathcal{T}_{\delta}$ be the set of even functions F , real on $\mathbb{R}$, analytic in a neighbourhood of the closed strip $S_{\delta} := \{| \text{Im } z | \leq \frac{1}{2} + \delta\}$, with $\|F\|_{\delta} := \sup_{S_{\delta}} (1 + |z|)^2 |F(z)| < \infty$. Put $\mathcal{T} := \bigcup_{\delta > 0} \mathcal{T}_{\delta}$.

Lemma I.2.2(a). *If $F \in \mathcal{T}_{\delta}$, then $\widehat{F}$ is real and even, and $|\widehat{F}(\xi)| \leq \pi \|F\|_{\delta} e^{-\pi(1+2\delta)|\xi|}$ for all real ξ .*

Proof. Let $\xi > 0$ and $c = \frac{1}{2} + \delta$. By Cauchy's theorem (the vertical connectors vanish because $|F(z)| \leq \|F\|_{\delta}(1 + |z|)^{-2}$), $\widehat{F}(\xi) = e^{-2\pi c \xi} \int_{\mathbb{R}} F(t - ic)e^{-2\pi i \xi t} dt$, and the integral is at most $\|F\|_{\delta} \int (1 + t^2)^{-1} dt = \pi \|F\|_{\delta}$ in absolute value. For $\xi < 0$ shift to $\text{Im } t = +c$. Reality and evenness follow from F being even and real on $\mathbb{R}$. $\square$

Lemma I.2.3 (Explicit formula on $\mathcal{T}$; [52], [7, (1.1)]). *For every $F \in \mathcal{T}$,*

$$(2.3) \quad \mathcal{A}(F) = \sum_{\rho} F(t_{\rho}) + \frac{1}{\pi} \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \widehat{F}(\xi_n),$$

where ρ runs over the non-trivial zeros of ζ , counted with multiplicity. Both series converge absolutely. No hypothesis on the zeros is made.

This is formula (1.1) of [7], which holds for functions analytic in $| \text{Im } z | < \frac{1}{2} + \varepsilon$ with $|f(z)| \ll (1 + |z|)^{-1-\eta}$ for some $\varepsilon, \eta > 0$, a class that contains $\mathcal{T}$. There the archimedean integral carries $\psi(\frac{1}{4} + \frac{it}{2})$ and the prime sum carries $\frac{1}{2\pi}(\widehat{F}(\xi_n) + \widehat{F}(-\xi_n))$; for even F , real on $\mathbb{R}$, these reduce to $\text{Re } \psi$ and to $\frac{1}{\pi}\widehat{F}(\xi_n)$. Absolute convergence follows from $|F(t_{\rho})| \leq \|F\|_{\delta}(1 + |\text{Im } \rho|)^{-2}$, the bound $O(T \log T)$ for the number of zeros up to height T , and Lemma I.2.2(a).

2.3. Admissible pairs, the slack and the conjectures.

Definition I.2.4 (Admissible pairs, cones, slack). An *admissible pair* is a pair (μ, ν) of positive Radon measures, μ even on $\mathbb{R}$ and ν on $[\xi_2, \infty)$, such that for every $F \in \mathcal{T}$ the integrals $\int F d\mu$ and $\int \widehat{F} d\nu$ converge absolutely and

$$\mathcal{A}(F) = \int_{\mathbb{R}} F d\mu + \frac{1}{\pi} \int_{[\xi_2, \infty)} \widehat{F} d\nu.$$

$\mathcal{K}$ denotes the set of admissible pairs, and $p_\zeta := (\mu_\zeta, \nu_\zeta)$ with $\mu_\zeta := \sum_{\gamma \in Z_\zeta} m_\gamma \delta_\gamma$ (m_γ the multiplicity) and $\nu_\zeta := \sum_{n \geq 2} \Lambda(n) n^{-1/2} \delta_{\xi_n}$. Put

$$\mathcal{C} := \{F \in \mathcal{T} : F \geq 0 \text{ on } \mathbb{R}, \widehat{F} \geq 0 \text{ on } [\xi_2, \infty)\}, \quad \mathcal{C}_{\text{OPS}} := \{F \in \mathcal{T} : F \geq 0 \text{ and } \widehat{F} \geq 0 \text{ on } \mathbb{R}\},$$

$$\kappa^* := \inf \left\{ \mathcal{A}(F) : F \in \mathcal{C}, \int F = 1 \right\}, \quad \kappa_{\text{OPS}}^* := \inf \left\{ \mathcal{A}(F) : F \in \mathcal{C}_{\text{OPS}}, \int F = 1 \right\}.$$

Since $\mathcal{C}_{\text{OPS}} \subseteq \mathcal{C}$, $\kappa^* \leq \kappa_{\text{OPS}}^*$. We consider the statements

$$(E) : \mathcal{K} \neq \emptyset, \quad (U) : \mathcal{K} \subseteq \{p_\zeta\}, \quad (S) : \kappa^* \leq 0.$$

By the explicit formula, p_ζ is admissible if and only if RH holds (Theorem I.2.9). For $q > 0$ let $\mathcal{A}_q(F) := \mathcal{A}(F) + \frac{\log q}{2\pi} \int F$, the archimedean functional of data with Gamma factor $\Gamma_{\mathbb{R}}$, one simple pole at $s = 1$ and conductor q . By Part I, Proposition 2.10, admissible pairs for $\mathcal{A}_q$ exist if and only if $q \geq q_{\min} := e^{-2\pi\kappa^*}$; so $q_{\min}$ is the best lower bound for the conductor that the explicit formula gives, over test functions in $\mathcal{C}$, from these data, and (S) says $q_{\min} \geq 1$. The cone $\mathcal{C}_{\text{OPS}}$ is the classical cone of Odlyzko, Poitou and Serre [40, 41, 44, 46], and κ_{OPS}^* gives the corresponding classical bound.

Conjecture U (Part I, Conjecture U). *Every admissible pair, if there is one, is ζ 's pair: $\mathcal{K} \subseteq \{p_\zeta\}$.*

Conjecture S (Part I, Conjecture S). $\kappa^* \leq 0$. *Equivalently, $q_{\min} \geq 1$.*

Definition I.2.11. An *exact magic function* is an $F \in \mathcal{C}$ with $\int F > 0$ and $\mathcal{A}(F) = 0$. For a function F on $\mathbb{R}$ with transform $\widehat{F}$ we write $Z(F) := F^{-1}(0) \cap \mathbb{R}$ and $\widehat{Z}(F) := \widehat{F}^{-1}(0) \cap [\xi_2, \infty)$.

If F is an exact magic function, then $F/\int F$ is admissible in the definition of κ^* , so $\kappa^* \leq 0$.

2.4. The results of Part I that are used.

Lemma I.2.6 (A priori bounds). *There is an absolute constant C such that every $(\mu, \nu) \in \mathcal{K}$ satisfies*

$$\mu([T-1, T+1]) \leq C \log(3 + |T|) \quad (T \in \mathbb{R}), \quad \nu([X-1, X+1]) \leq C e^{\pi X} \quad (X \in \mathbb{R}).$$

In particular, if $\nu^\sharp$ denotes the image of ν under $\xi \mapsto e^{2\pi\xi}$, then $\nu^\sharp([R, 2R]) \leq CR^{1/2}$ for every $R > 0$.

Proof. This is the proof of Part I, Appendix B.1.2. Let $g(t) = e^{-\pi t^2}$, so that $\widehat{g} = g$, and put $A_T := g(\cdot - T) + 2g + g(\cdot + T)$ and $B_X(t) := 2g(t)(1 + \cos 2\pi X t)$. Both are even, lie in $\mathcal{T}$ and are non-negative on $\mathbb{R}$, and $\widehat{A_T}(\xi) = 2g(\xi)(1 + \cos 2\pi T\xi) \geq 0$, $\widehat{B_X} = 2g + g(\cdot - X) + g(\cdot + X) \geq 0$; so both lie in $\mathcal{C}$. As $A_T \geq e^{-\pi}$ on $[T-1, T+1]$ and $\widehat{B_X} \geq e^{-\pi}$ on $[X-1, X+1]$, and both integrands in Definition I.2.4 are non-negative where the measures live, $\mu([T-1, T+1]) \leq e^\pi \mathcal{A}(A_T)$ and $\nu([X-1, X+1]) \leq \pi e^\pi \mathcal{A}(B_X)$. Finally $|A_T(\pm \frac{i}{2})| \leq 4e^{\pi/4}$, $B_X(\pm \frac{i}{2}) = 2e^{\pi/4}(1 + \cosh \pi X)$ and $|\Omega_\infty(t)| \leq C \log(2 + |t|)$ give $\mathcal{A}(A_T) \leq C \log(3 + |T|)$ and $\mathcal{A}(B_X) \leq C(1 + \cosh \pi X)$; for $X < -1$ the left side of the second bound vanishes, since ν lives on $[\xi_2, \infty)$. The last claim follows, as $[R, 2R]$ corresponds to an interval of length $\log 2/2\pi < 1$ starting at $X = \log R/2\pi$. $\square$

Theorem I.2.7(a) (Duality). *The following are equivalent: (i) $\mathcal{K} \neq \emptyset$; (ii) $\mathcal{A}(F) \geq 0$ for every $F \in \mathcal{C}$; (iii) $\kappa^* \geq 0$.*

Part I's statement also lists, as its condition (iii), condition (ii) for the even real combinations of Gaussian wave packets only; our (iii) is its (iv). The implication (i) $\Rightarrow$ (ii) is weak duality: for $(\mu, \nu) \in \mathcal{K}$ and $F \in \mathcal{C}$ both integrals in Definition I.2.4 are non-negative. (ii) $\Leftrightarrow$ (iii) is homogeneity (an $F \in \mathcal{C}$ with $\int F = 0$ vanishes identically). The implication (ii) $\Rightarrow$ (i) is conic linear-programming duality and is not reproduced here: Part I, Appendix B.1.1, proves it with Carathéodory's theorem in finite-dimensional subspaces of Gaussian wave packets, uniform bounds for positive solutions obtained as in the proof of Lemma I.2.6, and Helly's selection theorem.

Theorem I.2.9 (Logic; parts (a)–(c)). (a) *RH implies $p_\zeta \in \mathcal{K}$; hence RH implies (E).*

(b) *If $(\mu, \nu_\zeta) \in \mathcal{K}$ for some μ , then RH holds and $\mu = \mu_\zeta$.*

(c) (U) *holds if and only if both [(E) $\Rightarrow$ RH] and [RH $\Rightarrow \mathcal{K} = \{p_\zeta\}$] hold. In particular (U) implies RH $\Leftrightarrow$ (E) $\Leftrightarrow \kappa^* \geq 0$.*

Part (a) is Lemma I.2.3: under RH every t_ρ is real. Part (b) is not reproduced here; it is proved in Part I, Appendix B.1.2: if a zero $\rho_0 = \beta_0 + i\gamma_0$ were off the line, the Gaussians $g_u(t - \gamma_0) + g_u(t + \gamma_0)$, $g_u(t) = e^{-\pi u t^2}$, would make $\sum_\rho F(t_\rho)$ grow like $e^{\pi u(\beta_0 - \frac{1}{2})^2}$ along a sequence $u \rightarrow \infty$, while $\int F d\mu$ stays bounded. Part (c) follows from (a), (b) and Theorem I.2.7(a): under (U) and (E) the unique pair is p_ζ , hence RH by (b); under (U) and RH, $\mathcal{K} = \{p_\zeta\}$ by (a); and conversely the two implications give $\mathcal{K} = \{p_\zeta\}$ under RH and $\mathcal{K} = \emptyset$ under $\neg$ RH.

Lemma I.2.13 (Complementary slackness). *If $F \in \mathcal{C}$ and $\mathcal{A}(F) = 0$, then every $(\mu, \nu) \in \mathcal{K}$ has μ carried by $Z(F)$ and ν carried by $\widehat{Z}(F)$.*

Proof. $0 = \mathcal{A}(F) = \int F d\mu + \frac{1}{\pi} \int \widehat{F} d\nu$, with continuous integrands that are non-negative where the measures live. $\square$

Theorem I.3.6 (Zero-side support). *Let $(\mu, \nu) \in \mathcal{K}$, and assume that μ is carried by $Z_\zeta \cup \{0\}$. Then RH holds and $(\mu, \nu) = p_\zeta$.*

No hypothesis on ν is made beyond admissibility. The proof is in Part I, Appendix B.1.4, and is not reproduced here. *Idea.* The input is the Fourier interpolation basis of Bondarenko, Radchenko and Seip [7, Theorem 1.1], which assumes neither RH nor simplicity of the zeros, and the interpolation theorem of Radchenko and Viazovska [43]. With $\nu^\sharp$ as in Lemma I.2.6, the a priori bound $\nu^\sharp([R, 2R]) \leq CR^{1/2}$ makes $L(f) := \int \sqrt{x} \sum_{k \geq 1} f(kx) d\nu^\sharp(x)$ continuous on even Schwartz functions. Testing admissibility on the basis functions U_m of [7], which vanish at every zero of ζ and are test functions by (P5), computes L on the Radchenko–Viazovska basis; by Schwartz convergence of their interpolation series this gives, for every even self-dual Schwartz f with $f(0) = 0$, $L(f) = \sum_{n \geq 2} (\log n) f(n) - \frac{1}{2} \mu(\{0\}) \zeta(\frac{1}{2}) \int_0^\infty f(x) x^{-1/2} dx$. A limiting test with $g(t) = \sin^2(\pi t) / (\pi^2(t^2 - 1))$, which vanishes at the integers and is positive at all other points of $(1, \infty)$, and with its transform, forces $\mu(\{0\}) = 0$ and puts $\nu^\sharp$ on the integers $n \geq 2$; Möbius inversion then gives $\nu = \nu_\zeta$, and Theorem I.2.9(b) gives RH and $\mu = \mu_\zeta$.

Corollary I.3.9(a) (Magic-function principle). *Let $F \in \mathcal{C}$ with $\mathcal{A}(F) = 0$, and assume $Z(F) \subseteq Z_\zeta \cup \{0\}$. Then (U) holds. If (E) holds as well, then RH holds and $\mathcal{K} = \{p_\zeta\}$.*

Proof. Let $(\mu, \nu) \in \mathcal{K}$. By Lemma I.2.13, μ is carried by $Z(F) \subseteq Z_\zeta \cup \{0\}$, and Theorem I.3.6 gives RH and $(\mu, \nu) = p_\zeta$. So $\mathcal{K} \subseteq \{p_\zeta\}$; if moreover $\mathcal{K} \neq \emptyset$, then RH holds and $\mathcal{K} = \{p_\zeta\}$. $\square$

No condition on the zeros of $\widehat{F}$ is needed.

The last two inputs concern functions of the form $\Xi^2 H$, which vanish at every non-trivial zero of ζ , on or off the critical line.

Lemma 2.1 (The bound (4.1) of Part I; [48]). *For every $b \in (0, 1]$ there is C_b with*

$$(2.4) \quad |\Xi(t)| \leq C_b(1 + |\operatorname{Re} t|)^3 e^{-\pi|\operatorname{Re} t|/4} \quad (|\operatorname{Im} t| \leq b).$$

This follows from Stirling's formula for $\Gamma_{\mathbb{R}}$ and the convexity bound for ζ on $-\frac{1}{2} \leq \operatorname{Re} s \leq \frac{3}{2}$.

Corollary I.4.4 (The explicit formula for zero-killing functions). *Let $F = \Xi^2 H$ with H analytic on $|\operatorname{Im} t| \leq \frac{1}{2} + \delta$, and assume $F \in \mathcal{T}$. Then*

$$\mathcal{A}(F) = \frac{1}{\pi} \sum_{n \in \text{PP}} \Lambda(n) n^{-1/2} \widehat{F}(\xi_n).$$

No hypothesis on the zeros of ζ , on the growth of H , or on entireness is used.

Proof. Apply Lemma I.2.3. Every non-trivial zero ρ has $|\operatorname{Im} t_\rho| < \frac{1}{2}$ and $\Xi(t_\rho) = \xi(\rho) = 0$, so $F(t_\rho) = 0$ and the zero side vanishes. $\square$

3. REDUCTION TO A Γ -ONLY INTEGER-CRITICAL FUNCTION

A function $F = \Xi^2 H$ kills every zero of ζ (Corollary I.4.4), so $\mathcal{A}(F)$ is a sum over prime powers. This section removes the arithmetic from the prime side as well: on a line where the Dirichlet series of ζ^2 converges absolutely, $\widehat{F}$ is a positive combination of shifts of the transform of $\gamma_\infty^2 H$, which involves only the Gamma factor. The positive combination runs over the integers, which are closed under multiplication; so it suffices to make the Γ -only transform vanish at all integers $x \geq 2$ and be non-negative beyond the gap.

3.1. The growth class and the Voronoi decoupling. For H even, real on $\mathbb{R}$ and analytic on a neighbourhood of S_δ put

$$(3.1) \quad \begin{aligned} G_H(t) &:= \gamma_\infty(t)^2 H(t), & \widehat{G}_H(\xi) &:= \int_{\mathbb{R}} G_H(t) e^{-2\pi i \xi t} dt, \\ \widehat{G}_H(x) &:= \widehat{G}_H\left(\frac{\log x}{2\pi}\right) = \int_{\mathbb{R}} G_H(t) x^{-it} dt. \end{aligned}$$

We use the two readings of $\widehat{G}_H$, in ξ and in $x = e^{2\pi \xi}$, interchangeably; in particular $\widehat{G}_H(0)$ always means $\xi = 0$, i.e. $x = 1$. Note that $\widehat{G}_H$ is not even in ξ , since γ_∞ is not real on $\mathbb{R}$.

Definition 3.1 (The class $\mathcal{W}_\delta$). For $0 < \delta < 1$, $\mathcal{W}_\delta$ is the set of even functions H , real on $\mathbb{R}$, analytic on a neighbourhood of S_δ , such that

$$(C1) \quad \sup_{z \in S_\delta} (1 + |z|)^8 e^{-\pi|\operatorname{Re} z|/2} |H(z)| < \infty.$$

Condition (C1) is a bound in the strip S_δ only; it allows growth like $e^{\pi|\operatorname{Re} z|/2}$ in the strip and says nothing about the growth of H elsewhere. This is the boundary case: Part I's class $\mathcal{H}$ allows only growth like $e^{a|\operatorname{Re} z|}$ with $a < \pi/2$ in the strip, and at the rate $e^{\pi|\operatorname{Re} z|/2}$ this factor cancels exactly against the decay $|\Xi|^2 \ll e^{-\pi|\operatorname{Re} z|/2}$ of (2.4), so the power 8 is what makes $\Xi^2 H$ a test function.

Lemma 3.2 (Voronoi decoupling). *Let $0 < \delta < \frac{1}{2}$ and $H \in \mathcal{W}_\delta$. Then:*

- (a) $F := \Xi^2 H \in \mathcal{T}_\delta$.
- (b) $|G_H(z)| \leq C(1 + |\operatorname{Re} z|)^{-4+\delta}$ on S_δ ; in particular $\sup_{|y| \leq \frac{1}{2} + \delta} \int_{\mathbb{R}} |G_H(u + iy)| du < \infty$.
 $\widehat{G}_H$ is real and continuous, and $|\widehat{G}_H(\xi)| \leq C e^{-\pi(1+2\delta)|\xi|}$.

(c) For every real ξ ,

$$(3.2) \quad \widehat{F}(\xi) = \sum_{m \geq 1} d(m) m^{-1/2} \widehat{G}_H(\xi + \xi_m),$$

and for each $\eta \in (\frac{1}{2}, \frac{1}{2} + \delta]$ the m -th term is bounded by $C_\eta d(m) m^{-1/2 - \eta} e^{-2\pi\eta\xi}$; the series converges absolutely.

Proof. (a) Lemma 2.1 with $b = \frac{1}{2} + \delta < 1$ gives $|\Xi(z)|^2 \leq C(1 + |\operatorname{Re} z|)^6 e^{-\pi|\operatorname{Re} z|/2}$ on S_δ . With (C1), $|F(z)| \leq C(1 + |\operatorname{Re} z|)^6 (1 + |z|)^{-8} \leq C(1 + |z|)^{-2}$. F is even, real on $\mathbb{R}$ and analytic near S_δ .

(b) For $z \in S_\delta$ put $s = \frac{1}{2} + iz$; then $\operatorname{Re} s = \frac{1}{2} - \operatorname{Im} z \in [-\delta, 1 + \delta]$ and $\operatorname{Im} s = \operatorname{Re} z$. The function γ_∞ has no poles in S_δ . Stirling's formula [20, 5.11.9], uniformly for $\operatorname{Re} s$ in a compact interval and $|\operatorname{Im} s| \geq 1$, gives $|\gamma_\infty(z)|^2 = \frac{1}{4}|s|^2|s-1|^2 \pi^{-\operatorname{Re} s} |\Gamma(s/2)|^2 \asymp |\operatorname{Im} s|^{3+\operatorname{Re} s} e^{-\pi|\operatorname{Im} s|/2}$. With (C1), $|G_H(z)| \leq C(1 + |\operatorname{Re} z|)^{3+(1+\delta)-8}$. This gives uniform integrability on horizontal lines and uniform decay at their ends. For real t , $G_H(-t) = \overline{G_H(t)}$, so $\widehat{G}_H$ is real. The bound on $\widehat{G}_H$ follows by moving the line of integration to $\operatorname{Im} t = \mp(\frac{1}{2} + \delta)$ for $\pm\xi > 0$, as in Lemma I.2.2(a).

(c) Fix $\eta \in (\frac{1}{2}, \frac{1}{2} + \delta]$. By (a) and Cauchy's theorem, $\widehat{F}(\xi) = \int_{\operatorname{Im} w = -\eta} F(w) e^{-2\pi i \xi w} dw$. On this line $\operatorname{Re}(\frac{1}{2} + iw) = \frac{1}{2} + \eta > 1$, so $\zeta(\frac{1}{2} + iw)^2 = \sum_m d(m) m^{-1/2 - iw}$, with $\sum_m d(m) |m^{-1/2 - iw}| = \zeta(\frac{1}{2} + \eta)^2$. Since $F = G_H \zeta(\frac{1}{2} + i \cdot)^2$ and, by (b), $\int_{\operatorname{Im} w = -\eta} |G_H(w) e^{-2\pi i \xi w}| |dw| = e^{-2\pi\eta\xi} \int |G_H(u - i\eta)| du < \infty$, Fubini's theorem gives

$$\widehat{F}(\xi) = \sum_{m \geq 1} d(m) m^{-1/2} \int_{\operatorname{Im} w = -\eta} G_H(w) e^{-2\pi i (\xi + \xi_m) w} dw,$$

using $m^{-iw} = e^{-2\pi i \xi_m w}$. The function G_H is analytic on $-\eta \leq \operatorname{Im} w \leq 0$ and decays uniformly there by (b), so each line moves back to $\mathbb{R}$, and the m -th term is $d(m) m^{-1/2} \widehat{G}_H(\xi + \xi_m)$, of size at most $d(m) m^{-1/2} e^{-2\pi\eta(\xi + \xi_m)} \int |G_H(u - i\eta)| du$. $\square$

The double pole of ζ^2 at $w = -\frac{i}{2}$ is crossed only by F as a whole, where Ξ is entire, and never by a single term.

Proposition 3.3 (Exact magic functions from Γ -only data). *Let $0 < \delta < \frac{1}{2}$ and $H \in \mathcal{W}_\delta$, $H \not\equiv 0$, and assume*

- (C2) $H \geq 0$ on $\mathbb{R}$;
- (C3) $\widehat{G}_H(\xi_k) = 0$ for every integer $k \geq 2$;
- (C4) $\widehat{G}_H \geq 0$ on $[\xi_2, \infty)$.

Then $F := \Xi^2 H$ lies in $\mathcal{C}$, $\widehat{F}(\xi_n) = 0$ for every integer $n \geq 2$, $\int F = \widehat{F}(0) = \widehat{G}_H(0) > 0$ and $\mathcal{A}(F) = 0$. So F is an exact magic function and $\kappa^ \leq 0$. If moreover $\widehat{G}_H \geq 0$ on $[0, \infty)$, then $F \in \mathcal{C}_{\text{OPS}}$ and $\kappa_{\text{OPS}}^* \leq 0$.*

Proof. By Lemma 3.2(a), $F \in \mathcal{T}$, and $F \geq 0$ on $\mathbb{R}$ because Ξ is real there. If $\xi \geq \xi_2$, then every argument $\xi + \xi_m$ in (3.2) is $\geq \xi_2$, and the weights $d(m) m^{-1/2}$ are positive; so $\widehat{F}(\xi) \geq 0$ by (C4), and $F \in \mathcal{C}$. For $n \geq 2$, $\widehat{F}(\xi_n) = \sum_m d(m) m^{-1/2} \widehat{G}_H(\xi_{nm}) = 0$ by (C3). Corollary I.4.4 applies, since H is analytic on S_δ and $F \in \mathcal{T}$, and gives $\mathcal{A}(F) = \frac{1}{\pi} \sum_{n \in \text{PP}} \Lambda(n) n^{-1/2} \widehat{F}(\xi_n) = 0$. At $\xi = 0$, (3.2) and (C3) give $\widehat{F}(0) = \widehat{G}_H(0)$. Since $F \geq 0$ is continuous and not identically zero (Ξ has isolated zeros and $H \not\equiv 0$), $\int F > 0$; then $F/\int F \in \mathcal{C}$ has $\int = 1$ and $\mathcal{A} = 0$, so $\kappa^* \leq 0$. If $\widehat{G}_H \geq 0$ on $[0, \infty)$, the same argument gives $\widehat{F} \geq 0$ on $[0, \infty)$, and $\widehat{F}$ is even (Lemma I.2.2(a)); so $F \in \mathcal{C}_{\text{OPS}}$ and $\kappa_{\text{OPS}}^* \leq 0$. $\square$

No zero in (C3) needs to be double, and no condition on the zeros of $\widehat{G}_H$ or $\widehat{F}$ off the integers is used. Only the prime powers are needed in Corollary I.4.4; vanishing at all integers is what the decoupling requires.

Remark 3.4 (The converse direction; not used). Since $\sum_{d|m}(\mu * \mu)(d) d(m/d) = \delta_{m1}$, where μ is the Möbius function, (3.2) can be inverted at the integer nodes: $\widehat{G}_H(\xi_n) = \sum_m(\mu * \mu)(m) m^{-1/2} \widehat{F}(\xi_{mn})$, the double series converging absolutely by the decay in Lemma 3.2. So (C3) is equivalent to $\widehat{F}(\xi_n) = 0$ for all $n \geq 2$: the arithmetic-free problem at the integer nodes is equivalent to the problem for $\Xi^2 H$. We use only the direction proved in Proposition 3.3.

Following the terminology of the introduction, we call an $H \in \mathcal{W}_\delta$ with $H(0) = 1$ that satisfies (C2)–(C4) a Γ -only integer-critical function. The rest of the paper constructs one, with $H > 0$ on all of $\mathbb{R}$ and $\widehat{G}_H \geq 0$ on $[0, \infty)$.

4. THE AUTOMORPHIC INPUT

This section constructs the eigenfunction of Δ with eigenvalue $\frac{1}{4}$ from which the integer-critical function is lifted in Section 5: a spectral derivative of an odd Niebur–Poincaré series, made into an exact eigenfunction by a first-order operator at a double root, together with its symmetric-square partners. It is 1-periodic, odd and exponentially decaying at the cusp ∞ , but it is not $\mathrm{SL}_2(\mathbb{Z})$ -invariant (Remark 4.10); it is the Y^2 -component of a vector-valued function that is equivariant for the symmetric square (Lemma 4.8).

Notation. $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, $\Gamma_\infty = \{\pm \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}\}$, $\mathbb{H}$ the upper half-plane with $\tau = u + iv$, $e(x) = e^{2\pi ix}$, and $\Delta = -v^2(\partial_u^2 + \partial_v^2)$. $S = \begin{pmatrix} 0 & k \\ 1 & 0 \end{pmatrix}$ acts by $S\tau = -1/\tau$. $S(m, n; c) = \sum_{d \bmod c, (d, c)=1} e((m\bar{d} + nd)/c)$ is the Kloosterman sum; it is real, $S(1, -n; c) = S(-1, n; c)$, and we use only the trivial bound $|S(m, n; c)| \leq \varphi(c) \leq c$. The letter ν denotes a spectral parameter in this section and the next; it should not be confused with the prime measure of an admissible pair.

4.1. Bessel functions. We use the modified Bessel functions I_μ, K_μ , the Bessel functions J_μ, Y_μ , and the order derivatives

$$\dot{I}_2(z) := \partial_\mu I_\mu(z)|_{\mu=2}, \quad \dot{J}_2(z) := \partial_\mu J_\mu(z)|_{\mu=2} \quad (z > 0).$$

Lemma 4.1. (a) For $\mathrm{Re} \mu \geq 0$ and $y > 0$: $|I_\mu(y)|, |J_\mu(y)| \leq (y/2)^{\mathrm{Re} \mu} I_0(y)/|\Gamma(\mu + 1)| \leq (y/2)^{\mathrm{Re} \mu} e^y/|\Gamma(\mu + 1)|$.

(b) $\partial_\mu I_\mu|_{\mu=0} = -K_0$, $\partial_\mu K_\mu|_{\mu=0} = 0$, $\partial_\mu K_\mu|_{\mu=1} = K_0(z)/z$, and

$$(4.1) \quad \partial_\mu I_\mu(z)|_{\mu=1} = K_1(z) - \frac{I_0(z)}{z}.$$

(c) $\dot{I}_2(z) = -K_2(z) + 2I_0(z)/z^2 - 2I_1(z)/z$ and $\dot{J}_2(z) = \frac{\pi}{2}Y_2(z) + 2J_0(z)/z^2 + 2J_1(z)/z$.

Proof. (a) In the series $I_\mu(y) = (y/2)^\mu \sum_k (y^2/4)^k / (k! \Gamma(k + \mu + 1))$ write $\Gamma(k + \mu + 1) = \Gamma(\mu + 1)(\mu + 1)_k$ and use $|(\mu + 1)_k| \geq k!$; the series for J_μ has the same moduli. (b) The first three identities are [20, 10.38.3] with $n = 0$ and [20, 10.38.4] with $n = 0$ and $n = 1$. For (4.1), write $\dot{I}_\mu := \partial_\mu I_\mu$. Differentiating $I_{\nu-1} - I_{\nu+1} = (2\nu/z)I_\nu$ in ν at $\nu = 0$ gives $\dot{I}_{-1} - \dot{I}_1 = (2/z)I_0$; differentiating $I_{-\nu} = I_\nu + (2/\pi) \sin(\nu\pi)K_\nu$ at $\nu = 1$ gives $-\dot{I}_{-1} = \dot{I}_1 - 2K_1$. Eliminating $\dot{I}_{-1}$ gives $\dot{I}_1 = K_1 - I_0/z$. (c) These are [20, 10.38.3] and [20, 10.15.3] with $n = 2$. $\square$

The sign in (4.1) is load-bearing: it determines the growing modes in Proposition 4.9(d), hence the exponential decay of D_{rest} , the double pole in Theorem 5.4(d) and the value $\mathcal{G}(1) = -\frac{1}{2}$. Bounds for $\dot{I}_2$ and $\dot{J}_2$ are collected in Appendix B.

4.2. The odd Niebur–Poincaré series. For $\nu \in \mathbb{C}$ and $m \in \{\pm 1\}$ put $f_{\nu,m}(\tau) := \sqrt{v} I_\nu(2\pi v) e(mu)$ and

$$f_\nu := \frac{f_{\nu,1} - f_{\nu,-1}}{2i} = \sqrt{v} I_\nu(2\pi v) \sin(2\pi u), \quad P_\nu(\tau) := \sum_{\gamma \in \Gamma_\infty \backslash \Gamma} f_\nu(\gamma\tau) \quad (\operatorname{Re} \nu > \tfrac{1}{2}).$$

Each $f_{\nu,m}$ satisfies $\Delta f = (\frac{1}{4} - \nu^2)f$, since $W(v) = \sqrt{v} I_\nu(2\pi v)$ solves $v^2 W'' - 4\pi^2 v^2 W + (\frac{1}{4} - \nu^2)W = 0$.

Proposition 4.2. (a) *The series P_ν converges absolutely and uniformly on compact subsets of $\mathbb{H} \times \{\operatorname{Re} \nu > \frac{1}{2}\}$, together with the series of all its τ -derivatives.*

(b) *P_ν is Γ -invariant, real-analytic in τ , holomorphic in ν , satisfies $\Delta P_\nu = (\frac{1}{4} - \nu^2)P_\nu$, and is odd: $P_\nu(-\bar{\tau}) = -P_\nu(\tau)$.*

(c) *τ -derivatives commute with ∂_ν ; in particular ∂_ν commutes with Δ .*

Proof. (a) By Lemma 4.1(a), $|f_\nu(\tau)| \leq |\Gamma(\nu + 1)|^{-1} \pi^{\operatorname{Re} \nu} v^{\operatorname{Re} \nu + 1/2} e^{2\pi v}$, and, from $I'_\nu = I_{\nu+1} + (\nu/z)I_\nu$, the hyperbolic gradient norm $v|\nabla f_\nu|$ obeys a bound of the same form with an extra factor $(1+v)^2$. For $\gamma \notin \Gamma_\infty$ and τ in a compact $K \subset \mathbb{H}$, $\operatorname{Im} \gamma\tau \leq 1/v \leq 1/v_K$, so $|f_\nu(\gamma\tau)| + (\operatorname{Im} \gamma\tau)|\nabla f_\nu|(\gamma\tau) \leq C_{\nu,K} (\operatorname{Im} \gamma\tau)^{\operatorname{Re} \nu + 1/2}$, locally uniformly in ν . The Eisenstein series $\sum_{\Gamma_\infty \backslash \Gamma} (\operatorname{Im} \gamma\tau)^\sigma$ converges uniformly on K for $\sigma > 1$ [29, §3], and $\operatorname{Re} \nu + \frac{1}{2} > 1$. This gives (a) for the series and its first derivatives. Each term solves the elliptic equation $(\Delta - \frac{1}{4} + \nu^2)f = 0$ with analytic coefficients, so locally uniform convergence of the solutions implies locally uniform convergence of all their derivatives (interior Schauder estimates, [24, Theorems 6.2 and 6.17]). (b) Absolute convergence allows reindexing $\gamma \mapsto \gamma g$; the eigen-equation passes to the limit by (a); each term is entire in ν and the convergence is locally uniform. For oddness, $\varepsilon := \operatorname{diag}(-1, 1)$ normalises Γ and Γ_∞ , $(\varepsilon\gamma\varepsilon^{-1})(-\bar{\tau}) = -\bar{\gamma\tau}$, and $f_\nu(-\bar{Z}) = -f_\nu(Z)$. (c) Cauchy's integral formula in ν on a small circle, with the bounds of (a). $\square$

4.3. The Fourier expansion.

Lemma 4.3 (The Weyl-element integral). *Let $\operatorname{Re} \nu > 0$, $\varepsilon \in \{\pm 1\}$, $\kappa > 0$, and $g_\varepsilon(Z) := \sqrt{\operatorname{Im} Z} I_\nu(2\pi \operatorname{Im} Z) e(\varepsilon \operatorname{Re} Z)$. For $Y > 0$ the integral $\Psi_\varepsilon(\kappa, Y) := \int_{\mathbb{R}} g_\varepsilon(-\kappa/(U + iY)) e(-U) dU$ converges absolutely, and*

$$\Psi_+(\kappa, Y) = 2\sqrt{\kappa} J_{2\nu}(4\pi\sqrt{\kappa}) \sqrt{Y} K_\nu(2\pi Y), \quad \Psi_-(\kappa, Y) = 2\sqrt{\kappa} I_{2\nu}(4\pi\sqrt{\kappa}) \sqrt{Y} K_\nu(2\pi Y).$$

Proof. Put $w = U + iY$ and $G(w) := g_\varepsilon(-\kappa/w)$. Since $w \mapsto -\kappa/w$ is an isometry of $\mathbb{H}$ and $\operatorname{Im}(-\kappa/w) = \kappa Y/|w|^2$, Lemma 4.1(a) gives

$$(4.2) \quad |G(w)| + Y|\nabla G(w)| \leq C(\kappa Y)^{\operatorname{Re} \nu + 1/2} |w|^{-2\operatorname{Re} \nu - 1} \quad \text{whenever } \kappa Y \leq |w|^2.$$

Step 1: the shape in Y . G is a continuous eigenfunction of Δ with eigenvalue $\lambda := \frac{1}{4} - \nu^2$, and by (4.2) the function $w \mapsto \int_{\mathbb{R}} G(w+t)e(-t) dt$ converges absolutely, locally uniformly. Testing against $(\Delta - \lambda)\phi$ for compactly supported smooth ϕ and using Fubini and the translation invariance of Δ shows that it is a distributional, hence real-analytic, λ -eigenfunction. It equals $e(U)W(Y)$ with $W(Y) = \Psi_\varepsilon(\kappa, Y)$, so $Y^2 W'' - 4\pi^2 Y^2 W + \lambda W = 0$ and $W = \alpha\sqrt{Y} K_\nu(2\pi Y) + \beta\sqrt{Y} I_\nu(2\pi Y)$. By (4.2), $W(Y) = O(Y^{1/2})$ as $Y \rightarrow \infty$, which excludes the exponentially growing solution: $\beta = 0$. So $\Psi_\varepsilon(\kappa, Y) = a(\kappa)\sqrt{Y} K_\nu(2\pi Y)$.

Step 2: an ordinary differential equation in κ , for real $\nu > 1$. Then $K_\nu > 0$ on $(0, \infty)$. The translation $Z \mapsto Z + t$, conjugated by $Z = -\kappa/w$, is the flow $w \mapsto w/(1 - tw/\kappa)$, and $g_\varepsilon(Z + t) = e(\varepsilon t)g_\varepsilon(Z)$. Differentiating at $t = 0$ gives

$$(4.3) \quad (U^2 - Y^2) \partial_U G + 2UY \partial_Y G = 2\pi i \varepsilon \kappa G.$$

For $\nu > 1$, (4.2) makes $U^2 G$, $U^2 \partial_U G$ and $U \partial_Y G$ integrable in U , so we may apply the transform $\tilde{G}(\eta, Y) := \int G(U + iY) e(-\eta U) dU$ (η near 1) to (4.3). With $\widetilde{U}h = (i/2\pi)\partial_\eta \tilde{h}$ and $\widetilde{\partial_U}h = 2\pi i\eta \tilde{h}$,

$$\partial_\eta^2(\eta \tilde{G}) + 4\pi^2 Y^2 \eta \tilde{G} - 2Y \partial_\eta \partial_Y \tilde{G} + 4\pi^2 \varepsilon \kappa \tilde{G} = 0.$$

The substitution $U = U'/\eta$ gives $\tilde{G}(\eta, Y) = \eta^{-1} a(\kappa\eta) W_0(\eta Y)$ with $W_0(y) = \sqrt{y} K_\nu(2\pi y)$ (so $a \in C^2$). With $k = \kappa\eta$ and $y = \eta Y$ one has $\partial_\eta^2(\eta \tilde{G}) = \kappa^2 a'' W_0 + 2\kappa Y a' W_0' + Y^2 a W_0''$ and $\partial_\eta \partial_Y \tilde{G} = \kappa a' W_0' + Y a W_0''$, so the terms with W_0' cancel and, after multiplication by η^2 ,

$$k^2 a''(k) W_0(y) + a(k) [-y^2 W_0''(y) + 4\pi^2 y^2 W_0(y)] + 4\pi^2 \varepsilon k a(k) W_0(y) = 0.$$

Since $-y^2 W_0'' + 4\pi^2 y^2 W_0 = \lambda W_0$ and $W_0 > 0$,

$$(4.4) \quad k^2 a''(k) + (4\pi^2 \varepsilon k + \tfrac{1}{4} - \nu^2) a(k) = 0.$$

With $a(k) = \sqrt{k} b(z)$, $z = 4\pi\sqrt{k}$, this becomes $z^2 b'' + z b' + (\varepsilon z^2 - 4\nu^2) b = 0$: Bessel's equation of order 2ν ($\varepsilon = 1$) or the modified one ($\varepsilon = -1$). So $a(k) = \sqrt{k} [\alpha J_{2\nu} + \beta \tilde{Y}] (4\pi\sqrt{k})$ with $\tilde{Y} \in \{J_{-2\nu}, Y_{2\nu}\}$, respectively $a(k) = \sqrt{k} [\alpha I_{2\nu} + \beta K_{2\nu}] (4\pi\sqrt{k})$.

Step 3: normalisation as $\kappa \rightarrow 0$, for real $\nu > 1$. Fix $Y = 1$. As $\kappa \rightarrow 0$, $\kappa^{-\nu-1/2} g_\varepsilon(-\kappa/(U+i)) \rightarrow \pi^\nu \Gamma(\nu+1)^{-1} (U^2+1)^{-\nu-1/2}$, dominated by $C(U^2+1)^{-\nu-1/2}$. By $\int_{\mathbb{R}} (U^2+1)^{-\nu-1/2} e(-U) dU = 2\pi^{\nu+1/2} K_\nu(2\pi)/\Gamma(\nu+\frac{1}{2})$ [20, 10.32.11] and Legendre's duplication formula [20, 5.5.5], $a(\kappa) = 2(2\pi)^{2\nu} \kappa^{\nu+1/2} \Gamma(2\nu+1)^{-1} (1+o(1))$ for both signs. The second solutions are $\asymp \kappa^{1/2-\nu}$, which is incompatible with $a = O(\kappa^{\nu+1/2})$; so $\beta = 0$. Since $\sqrt{k} J_{2\nu}(4\pi\sqrt{k})$ and $\sqrt{k} I_{2\nu}(4\pi\sqrt{k})$ are both $(2\pi)^{2\nu} k^{\nu+1/2} \Gamma(2\nu+1)^{-1} (1+O(k))$, we get $\alpha = 2$.

Step 4: all $\operatorname{Re} \nu > 0$. Both sides of the claim are holomorphic in ν on $\operatorname{Re} \nu > 0$ (for the left side, by (4.2) and dominated convergence), and they agree for real $\nu > 1$. $\square$

Theorem 4.4 (Fourier expansion). *For $\operatorname{Re} \nu > \frac{1}{2}$ and $\tau \in \mathbb{H}$,*

$$P_\nu(\tau) = \sqrt{v} I_\nu(2\pi v) \sin(2\pi u) + \sum_{n \geq 1} A_n(\nu) \sqrt{v} K_\nu(2\pi n v) \sin(2\pi n u),$$

$$A_n(\nu) = 2 \sum_{c \geq 1} \frac{1}{c} \left[S(1, n; c) J_{2\nu} \left(\frac{4\pi\sqrt{n}}{c} \right) - S(-1, n; c) I_{2\nu} \left(\frac{4\pi\sqrt{n}}{c} \right) \right].$$

The c -series converges absolutely and locally uniformly in ν , so A_n is holomorphic on $\operatorname{Re} \nu > \frac{1}{2}$ and may be differentiated termwise; $|A_n(\nu)| \leq 4\zeta(2\operatorname{Re} \nu) (2\pi\sqrt{n})^{2\operatorname{Re} \nu} e^{4\pi\sqrt{n}} / |\Gamma(2\nu+1)|$; and the n -series converges absolutely and locally uniformly on $\mathbb{H} \times \{\operatorname{Re} \nu > \frac{1}{2}\}$. In particular P_ν has no constant term.

Proof. Representatives of $\Gamma_\infty \backslash \Gamma$ are the identity and, for $c \geq 1$ and $(c, d) = 1$, matrices $\gamma_{c,d}$ with bottom row (c, d) . Writing $d = d_0 + kc$ with d_0 modulo c , $\gamma_{c,d} \tau = a/c - 1/(c^2(\tau + d_0/c + k))$ with $a \equiv \bar{d}_0 \pmod{c}$, and $f_{\nu,m}(Z + a/c) = e(ma/c) f_{\nu,m}(Z)$. By (4.2) (with $\kappa = c^{-2}$) Poisson summation applies to each block, and since the whole series converges absolutely, uniformly in u (Proposition 4.2), the Fourier coefficients may be taken block by block. The $e(nu)$ -coefficient of the blocks with modulus c of $\sum_\gamma f_{\nu,m}(\gamma\tau)$ is $S(m, n; c) B_c(m, n; v)$ with $B_c(m, n; v) := \int_{\mathbb{R}} f_{\nu,m}(-1/(c^2(U+iv))) e(-nU) dU$. For $n \geq 1$, $f_{\nu,m}(Z) = g_{\operatorname{sgn} m}(Z)$, and the substitution $U = U'/n$ turns B_c into $n^{-1} \Psi_{\operatorname{sgn} m}(n/c^2, nv)$; Lemma 4.3 gives $B_c(m, n; v) = (2/c) \sqrt{v} K_\nu(2\pi n v)$ times $J_{2\nu}(4\pi\sqrt{n}/c)$ if $m = 1$ and $I_{2\nu}(4\pi\sqrt{n}/c)$ if $m = -1$. For $n \leq -1$, $u \mapsto -u$ gives $B_c(m, n; v) = B_c(-m, -n; v)$. Assembling $f_\nu = (f_{\nu,1} - f_{\nu,-1})/(2i)$ with $S(1, -n; c) = S(-1, n; c)$ gives the stated sine series; the identity coset contributes f_ν , and the constant term vanishes because P_ν is odd. The bound follows from $|S| \leq c$ and Lemma 4.1(a) at order 2ν ; with $K_\nu(2\pi n v) \leq C_\nu e^{-2\pi n v} (1 + (nv)^{-\operatorname{Re} \nu})$ the n -series converges. $\square$

Remark 4.5 (The literature). Series of this kind were introduced by Niebur [38]; see also Fay [22]. The expansion of Theorem 4.4, with exactly the constant 2 and the assignment of J to $mn > 0$ and I to $mn < 0$, is stated by Bringmann, Jorgenson and Smajlović [9, §2.4] (with $s = \nu + \frac{1}{2}$), who note that they correct misprints in Niebur's Theorem 1. Lemma 4.3 proves it from scratch for every $\operatorname{Re} \nu > 0$, so no normalisation is taken from the literature; the termwise ν -derivative below relies on this.

4.4. The double root and the eigenvalue- $\frac{1}{4}$ eigenfunction.

Lemma 4.6 (A Jordan block). *If $\phi \in C^3(\mathbb{H})$ satisfies $\Delta\phi = \lambda\phi$, then*

$$\left(\Delta - \frac{1}{4}\right)\left(2\partial_v\phi + \left(\frac{1}{4} - \lambda\right)\frac{\phi}{v}\right) = -\left(\lambda + \frac{3}{4}\right)^2\frac{\phi}{v}.$$

Proof. $\Delta(\partial_v\phi) = \partial_v(\Delta\phi) - (2/v)\Delta\phi = \lambda\partial_v\phi - 2\lambda\phi/v$ and $\Delta(\phi/v) = \lambda\phi/v + 2\partial_v\phi - 2\phi/v$. Hence $\Delta(2\partial_v\phi + (\frac{1}{4} - \lambda)\phi/v) = \frac{1}{2}\partial_v\phi - (\lambda^2 + \frac{7}{4}\lambda + \frac{1}{2})\phi/v$; subtracting $\frac{1}{4}$ times the argument leaves $-(\lambda^2 + \frac{3}{2}\lambda + \frac{9}{16})\phi/v$. $\square$

For $\phi = P_\nu$ we have $\lambda = \frac{1}{4} - \nu^2$, so $\frac{1}{4} - \lambda = \nu^2$ and $\lambda + \frac{3}{4} = 1 - \nu^2$: the right side of Lemma 4.6 has a double zero at $\nu = 1$, where the eigenvalue of P_ν is $-\frac{3}{4}$. Put $P_1 := P_\nu|_{\nu=1}$, $\dot{P} := \partial_\nu P_\nu|_{\nu=1}$,

$$(4.5) \quad \tilde{\Phi} := \partial_\nu \left[\left(2\partial_v + \frac{\nu^2}{v} \right) P_\nu \right]_{\nu=1} = 2\partial_v \dot{P} + \frac{\dot{P}}{v} + \frac{2P_1}{v},$$

and, for $n \geq 1$,

$$(4.6) \quad \mathcal{S}_n := \sum_{c \geq 1} \frac{1}{c} \left[S(1, n; c) j_2 \left(\frac{4\pi\sqrt{n}}{c} \right) - S(-1, n; c) i_2 \left(\frac{4\pi\sqrt{n}}{c} \right) \right], \quad a_n := n\mathcal{S}_n + \frac{1}{4}\delta_{n1}.$$

Since $\partial_\nu J_{2\nu} = 2(\partial_\mu J_\mu)|_{\mu=2\nu}$, termwise differentiation in Theorem 4.4 gives $A'_n(1) = 4\mathcal{S}_n$.

Proposition 4.7. (a) $(\Delta + \frac{3}{4})\dot{P} = -2P_1$.

(b) $(\Delta - \frac{1}{4})\tilde{\Phi} = 0$.

(c) $\tilde{\Phi}$ is 1-periodic and odd, $\tilde{\Phi}(-\bar{\tau}) = -\tilde{\Phi}(\tau)$; in particular $\tilde{\Phi} = 0$ on the lines $\operatorname{Re} \tau \in \frac{1}{2}\mathbb{Z}$.

(d) $\tilde{\Phi}(\tau) = \sum_{n \geq 1} \beta_n \sqrt{v} K_0(2\pi n v) \sin(2\pi n u)$ with $\beta_n = -4\pi(nA'_n(1) + \delta_{n1}) = -16\pi a_n$.

(e) $|A_n(1)| \leq \frac{4\pi^4}{3} n e^{4\pi\sqrt{n}}$, $|\mathcal{S}_n| \leq 1.02 e^{4\pi\sqrt{n}}$ and $|\beta_n| \leq 52 n e^{4\pi\sqrt{n}}$ for all $n \geq 1$. Consequently $\tilde{\Phi}$ and all its derivatives are $O_{v_0}(e^{-2\pi v})$ for $v \geq v_0 > 0$, uniformly in u : $\tilde{\Phi}$ decays exponentially at the cusp ∞ .

Proof. (a) Differentiate $(\Delta - \frac{1}{4} + \nu^2)P_\nu = 0$ in ν at $\nu = 1$ (Proposition 4.2(c)). (b) By Lemma 4.6, $(\Delta - \frac{1}{4})[(2\partial_v + \nu^2/v)P_\nu] = -(1 - \nu^2)^2 P_\nu/v$, whose ν -derivative vanishes at $\nu = 1$. (c) P_ν is 1-periodic and odd, and $2\partial_v + \nu^2/v$ preserves both properties. (d) From $I'_\nu = I_{\nu-1} - (\nu/z)I_\nu$ and $K'_\nu = -K_{\nu-1} - (\nu/z)K_\nu$, the n -th sine coefficient of $(2\partial_v + \nu^2/v)P_\nu$ is $\delta_{n1}[4\pi\sqrt{v}I_{\nu-1}(2\pi v) + (1 - \nu)^2 I_\nu(2\pi v)/\sqrt{v}] + A_n(\nu)[-4\pi n\sqrt{v}K_{\nu-1}(2\pi n v) + (1 - \nu)^2 K_\nu(2\pi n v)/\sqrt{v}]$. The coefficients are holomorphic in ν and commute with ∂_ν . At $\nu = 1$, Lemma 4.1(b) gives the ν -derivative $-4\pi\sqrt{v}K_0(2\pi v)\delta_{n1} - 4\pi n A'_n(1)\sqrt{v}K_0(2\pi n v)$. (e) The first bound is Theorem 4.4 at $\nu = 1$. For $\mathcal{S}_n$, split the c -sum at $c_0 = 2\pi\sqrt{n}$: by Lemma B.3 the terms with $z_c = 4\pi\sqrt{n}/c \geq 2$ contribute at most $e^{4\pi\sqrt{n}} + 2\pi\sqrt{n}e^{2\pi\sqrt{n}} + 2\pi\sqrt{n}(\frac{\pi}{2} + 1)$, and those with $c > c_0$ at most $50\sqrt{n}$. For the latter, $z_c < 2$, and Lemma B.3(c) with $|S| \leq c$, $e^{z_c^2/4} < e$, $z_c^2/12 < \frac{1}{3}$ and $\log(2/z_c) = \log(c/c_0)$ bounds the c -th term by $e(c_0/c)^2[\log(c/c_0) + \psi(3) + \frac{1}{3}]$; since $\sum_{c > c_0} c^{-2} \leq c_0^{-1} + c_0^{-2}$ and $\sum_{c > c_0} c^{-2} \log(c/c_0) \leq \int_{c_0}^\infty c^{-2} \log(c/c_0) dc + \max_{c \geq c_0} c^{-2} \log(c/c_0) = c_0^{-1} + (2ec_0^2)^{-1}$, their sum is at most $e c_0(\frac{4}{3} + \psi(3)) + \frac{1}{2} + e(\psi(3) + \frac{1}{3}) < 38.6\sqrt{n} + 4 \leq 50\sqrt{n}$, as $c_0 = 2\pi\sqrt{n}$ and $\psi(3) = \frac{3}{2} - \gamma_E$.

Finally $e^{-4\pi\sqrt{n}}(2\pi\sqrt{n}e^{2\pi\sqrt{n}} + 67\sqrt{n}) < 0.012$. Then $|\beta_n| = 16\pi|n\mathcal{S}_n + \frac{1}{4}\delta_{n1}| \leq 52ne^{4\pi\sqrt{n}}$. The decay follows from $K_0(y) \leq \sqrt{\pi/2y}e^{-y}$ [20, §10.40(ii)] and $\sum_n n^{j+1}e^{4\pi\sqrt{n}-2\pi nv_0} < \infty$. $\square$

The bounds in (e) are crude: in fact $a_n \sim \bar{a}_n := n^{1/4}e^{4\pi\sqrt{n}}/(4\sqrt{2}\pi^2)$ (Theorem 6.1). They are used only for absolute convergence, which keeps the existence part of the argument independent of the positivity of the a_n .

4.5. The symmetric-square structure.

Lemma 4.8 (Symmetric-square equivariance). *Let $\phi \in C^2(\mathbb{H})$ be Γ -invariant and $\nu \in \mathbb{C}$. Define Ψ_0, Ψ_1, Ψ_2 by*

$$2 \operatorname{Re} [(2i\partial_\tau\phi)(\tau X + Y)^2] + \nu^2\phi \frac{|\tau X + Y|^2}{v} = \Psi_0(\tau)Y^2 + \Psi_1(\tau)XY + \Psi_2(\tau)X^2.$$

Then $\Psi_0 = 2\partial_v\phi + \nu^2\phi/v$, $\Psi_1 = 2u\Psi_0 - 4v\partial_u\phi$ and $\Psi_2 = u^2\Psi_0 - 4uv\partial_u\phi + \nu^2v\phi - 2v^2\partial_v\phi$. Moreover $\Psi_2 = \Psi_0 \circ S$, $\Psi_0(\tau+1) = \Psi_0(\tau)$, $\Psi_1(\tau+1) = \Psi_1(\tau) + 2\Psi_0(\tau)$, $\Psi_2(\tau+1) = \Psi_2(\tau) + \Psi_1(\tau) + \Psi_0(\tau)$; hence

$$(4.7) \quad \Psi_2(\tau+1) - 2\Psi_2(\tau) + \Psi_2(\tau-1) = 2\Psi_0(\tau).$$

Proof. For $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$: $(\partial_\tau\phi)(\gamma\tau) = (c\tau + d)^2\partial_\tau\phi(\tau)$, $(\gamma\tau)X + Y = (\tau X' + Y')/(c\tau + d)$ with $(X', Y') = (X, Y)\gamma$, and $\operatorname{Im} \gamma\tau = v/|c\tau + d|^2$. So the left side $\mathfrak{F}(\tau; X, Y)$ satisfies $\mathfrak{F}(\gamma\tau; (X, Y)) = \mathfrak{F}(\tau; (X, Y)\gamma)$. The components follow from $2i\partial_\tau\phi = \partial_v\phi + i\partial_u\phi$ and $|\tau X + Y|^2 = |\tau|^2X^2 + 2uXY + Y^2$. For $\gamma = S$, $(X, Y)S = (Y, -X)$, which gives $\Psi_2 \circ S = \Psi_0$; for $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $(X, Y)T = (X, X + Y)$, which gives the translation rules, and (4.7) follows by applying them at τ and $\tau - 1$. $\square$

Apply Lemma 4.8 to $\phi = P_\nu$ and differentiate in ν at $\nu = 1$. Then $\partial_\nu\Psi_0|_{\nu=1} = \tilde{\Phi}$, and we put

$$(4.8) \quad \Phi := \partial_\nu\Psi_2|_{\nu=1}, \quad E_1 := -4v\partial_u\dot{P}, \quad E_0 := 2vP_1 + v\dot{P} - 2v^2\partial_v\dot{P},$$

so that $\Phi = u^2\tilde{\Phi} + uE_1 + E_0$ with E_1, E_0 1-periodic.

- Proposition 4.9.** (a) $\Phi = \tilde{\Phi} \circ S$. Φ is an eigenfunction of Δ with eigenvalue $\frac{1}{4}$, it is odd, and $\Phi(iv) = 0$ for $v > 0$.
 (b) $\Phi(\tau+1) - 2\Phi(\tau) + \Phi(\tau-1) = 2\tilde{\Phi}(\tau)$.
 (c) $\dot{P} = \sqrt{v}[K_1(2\pi v) - \frac{I_0(2\pi v)}{2\pi v}] \sin 2\pi u + \sum_{n \geq 1} \left[A'_n(1)\sqrt{v}K_1(2\pi nv) + A_n(1)\frac{K_0(2\pi nv)}{2\pi n\sqrt{v}} \right] \sin 2\pi nu$ and $P_1 = \sqrt{v}I_1(2\pi v) \sin 2\pi u + \sum_{n \geq 1} A_n(1)\sqrt{v}K_1(2\pi nv) \sin 2\pi nu$.
 (d) (Growing modes.) $E_1 = 4\sqrt{v}I_0(2\pi v) \cos 2\pi u + (\text{decaying})$ and $E_0 = [4v^{3/2}I_1(2\pi v) - \frac{1}{\pi}\sqrt{v}I_0(2\pi v)] \sin 2\pi u + (\text{decaying})$, where “decaying” is a series $\sum_{n \geq 1} c_n(v) \cos 2\pi nu$, respectively $\sum_{n \geq 1} c_n(v) \sin 2\pi nu$, with $|c_n(v)| + |c'_n(v)| \leq Cn^3e^{4\pi\sqrt{n}}(1+v)^2e^{-2\pi nv}$ for $v \geq \frac{1}{2}$.
 (e) (No constant terms.) E_1 and E_0 have zero mean in u for every v .
 (f) For $v \geq 1$ and $|u| \leq 2$, $|\Phi(\tau)| + |\nabla\Phi(\tau)| \leq C_\Phi v^2e^{2\pi v}$.

Proof. (a) $\Psi_2 = \Psi_0 \circ S$ for every ν ; transport Proposition 4.7(b),(c) by the isometry S , which commutes with $\tau \mapsto -\bar{\tau}$; and $\tilde{\Phi}(i/v) = 0$. (b) is (4.7) differentiated in ν . (c) follows from Theorem 4.4 by termwise differentiation and Lemma 4.1(b). (d) The growing part of $\dot{P}$ is $-(2\pi)^{-1}v^{-1/2}I_0(2\pi v) \sin 2\pi u$, by (4.1). Then $-4v\partial_u$ of it is $4\sqrt{v}I_0(2\pi v) \cos 2\pi u$, and, with $I'_0 = I_1$, the growing part of E_0 is $2v^{3/2}I_1 - (2\pi)^{-1}\sqrt{v}I_0 - (2\pi)^{-1}\sqrt{v}I_0 + 2v^{3/2}I_1$. The remaining terms are the K -modes and the K_1 part of the first mode; the bounds follow from Proposition 4.7(e) and $K_j(y) \leq C(1+y^{-j})e^{-y}$. (e) Every term in (c) carries $\sin 2\pi nu$ or (after ∂_u) $\cos 2\pi nu$ with $n \geq 1$; this is where the oddness of the seed is used. (f) follows from $\Phi = u^2\tilde{\Phi} + uE_1 + E_0$, (d) and $I_j(y) \leq e^y$. $\square$

Remark 4.10 ($\tilde{\Phi}$ is not Γ -invariant). If $\tilde{\Phi} \circ S = \tilde{\Phi}$, then $\Phi = \tilde{\Phi}$ would be 1-periodic, the left side of Proposition 4.9(b) would vanish, and $\tilde{\Phi} = 0$; but $\beta_1 = -16\pi a_1 \neq 0$ (Theorem 6.1). So $\tilde{\Phi}$ is a 1-periodic, odd eigenfunction of Δ with eigenvalue $\frac{1}{4}$ that decays exponentially at the cusp ∞ , but it is not an automorphic form for Γ , in particular not a Maass cusp form; the Γ -equivariant object is the triple (Ψ_0, Ψ_1, Ψ_2) of Lemma 4.8, differentiated in ν . Nothing below uses Γ -invariance of $\tilde{\Phi}$.

Writing the growing modes as $E_1 \ni q\sqrt{v}I_0(2\pi v)\cos 2\pi u$ and $E_0 \ni (qv^{3/2}I_1(2\pi v) + q'\sqrt{v}I_0(2\pi v))\sin 2\pi u$, we have $(q, q') = (4, -1/\pi)$ exactly, so $q'/q = -1/(4\pi)$; this ratio is what makes the double pole in Theorem 5.4(d) residue-free.

The density on the imaginary axis. Put $D(v) := -\partial_u \Phi(iv)$ for $v > 0$. Since $\Phi = \tilde{\Phi} \circ S$,

$$(4.9) \quad D(v) = v^{-2}(\partial_u \tilde{\Phi})(i/v) = v^{-5/2} \sum_{n \geq 1} 2\pi n \beta_n K_0(2\pi n/v),$$

which is $O(e^{-2\pi/v})$ as $v \rightarrow 0$. On the other hand $D = -E_1(iv) - \partial_u E_0(iv)$, so Proposition 4.9(d) gives

$$(4.10) \quad D = D_{\text{grow}} + D_{\text{rest}}, \quad D_{\text{grow}}(v) = -2\sqrt{v}I_0(2\pi v) - 8\pi v^{3/2}I_1(2\pi v),$$

with $|D_{\text{rest}}(v)| \leq C_D(1+v)^2 e^{-2\pi v}$ for $v \geq 1$ and $|D_{\text{rest}}(v)| \leq C_D \sqrt{v}$ for $0 < v \leq 1$. The factor $(1+v)^2$ cannot be removed: the first mode gives $D_{\text{rest}}(v) = -4\pi^2(1+4\mathcal{S}_1)v^2 e^{-2\pi v}(1+O(1/v))$.

Remark 4.11 (Representation-theoretic meaning). Tensoring a principal series representation π_s of $\text{SL}_2(\mathbb{R})$ with Sym^2 produces constituents with the infinitesimal characters of $\pi_{s-1}, \pi_s, \pi_{s+1}$ (Bernstein–Gelfand [5]). For Maass forms, Alfes, Burban and Raum compute the Laplacian on the Sym^m -valued combinations of an eigenfunction with its iterated raising and lowering images, for an arbitrary eigenvalue, in a lemma of [1, §4.4] (section numbering of the arXiv version); they use it at eigenvalue 0. The Δ -eigenvalue $(s-1)(2-s)$ of the constituent π_{s-1} is stationary at $s = \frac{3}{2}$ ($\nu = 1$), where it equals $\frac{1}{4}$. Our double root (Lemma 4.6) is the case $m = 2$ of that lemma, at the point where this shifted eigenvalue is stationary, and the ν -derivative of the Y^2 -component in Lemma 4.8 is the resulting eigenvalue- $\frac{1}{4}$ eigenfunction. This interpretation is heuristic and is not used in the proofs.

5. THE GREEN-FLUX LIFT

In this section $\tilde{\Phi}$, $\Phi = \tilde{\Phi} \circ S$ and D are as in Section 4, and

$$(5.1) \quad f := \partial_u \tilde{\Phi}, \quad f(u+iv) = \sum_{n \geq 1} 2\pi n \beta_n \sqrt{v} K_0(2\pi n v) \cos(2\pi n u).$$

By Proposition 4.7, f is even in u , 1-periodic, and $|\tilde{\Phi}(u+iv)| + |\nabla \tilde{\Phi}(u+iv)| + |f(u+iv)| \leq C_0 e^{-2\pi v}$ for $v \geq \frac{1}{2}$; by (4.9), $D(v) = v^{-2}f(i/v)$. We lift $\tilde{\Phi}$ to a function of $x > 0$ by integrating a Green form against a family of Whittaker functions, in the manner of the contour integrals of Viazovska [50], Cohn–Kumar–Miller–Radchenko–Viazovska [14, 15] and Cohn–Gonçalves [13]; since $\tilde{\Phi}$ is not holomorphic, Green’s identity replaces Cauchy’s theorem.

5.1. The kernel. For $\tau = u + iv \in \mathbb{H}$ and $\text{Re } x > 0$ (principal square root) put

$$(5.2) \quad k_\tau(x) := \sqrt{xv} K_0(2\pi xv) \cos(2\pi ux).$$

(K1) For fixed x , $\Delta_\tau k_\tau(x) = \frac{1}{4}k_\tau(x)$, since $\sqrt{v}K_0(2\pi xv)e^{\pm 2\pi iux}$ are Whittaker functions of eigenvalue $\frac{1}{4}$.

(K2) With $(m_{-1}, m_0, m_1) = (1, -2, 1)$: $\sum_c m_c k_{\tau+c}(x) = -4 \sin^2(\pi x) k_\tau(x)$.

(K3) $k_{\lambda\tau}(x) = k_\tau(\lambda x)$ for $\lambda > 0$, and $k_{-\bar{\tau}}(x) = k_\tau(x)$.

(K4) For real $\sigma > -\frac{1}{2}$, with $\gamma_0(w) := \Gamma_{\mathbb{R}}(w + \frac{1}{2})^2 = \pi^{-w-1/2} \Gamma(\frac{w+1/2}{2})^2$, $\int_0^\infty |k_\tau(x)| x^{\sigma-1} dx \leq \frac{1}{4} \gamma_0(\sigma) v^{-\sigma}$, and $\int_0^\infty |\nabla_\tau k_\tau(x)| x^{\sigma-1} dx \leq c_1(\sigma) v^{-\sigma-1}$ with $c_1(\sigma) < \infty$.

(K4) follows from $|k_\tau(x)| \leq \sqrt{xv} K_0(2\pi xv)$, the analogous bounds for the derivatives, and $\int_0^\infty y^{\mu-1} K_\nu(2\pi y) dy = \frac{1}{4} \pi^{-\mu} \Gamma(\frac{\mu-\nu}{2}) \Gamma(\frac{\mu+\nu}{2})$ [20, 10.43.19].

Lemma 5.1 (The Mellin transform of the kernel). *For $\operatorname{Re} w > -\frac{1}{2}$ put $m_\tau(w) := \gamma_0(w)^{-1} \int_0^\infty k_\tau(x) x^{w-1} dx$. Then:*

- (i) $m_{iv}(w) = \frac{1}{4} v^{-w}$;
- (ii) $\tau \mapsto m_\tau(w)$ is an eigenfunction of Δ with eigenvalue $\frac{1}{4}$, even in u ;
- (iii) writing $\tau = re^{i\theta}$, $m_\tau(w) = r^{-w} f_w(\theta)$, where f_w solves

$$(5.3) \quad f_w''(\theta) + \left(w^2 + \frac{1}{4 \sin^2 \theta} \right) f_w(\theta) = 0 \quad (0 < \theta < \pi), \quad f_w(\pi - \theta) = f_w(\theta), \quad f_w(\frac{\pi}{2}) = \frac{1}{4};$$

- (iv) for $|\operatorname{Re} w| < \frac{1}{2}$, $m_{S\tau}(w) = m_\tau(-w)$.

Proof. (i) $\int_0^\infty \sqrt{x} K_0(2\pi x) x^{w-1} dx = \frac{1}{4} \gamma_0(w)$ by [20, 10.43.19], and $k_{iv}(x) = \sqrt{xv} K_0(2\pi xv)$. (ii) Differentiate under the integral (the second τ -derivatives of $k_\tau(x)$ are bounded by $C(1 + x^2)$ times the bounds of (K4), locally uniformly in τ) and use (K1) and (K3). (iii) By (K3), $m_{re^{i\theta}}(w) = r^{-w} m_{e^{i\theta}}(w)$. In polar coordinates $\Delta = -\sin^2 \theta (r^2 \partial_r^2 + r \partial_r + \partial_\theta^2)$, so (ii) gives the differential equation; $e^{i(\pi-\theta)} = -\overline{e^{i\theta}}$ and (K3) give the symmetry, and (i) gives the value at $\pi/2$. (iv) The equation (5.3) depends on w only through w^2 and is regular at $\theta = \frac{\pi}{2}$; its solutions with $f'(\frac{\pi}{2}) = 0$ form a line, fixed by the value at $\frac{\pi}{2}$. For $|\operatorname{Re} w| < \frac{1}{2}$ both f_w and f_{-w} are defined, symmetric and equal to $\frac{1}{4}$ at $\frac{\pi}{2}$, so $f_w = f_{-w}$. Since $S(re^{i\theta}) = r^{-1} e^{i(\pi-\theta)}$, $m_{S\tau}(w) = r^w f_w(\pi - \theta) = r^w f_{-w}(\theta) = m_\tau(-w)$. $\square$

The one-sided kernel $\sqrt{v} K_0(2\pi xv) e^{2\pi i u x}$ does not have property (iv) off the imaginary axis; only the even kernel (5.2) is used.

5.2. The Green form and the four-path functional. For two eigenfunctions F, g of Δ with eigenvalue $\frac{1}{4}$ on an open set, put $\omega(F, g) := F \star dg - g \star dF$, where $\star du = dv$ and $\star dv = -du$. Then $d\omega(F, g) = (F \Delta_0 g - g \Delta_0 F) du \wedge dv = 0$, where $\Delta_0 = \partial_u^2 + \partial_v^2 = -v^{-2} \Delta$. The Hodge star on 1-forms is conformally invariant, so $\omega(F \circ \varphi, g \circ \varphi) = \varphi^* \omega(F, g)$ for every holomorphic φ . Along an oriented curve with unit tangent $\hat{t} = (t_u, t_v)$ and right normal $\hat{n} = (t_v, -t_u)$ we have $\star dF(\hat{t}) = \partial_{\hat{n}} F$; so

$$(5.4) \quad \omega(F, g) = -g \partial_{\hat{n}} F ds \quad \text{wherever } F = 0 \text{ on the curve,}$$

and no derivative of g appears.

Paths and weights. γ_1 is the arc of the unit circle from -1 to i , γ_2 the arc from 1 to i , γ_3 the segment from 0 to i , and γ_4 the ray from i to $i\infty$. The weights are

$$\Phi_1 := -\frac{1}{4} \Phi(\cdot + 1), \quad \Phi_2 := -\frac{1}{4} \Phi(\cdot - 1), \quad \Phi_3 := \frac{1}{2} \Phi, \quad \Phi_4 := -\frac{1}{2} \tilde{\Phi}.$$

For $Y \geq \frac{1}{2}$ let $\tau_+(Y) := (Y^2 - \frac{1}{4} + iY)/(Y^2 + \frac{1}{4}) \in \gamma_2$ and $\tau_-(Y) := -\overline{\tau_+(Y)} \in \gamma_1$; then $|\tau_\pm| = 1$, $\arg \tau_+(Y) = \theta(Y) := 2 \operatorname{arccot}(2Y) \in (0, \frac{\pi}{2}]$, $S\tau_\pm = \tau_\mp$, and $-1/(\tau_+(Y) - 1) = \frac{1}{2} + iY$.

Lemma 5.2. $\Phi_j = 0$ on γ_j for $j = 1, \dots, 4$.

Proof. $\gamma_3, \gamma_4 \subset i\mathbb{R}_+$, where $\Phi = \tilde{\Phi} = 0$ (Propositions 4.7(c) and 4.9(a)). For $\tau = e^{i\theta}$, $-1/(\tau - 1) = \frac{1}{2} + \frac{i}{2} \cot \frac{\theta}{2}$ and $-1/(\tau + 1) = -\frac{1}{2} + \frac{i}{2} \tan \frac{\theta}{2}$; so $\Phi(\tau \mp 1) = \tilde{\Phi}(\pm \frac{1}{2} + iY) = 0$ on the unit circle. $\square$

Lemma 5.3 (The four-path functional). *For every function g that is C^1 on a neighbourhood of $\gamma_1 \cup \dots \cup \gamma_4$ and for which the integrals below converge absolutely,*

$$(5.5) \quad \mathcal{L}[g] := \sum_{j=1}^4 \int_{\gamma_j} \omega(\Phi_j, g) = \frac{1}{2} \int_1^\infty f(iY) [g(iY) + g(i/Y)] dY \\ - \frac{1}{4} \int_{1/2}^\infty f(\tfrac{1}{2} + iY) [g(\tau_+(Y)) + g(\tau_-(Y))] dY.$$

For g merely continuous on the paths we define $\mathcal{L}[g]$ by the right side. Moreover $\mathcal{L}[g \circ S] = \mathcal{L}[g]$ whenever both sides converge absolutely.

Proof. By Lemma 5.2 and (5.4), $\int_{\gamma_j} \omega(\Phi_j, g) = -\int_{\gamma_j} g \partial_{\hat{n}} \Phi_j ds$. On γ_4 ($\tau = iY$, $Y \geq 1$, $\hat{n} = +\partial_u$) this is $\frac{1}{2} \int_1^\infty f(iY) g(iY) dY$. On γ_3 ($\tau = iv$, $0 < v \leq 1$) it is $\frac{1}{2} \int_0^1 g(iv) D(v) dv = \frac{1}{2} \int_1^\infty g(i/Y) f(iY) dY$, by $D(v) = v^{-2} f(i/v)$. On γ_2 ($\tau = e^{i\theta}$, θ from 0 to $\frac{\pi}{2}$, $\hat{n}$ the outward radial direction): $\Phi(\tau - 1) = \tilde{\Phi}(\sigma(\tau))$ with $\sigma(\tau) = -1/(\tau - 1)$ and $\sigma'(\tau) e^{i\theta} = -1/(4 \sin^2 \frac{\theta}{2})$, which is real; so $\partial_r[\Phi(\tau - 1)] = -f(\sigma(\tau))/(4 \sin^2 \frac{\theta}{2})$ and $\partial_{\hat{n}} \Phi_2 = f(\sigma)/(16 \sin^2 \frac{\theta}{2})$. The substitution $Y = \frac{1}{2} \cot \frac{\theta}{2}$, $d\theta/(4 \sin^2 \frac{\theta}{2}) = -dY$, gives $-\frac{1}{4} \int_{1/2}^\infty g(\tau_+(Y)) f(\frac{1}{2} + iY) dY$. On γ_1 (θ from π to $\frac{\pi}{2}$, $\hat{n}$ the inward radial direction) the same computation with $-1/(\tau + 1)$, $\sigma' e^{i\theta} = 1/(4 \cos^2 \frac{\theta}{2})$, $Y = \frac{1}{2} \tan \frac{\theta}{2}$ and $f(-\frac{1}{2} + iY) = f(\frac{1}{2} + iY)$ gives $-\frac{1}{4} \int_{1/2}^\infty g(\tau_-(Y)) f(\frac{1}{2} + iY) dY$. Finally S exchanges iY and i/Y and exchanges $\tau_+(Y)$ and $\tau_-(Y)$, and both brackets in (5.5) are symmetric. $\square$

5.3. Class membership. Define, for $\operatorname{Re} x > 0$,

$$(5.6) \quad \mathcal{G}(x) := \mathcal{L}[k.(x)] = \frac{1}{2} \int_1^\infty f(iY) [k_{iY}(x) + k_{i/Y}(x)] dY - \frac{1}{2} \int_{1/2}^\infty f(\tfrac{1}{2} + iY) k_{\tau_+(Y)}(x) dY,$$

where the two arc terms coincide because $k_{-\bar{\tau}} = k_{\tau}$. Put $R_L(x) := -\frac{1}{2\pi^2(x^2-1)} - \frac{2}{\pi^2(x^2-1)^2}$.

Theorem 5.4 (The lift). (a) *The integrals in (5.6) converge absolutely, locally uniformly on $\{\operatorname{Re} x > 0\}$; $\mathcal{G}$ is analytic there and real on $(0, \infty)$.*

(b) *For $x > 1$, $\mathcal{G}(x) = \sin^2(\pi x) \int_0^\infty D(v) k_{iv}(x) dv$.*

(c) *For $x > 1$,*

$$\mathcal{G}(x) = \pi \sin^2(\pi x) \sqrt{x} \sum_{n \geq 1} \beta_n K_0(4\pi \sqrt{nx}) = -16\pi^2 \sin^2(\pi x) \sqrt{x} \sum_{n \geq 1} a_n K_0(4\pi \sqrt{nx}),$$

the series converging absolutely and locally uniformly on $\{\operatorname{Re} \sqrt{x} > 1\}$.

(d) *For $\operatorname{Re} x > 0$, $\mathcal{G}(x) = \sin^2(\pi x) [\sqrt{x} R_L(x) + \mathcal{E}(x)]$ with $\mathcal{E}(x) := \int_0^\infty D_{\text{rest}}(v) k_{iv}(x) dv$ analytic on $\operatorname{Re} x > 0$ (the right side has a removable singularity at $x = 1$). In particular $\mathcal{G}(1) = -\frac{1}{2}$.*

(e) *For $\sigma = \operatorname{Re} w > -\frac{1}{2}$: $\int_0^\infty |\mathcal{G}(x)| x^{\sigma-1} dx < \infty$ and $\int_0^\infty \mathcal{G}(x) x^{w-1} dx = \gamma_0(w) R(w)$ with $R(w) := \mathcal{L}[m.(w)]$, which is analytic on $\operatorname{Re} w > -\frac{1}{2}$. For $|\operatorname{Re} w| < \frac{1}{2}$, $R(-w) = R(w)$.*

(f) $\mathcal{G}(n) = \mathcal{G}'(n) = 0$ for every integer $n \geq 2$.

Proof. (a) On a compact set $K \subset \{\operatorname{Re} x > 0\}$: $|\cos(2\pi ux)| \leq C_K$ for $|u| \leq 1$, $|K_0(z)| \leq C(1 + |\log |z||)$ for $|z| \leq 1$ and $|K_0(z)| \leq C_K |z|^{-1/2} e^{-\operatorname{Re} z}$ for $|z| \geq 1$, $|\arg z| \leq \frac{\pi}{2} - \varepsilon_K$ [20, 10.40.2, 10.40.10]. On the paths $\operatorname{Im}(iY) = Y \geq 1$, $\operatorname{Im}(i/Y) = 1/Y$ and $\operatorname{Im} \tau_+(Y) = Y/(Y^2 + \frac{1}{4}) \in [\frac{1}{2Y}, 1]$, so $|k_\tau(x)| \leq C_K(1 + \log(1 + Y))$ there, while $|f(iY)|, |f(\frac{1}{2} + iY)| \leq C_0 e^{-2\pi Y}$. Dominated convergence gives (a).

(b) Fix $x > 1$. *Rays.* For $c \in \{-1, 0, 1\}$ the function $\Phi(\cdot - c)$ vanishes on the ray $c + i\mathbb{R}_+$, so by (5.4) $\int_{c+i\mathbb{R}_+} \omega(\Phi(\cdot - c), k.(x)) = \int_0^\infty k_{c+iv}(x) D(v) dv$, which converges absolutely because

$|D(v)| \leq Cv^{3/2}e^{2\pi v}$ for $v \geq 1$ (by (4.10)), $k_{iv}(x) = O(e^{-2\pi xv})$, and $D(v) = O(e^{-2\pi/v})$ as $v \rightarrow 0$. As $k_{c+iv}(x) = \sqrt{xv}K_0(2\pi xv) \cos(2\pi cx)$, (K2) gives

$$(5.7) \quad \sin^2(\pi x) \int_0^\infty D(v)k_{iv}(x) dv = -\frac{1}{4} \sum_c m_c \int_{c+i\mathbb{R}_+} \omega(\Phi(\cdot - c), k.(x)).$$

Regions. For $T > 1$ and small $\varepsilon > 0$ let $\Omega_1 := \{0 < \operatorname{Re} \tau < 1, |\tau| > 1, \varepsilon < \operatorname{Im} \tau < T\}$ and $\Omega_{-1} := -\overline{\Omega_1}$. On $\Omega_{\pm 1}$ the form $\omega_{\pm 1} := \omega(\Phi(\cdot \mp 1), k.(x))$ is closed, and Stokes' theorem gives, with all vertical segments oriented upwards,

$$\begin{aligned} \int_{1+i(\varepsilon, T]} \omega_1 &= \int_{\gamma_2} \omega_1 + \int_{[i, iT]} \omega_1 + \int_{h_1(T)} \omega_1 + o(1), \\ \int_{-1+i(\varepsilon, T]} \omega_{-1} &= \int_{\gamma_1} \omega_{-1} + \int_{[i, iT]} \omega_{-1} + \int_{h_{-1}(T)} \omega_{-1} + o(1) \end{aligned}$$

as $\varepsilon \rightarrow 0$, where $h_{\pm 1}(T)$ is the segment from iT to $\pm 1 + iT$ and $o(1)$ is the contribution of the bottom cut $h_{\pm 1}(\varepsilon)$ near ± 1 .

Horn estimate (at $+1$; -1 is symmetric). The map $\sigma(\tau) = -1/(\tau - 1)$ sends the lines $\operatorname{Re} \tau = 1$ and $|\tau| = 1$ to $\operatorname{Re} \sigma = 0$ and $\operatorname{Re} \sigma = \frac{1}{2}$. On the cut $\{u + i\varepsilon : \sqrt{1 - \varepsilon^2} < u < 1\}$ we have $|\tau - 1|^2 \leq 2\varepsilon^2$, so $|\sigma| \geq 1/(\sqrt{2}\varepsilon)$ and $\operatorname{Im} \sigma \geq 1/(2\varepsilon)$ for $\varepsilon \leq \frac{1}{2}$; hence $|\Phi(\tau - 1)| = |\tilde{\Phi}(\sigma)| \leq C_0 e^{-\pi/\varepsilon}$ and $|\nabla \Phi(\tau - 1)| \leq C_0 \varepsilon^{-2} e^{-\pi/\varepsilon}$, while $|k_\tau(x)| + |\nabla k_\tau(x)| \leq C_x \varepsilon^{-1/2} (1 + |\log \varepsilon|)$ and the cut has length at most ε^2 . So the cut contributes $O(e^{-\pi/\varepsilon})$, and the same estimates show that all integrands are super-exponentially small at $\tau = \pm 1$.

Connectors. On $h_{\pm 1}(T)$, $|\Phi(\tau \mp 1)| + |\nabla \Phi(\tau \mp 1)| \leq C_\Phi T^2 e^{2\pi T}$ (Proposition 4.9(f)) and $|k_\tau(x)| + |\nabla k_\tau(x)| \leq C_x \sqrt{T} e^{-2\pi x T}$, so $\int_{h_{\pm 1}(T)} \omega_{\pm 1} = O(T^{5/2} e^{-2\pi(x-1)T}) \rightarrow 0$. The connectors and the convergence of the ray integrals in (5.7) are the only places where $x > 1$ is used, and this cannot be avoided: $\Phi(\tau \pm 1)$ and D grow like $ve^{2\pi v}$, by the growing modes of Proposition 4.9(d).

Assembly. The ray $c = 0$ is $\gamma_3 \cup [i, i\infty)$. Insert the three decompositions into (5.7); the weights $-\frac{1}{4}m_c$ turn $\omega(\Phi(\cdot \mp 1), \cdot)$ on γ_2, γ_1 and $\omega(\Phi, \cdot)$ on γ_3 into $\omega(\Phi_j, \cdot)$, $j = 2, 1, 3$. On $[i, iT]$ the three weights combine, by bilinearity and Proposition 4.9(b), to $-\frac{1}{4}[\Phi(\cdot - 1) - 2\Phi + \Phi(\cdot + 1)] = -\frac{1}{2}\tilde{\Phi} = \Phi_4$. Letting $T \rightarrow \infty$ gives $\sum_j \int_{\gamma_j} \omega(\Phi_j, k.(x)) = \mathcal{G}(x)$.

(c) By (4.9) and Lemma 5.5 below, $\sum_n 2\pi n |\beta_n| \int_0^\infty v^{-5/2} K_0(2\pi n/v) \sqrt{xv} K_0(2\pi xv) dv = \pi \sqrt{x} \sum_n |\beta_n| K_0(4\pi \sqrt{nx}) < \infty$ for $x > 1$, since $|\beta_n| \leq 52ne^{4\pi \sqrt{n}}$ and $K_0(y) \leq \sqrt{\pi/2y} e^{-y}$ [20, §10.40(ii)]. Tonelli's theorem allows termwise integration in (b). Only $|\beta_n|$ is used, not the sign of the a_n . The same bound gives local uniformity on $\{\operatorname{Re} \sqrt{x} > 1\}$.

(d) For $0 < a < b$, $\int_0^\infty t I_0(at) K_0(bt) dt = 1/(b^2 - a^2)$: insert $I_0(at) = \sum_k (at/2)^{2k}/k!$ and $\int_0^\infty t^{2k+1} K_0(bt) dt = 2^{2k} k!^2 b^{-2k-2}$ [20, 10.43.19] (all terms positive). Differentiating in a , $\int_0^\infty t^2 I_1(at) K_0(bt) dt = 2a/(b^2 - a^2)^2$. With (4.10), for $x > 1$, $\int_0^\infty D_{\text{grow}}(v) k_{iv}(x) dv = \sqrt{x} R_L(x)$. The function $\mathcal{E}$ is analytic on $\operatorname{Re} x > 0$: on $v \geq 1$ the integrand is at most $C(1 + v)^{5/2} e^{-2\pi(1+\operatorname{Re} x)v} (1 + |\log v|)$ locally uniformly in x , and on $(0, 1]$ it is at most $Cv(1 + |\log(|x|v)|)$. So $V(x) := \sin^2(\pi x)[\sqrt{x} R_L(x) + \mathcal{E}(x)]$ is analytic on $\operatorname{Re} x > 0$ (the double pole of R_L at $x = 1$ is cancelled by $\sin^2$), and $V = \mathcal{G}$ on $(1, \infty)$ by (b); by (a) and the identity theorem $V = \mathcal{G}$ on $\operatorname{Re} x > 0$. Near $x = 1$, $\sqrt{x} R_L(x) = -\frac{1}{2\pi^2(x-1)^2} + O(1)$ (the terms in $(x-1)^{-1}$ cancel) and $\sin^2(\pi x) = \pi^2(x-1)^2 + O((x-1)^4)$; so $\mathcal{G}(1) = -\frac{1}{2}$.

(e) Let $\sigma > -\frac{1}{2}$. By (K4), $\int_0^\infty |k_\tau(x)| x^{\sigma-1} dx \leq \frac{1}{4} \gamma_0(\sigma) (\operatorname{Im} \tau)^{-\sigma}$, and on the parametrised paths $(\operatorname{Im} \tau)^{-\sigma} \leq 2^{|\sigma|} \max(Y^\sigma, Y^{-\sigma})$, while $|f| \leq C_0 e^{-2\pi Y}$. So $\int_0^\infty \int_{\text{paths}} |f| |k_\tau(x)| x^{\sigma-1} dY dx < \infty$. The theorems of Tonelli and Fubini give the absolute convergence and

$$\int_0^\infty \mathcal{G}(x) x^{w-1} dx = \mathcal{L}[\gamma_0(w) m.(w)] = \gamma_0(w) R(w),$$

and R is analytic by dominated convergence. For $|\operatorname{Re} w| < \frac{1}{2}$, Lemma 5.1(iv) and the S -invariance in Lemma 5.3 give $R(-w) = \mathcal{L}[m.(-w)] = \mathcal{L}[\tilde{m}_S(w)] = \mathcal{L}[m.(w)] = R(w)$.

(f) By (c), on $(1, \infty)$ we have $\mathcal{G} = \sin^2(\pi x) \tilde{L}(x)$ with $\tilde{L}$ analytic on a complex neighbourhood of $(1, \infty)$; at an integer $n \geq 2$, $\sin^2(\pi x)$ has a double zero. $\square$

Lemma 5.5. *For $Z > 0$, $\int_0^\infty K_0(Zt)K_0(Z/t) dt = (\pi/Z)K_0(2Z)$. Consequently, for $n \geq 1$ and $x > 0$, $\int_0^\infty v^{-2}K_0(2\pi n/v)K_0(2\pi xv) dv = K_0(4\pi\sqrt{nx})/(2n)$.*

Proof. Both sides of the first identity are smooth and decay exponentially in Z , so it suffices to compare Mellin transforms in Z for $\operatorname{Re} w > 1$. With $a = Zt$, $b = Z/t$ (so $dZ dt = da db/(2b)$), the left side has Mellin transform $\frac{1}{2}M[K_0](\frac{w+1}{2})M[K_0](\frac{w-1}{2})$ with $M[K_0](s) = 2^{s-2}\Gamma(s/2)^2$, which by the duplication formula [20, 5.5.5] equals $\frac{\pi}{4}\Gamma(\frac{w-1}{2})^2$, the Mellin transform of $(\pi/Z)K_0(2Z)$. The second identity follows with $Z = 2\pi\sqrt{nx}$, $v = \sqrt{n/x}w$ and $w \mapsto 1/w$. $\square$

We normalise by

$$(5.8) \quad \hat{G}_{\text{raw}} := -\frac{\mathcal{G}}{16\pi^2}, \quad \text{so that} \\ \hat{G}_{\text{raw}}(x) = \sin^2(\pi x)\sqrt{x} \sum_{n \geq 1} a_n K_0(4\pi\sqrt{nx}) \quad (x > 1), \quad \hat{G}_{\text{raw}} = \frac{1}{32\pi^2} \text{ at } x = 1.$$

In particular $\hat{G}_{\text{raw}}(x)/\sin^2(\pi x)$ has a double pole at $x = 1$, with leading coefficient $\frac{1}{32\pi^4}$ and zero residue, by (d); this is the shadow, in the Γ -only transform, of the evenness of the full transform $\hat{F}$.

5.4. Small x , entireness and the growth bound.

Lemma 5.6 (Small x). *For $0 < x < 1$, $\mathcal{G}(x) = \sqrt{x}[\varphi_1(x^2) + \varphi_2(x^2)\log x]$ with φ_1, φ_2 analytic on the unit disc and $\varphi_1(0) = \varphi_2(0) = 0$. Consequently $(x\partial_x)^j \mathcal{G}(x) = O_j(x^{5/2}(1 + \log(1/x)))$ as $x \rightarrow 0^+$, for every $j \geq 0$.*

Proof. By Theorem 5.4(d), $\mathcal{G} = \sin^2(\pi x)[\sqrt{x}R_L(x) + \mathcal{E}(x)]$. Insert

$$K_0(z) = \sum_{k \geq 0} \frac{(z/2)^{2k}}{k!^2} \left[h_k - \gamma_E - \log \frac{z}{2} \right], \quad h_k := \sum_{j \leq k} \frac{1}{j},$$

[20, 10.31.2] with $z = 2\pi xv$ into $\mathcal{E}$. Termwise integration is justified by dominated convergence: $|D_{\text{rest}}(v)| \leq C_D m(v)$ with $m(v) = \sqrt{v}$ on $(0, 1]$ and $(1+v)^2 e^{-2\pi v}$ on $(1, \infty)$, and $\sum_k (\pi xv)^{2k} k!^{-2} (h_k + \gamma_E + |\log(\pi xv)|) \leq I_0(2\pi xv)(2 + \log(1 + \pi xv) + |\log(\pi xv)|)$, which is integrable against $m(v)\sqrt{v}$ for $0 < x < 1$. (With $y = \pi xv$, the weights $w_k = y^{2k}/k!^2$ have sum $I_0(2y)$ and mean $\sum_k k w_k / I_0(2y) = y I_1(2y) / I_0(2y) \leq y$; as $h_k \leq 1 + \log(1+k)$, $\gamma_E < 1$ and $\log(1+k)$ is concave in k , Jensen's inequality gives the bound.) With $\vartheta_k := \int_0^\infty D_{\text{rest}}(v) v^{2k+1/2} dv$ and $\tilde{\vartheta}_k := \int_0^\infty D_{\text{rest}}(v) v^{2k+1/2} \log v dv$, which satisfy $|\vartheta_k|, |\tilde{\vartheta}_k| \leq C(1 + \Gamma(2k + \frac{9}{2}))(2\pi)^{-2k}(1 + \log(2k + 5))$,

$$\mathcal{E}(x)/\sqrt{x} = \sum_{k \geq 0} \frac{(\pi x)^{2k}}{k!^2} [(h_k - \gamma_E - \log \pi - \log x)\vartheta_k - \tilde{\vartheta}_k] \quad (|x| < 1),$$

and the series converge for $|x| < 1$, since $(2k)!/(4^k k!^2) \asymp k^{-1/2}$. This exhibits $\mathcal{E}(x)/\sqrt{x}$ as $\psi_1(x^2) + \psi_2(x^2)\log x$ with ψ_1, ψ_2 analytic on the unit disc. Also R_L is even and analytic on the unit disc. The factor $\sin^2(\pi x)$ is even, entire, and has a double zero at 0. Multiplying out gives φ_1, φ_2 with $\varphi_i(0) = 0$; the statement about $(x\partial_x)^j$ follows by termwise differentiation. $\square$

The absence of constant terms (Proposition 4.9(e)) is used exactly here: it gives the exponential decay of D_{rest} . A mode $e_1\sqrt{v} + e_2\sqrt{v}\log v$ in E_1 would add to $\mathcal{G}$ a term $\sin^2(\pi x)x^{-3/2}(a + b\log x) \sim \pi^2\sqrt{x}(a + b\log x)$, and the corresponding H would have poles at $t = \pm\frac{i}{2}$.

Lemma 5.7. (a) $\int_0^\infty \mathcal{G}(x)x^{w-1}dx$ converges absolutely and is analytic for $\operatorname{Re} w > -\frac{5}{2}$, and extends meromorphically to $\mathbb{C}$ with at most double poles, all in $\{-\frac{1}{2} - 2k : k \geq 1\}$.
 (b) R extends to an entire even function with $R(\frac{1}{2}) = R'(\frac{1}{2}) = R(-\frac{1}{2}) = R'(-\frac{1}{2}) = 0$ and $R(\bar{w}) = \overline{R(w)}$.

Proof. (a) Split at $r \in (0, 1)$. On $[r, \infty)$, $\mathcal{G}$ is continuous and decays faster than any power (Theorem 5.4(c)), so $\int_r^\infty$ is entire in w . On $(0, r]$, Lemma 5.6 gives $\mathcal{G} = \sum_{k \geq 1} x^{2k+1/2}(c_k + d_k \log x)$ with $\sum_k (|c_k| + |d_k|)r^{2k} < \infty$, and integrating termwise gives a series of the functions $r^{w+2k+1/2}/(w+2k+\frac{1}{2})$ and $r^{w+2k+1/2}/(w+2k+\frac{1}{2})^2$ (times constants and $\log r$), which converges locally uniformly off $\{-\frac{1}{2} - 2k\}$. (b) γ_0 has no zeros and has double poles exactly at $w = -\frac{1}{2} - 2k$, $k \geq 0$. At $w = -\frac{1}{2}$ the Mellin transform of $\mathcal{G}$ is analytic, so R has a double zero there; at $w = -\frac{1}{2} - 2k$, $k \geq 1$, the poles cancel. So R is entire. By Theorem 5.4(e), $R(-w) = R(w)$ on $|\operatorname{Re} w| < \frac{1}{2}$, hence on $\mathbb{C}$; so the double zero at $-\frac{1}{2}$ is also one at $\frac{1}{2}$. Reality: $\mathcal{G}$ is real on $(0, \infty)$ and $\gamma_0(\bar{w}) = \overline{\gamma_0(w)}$. $\square$

Definition 5.8. $H_{\text{raw}}(t) := -\frac{R(it)}{8\pi^3(t^2 + \frac{1}{4})^2}$.

Proposition 5.9. (a) H_{raw} is entire, even, and real on $\mathbb{R}$.
 (b) For every $\delta \in (0, 1)$ there is C_δ with $|H_{\text{raw}}(z)| \leq C_\delta(1 + |z|)^{-9+\delta}e^{\pi|\operatorname{Re} z|/2}$ on S_δ . In particular $H_{\text{raw}} \in \mathcal{W}_\delta$ for every $\delta \in (0, 1)$.
 (c) The Γ -only transform of H_{raw} in the sense of (3.1) is $\hat{G}_{\text{raw}}$: for every real ξ , $\int_{\mathbb{R}} \gamma_\infty(t)^2 H_{\text{raw}}(t) e^{-2\pi i \xi t} dt = \hat{G}_{\text{raw}}(e^{2\pi i \xi})$, the integral converging absolutely.

Proof. Throughout, $s = \frac{1}{2} + it$ and $w = s - \frac{1}{2} = it$, so $(t^2 + \frac{1}{4})^2 = s^2(s-1)^2$. (a) R is entire with double zeros at $w = \pm\frac{1}{2}$ (Lemma 5.7), and $s^2(s-1)^2$ has double zeros exactly at $w = \mp\frac{1}{2}$; so H_{raw} is entire. Evenness is $R(-w) = R(w)$, and reality on $\mathbb{R}$ follows from $R(it) = R(-it) = \overline{R(it)}$.

(b) Put $\mathcal{M}(s) := \int_0^\infty \hat{G}_{\text{raw}}(x)x^{s-3/2}dx = -\gamma_0(s-\frac{1}{2})R(s-\frac{1}{2})/(16\pi^2)$. Since $\gamma_\infty(t)^2 = \frac{1}{4}s^2(s-1)^2\gamma_0(s-\frac{1}{2})$,

$$(5.9) \quad 2\pi\gamma_\infty(t)^2 H_{\text{raw}}(t) = \mathcal{M}(\frac{1}{2} + it) \quad \text{as meromorphic functions of } t.$$

For $j \geq 0$ and $\sigma > -2$ let $N_j(\sigma) := \int_0^\infty |(x\partial_x)^j \hat{G}_{\text{raw}}(x)|x^{\sigma-3/2}dx$, finite by Lemma 5.6 and the decay at infinity. In $y = \log x$, $\mathcal{M}(s) = \int_{\mathbb{R}} \hat{G}_{\text{raw}}(e^y)e^{(s-1/2)y}dy$; integrating by parts j times (the boundary terms vanish for $\operatorname{Re} s > -2$ by Lemma 5.6 and the decay at infinity) gives $|\mathcal{M}(s)| \leq N_j(\operatorname{Re} s)|s - \frac{1}{2}|^{-j}$. For $z \in S_\delta$ put $s = \frac{1}{2} + iz$; then $\operatorname{Re} s \in [-\delta, 1+\delta]$ and $\operatorname{Im} s = \operatorname{Re} z$. Stirling's formula, uniformly in that range [20, 5.11.9], gives $|\gamma_\infty(z)|^2 \geq c_\delta |\operatorname{Re} z|^{3+\operatorname{Re} s} e^{-\pi|\operatorname{Re} z|/2}$ for $|\operatorname{Re} z| \geq T_\delta$. With $j = 6$, $|H_{\text{raw}}(z)| \leq (2\pi c_\delta)^{-1} \sup_{[-\delta, 1+\delta]} N_6 \cdot |\operatorname{Re} z|^{-9-\operatorname{Re} s} e^{\pi|\operatorname{Re} z|/2}$ for $|\operatorname{Re} z| \geq T_\delta$ (N_6 is convex in σ , so its supremum is attained at an endpoint); and on the compact rest of S_δ the entire function H_{raw} is bounded.

(c) On $\operatorname{Re} s = \frac{1}{2}$, with $g(\xi) := \hat{G}_{\text{raw}}(e^{2\pi i \xi})$, $\mathcal{M}(\frac{1}{2} + it) = 2\pi \int_{\mathbb{R}} g(\xi)e^{2\pi i \xi t} d\xi$. The function g is continuous and integrable ($O(|\xi|e^{5\pi\xi})$ as $\xi \rightarrow -\infty$ by Lemma 5.6, super-exponentially small as $\xi \rightarrow +\infty$), so by (5.9) $\gamma_\infty^2 H_{\text{raw}}$ is its inverse Fourier transform; it is integrable by (b), and Fourier inversion gives the claim. $\square$

Remark 5.10 (Growth along the imaginary axis; not used). One can show that H_{raw} is entire of order 1 but not of exponential type. For the second point: for real $w \rightarrow +\infty$ the Mellin integral

of $\mathcal{G}$ is dominated by large x , where $\mathcal{G} = -16\pi^2 \sin^2(\pi x) \sqrt{x} \sum_n a_n K_0(4\pi\sqrt{nx})$ has one sign once the a_n are known to be positive (Theorem 6.1); this gives $|R(w)| \asymp \Gamma(w)(2\pi)^{-w}$ up to powers of w , hence $|H_{\text{raw}}(iy)| = e^{y \log(y/2\pi e) + O(\log y)}$ as $y \rightarrow +\infty$. We do not write out the details of this sketch, or of the bound of order 1, since nothing below uses them. What we prove and use about the growth of H_{raw} is the strip bound of Proposition 5.9(b), which allows growth like $e^{\pi|\text{Re } z|/2}$ in the strip; on the real line, Theorem 7.4 gives $H_{\text{raw}}(t) \geq 0.2554 e^{0.6435|t|}/(2\pi^2(t^2 + \frac{1}{4})^2)$ for $|t| \geq 8$.

5.5. Integer zeros and a normalisation identity.

Proposition 5.11. (a) $\widehat{G}_{\text{raw}}(\xi_n) = \partial_\xi \widehat{G}_{\text{raw}}(\xi_n) = 0$ for every integer $n \geq 2$, and $\widehat{G}_{\text{raw}}(0) = \frac{1}{32\pi^2}$.

$$(b) \int_{\mathbb{R}} \Xi(t)^2 H_{\text{raw}}(t) dt = \frac{1}{32\pi^2}.$$

Proof. (a) is Theorem 5.4(f),(d) and (5.8), since $x = e^{2\pi\xi}$ is a real-analytic change of variable. (b) By Proposition 5.9, $H_{\text{raw}} \in \mathcal{W}_\delta$ for $\delta < \frac{1}{2}$, and its Γ -only transform is $\widehat{G}_{\text{raw}}$. Lemma 3.2(c) at $\xi = 0$ and (a) give $\int \Xi^2 H_{\text{raw}} = \sum_m d(m) m^{-1/2} \widehat{G}_{\text{raw}}(\xi_m) = \widehat{G}_{\text{raw}}(0)$. $\square$

The identity (b) is exact and involves no normalising constant; it will fix the sign of the normalisation in Section 8.

6. POSITIVITY ON THE FOURIER SIDE

On $(1, \infty)$ the Γ -only transform is $\sin^2(\pi x) \sqrt{x} \sum_n a_n K_0(4\pi\sqrt{nx})$ (Theorem 5.4(c)), so the sign of $\widehat{G}_{\text{raw}}$ beyond $x = 1$ is decided by the signs of the coefficients $a_n = n\mathcal{S}_n + \frac{1}{4}\delta_{n1}$ of (4.6). This section proves that every a_n is positive, with explicit two-sided bounds, using only the trivial bound for Kloosterman sums; and it records the certified value of the normalising constant.

6.1. The coefficients are positive. Put $z_n := 4\pi\sqrt{n}$, $\mathbf{e}(w) := e^w/\sqrt{2\pi w}$ and $\bar{a}_n := n \cdot \frac{2}{z_n} \mathbf{e}(z_n) = \frac{n^{1/4} e^{4\pi\sqrt{n}}}{4\sqrt{2}\pi^2}$. With $\kappa_1 := 1 + e/3$, $\kappa_2 := 2e/9$ and $\kappa_3 := \kappa_1\psi(3) + \kappa_2 = 2.36297\dots$, define for $z \geq 4\pi$

$$\begin{aligned} \lambda_0(z) &:= \left(1 - \frac{1}{z}\right) \left(1 - \frac{3}{8z} - \frac{15}{32z^2}\right) - \frac{\pi^3}{16z^2}, \\ \varrho(z) &:= \frac{(\pi+2)z}{8\mathbf{e}(z)} + \sqrt{2}e^{-z/2} + \frac{(\pi+2)z^2}{8\mathbf{e}(z)} + \frac{\sqrt{3}z^2}{4}e^{-2z/3} + \left[\kappa_3 + \frac{z}{2}(\kappa_1 + \kappa_3)\right] \frac{z}{2\mathbf{e}(z)}, \\ \lambda_-(z) &:= \lambda_0(z) - \frac{(\pi+1)z}{4\mathbf{e}(z)} - \varrho(z), \quad \lambda_+(z) := 1 - \frac{1}{z} + \left(\frac{\pi+1}{2} + \frac{2}{z^2}\right) \frac{z}{2\mathbf{e}(z)} + \varrho(z). \end{aligned}$$

Theorem 6.1 (Positivity of the coefficients). *For every $n \geq 1$,*

$$\frac{2}{z_n} \mathbf{e}(z_n) \lambda_-(z_n) \leq \mathcal{S}_n \leq \frac{2}{z_n} \mathbf{e}(z_n) \lambda_+(z_n),$$

and λ_- is increasing on $[4\pi, \infty)$ with $\lambda_-(4\pi) = 0.85010\dots$. Hence $\mathcal{S}_n > 0$ and $a_n > 0$ for every $n \geq 1$. The proof uses only the trivial bound $|S(\pm 1, n; c)| \leq \varphi(c)$; it involves no check over finitely many n , and its only numerical input is the evaluation of λ_- at the single point $z = 4\pi$.

Theorem 6.2 (Two-sided bounds). *For every $n \geq 1$, $(1 - 2/z_n) \bar{a}_n \leq a_n \leq \bar{a}_n$.*

The decisive numbers are elementary evaluations at the single point $z = 4\pi$: $\lambda_0(4\pi) = 0.877952$, $(\pi+1)z/(4\mathbf{e}(z)) = 0.000403$, and the five summands of $\varrho(4\pi)$ are 0.000250, 0.002641, 0.003145, 0.015725, 0.005683, so $\lambda_-(4\pi) = 0.850105$. We evaluated all of them in ball arithmetic as a check (Appendix A); a pocket calculator suffices.

Proof of Theorem 6.1. Fix n and write $z = z_n \geq 4\pi$, $M(w) := |\dot{J}_2(w)| + |\dot{I}_2(w)|$, and $\mathcal{S}_n = T_1 + R_n$ with $T_1 := \dot{J}_2(z) - \dot{I}_2(z)$ the term $c = 1$. By the trivial bound, $|R_n| \leq \sum_{c \geq 2} (\varphi(c)/c) M(z/c)$. We split the sum, using the Bessel bounds of Lemmas B.3–B.5 in Appendix B.

- (i) $c = 2$: $\varphi(2)/2 = \frac{1}{2}$ and $w = z/2 \geq 2\pi$, so $M(z/2) \leq \frac{\pi+2}{2} + \frac{4}{z}e(z/2)$; since $e(z/2)/e(z) = \sqrt{2}e^{-z/2}$, the contribution divided by $\frac{2}{z}e(z)$ is at most $\frac{(\pi+2)z}{8e(z)} + \sqrt{2}e^{-z/2}$.
- (ii) $3 \leq c \leq z/2$: $w = z/c \geq 2$ and $\varphi(c)/c \leq 1$, so $M(z/c) \leq \frac{\pi+2}{2} + \frac{2c}{z}e(z/c) \leq \frac{\pi+2}{2} + e(z/3)$, as e is increasing on $[\frac{1}{2}, \infty)$. There are at most $z/2$ such c , and $e(z/3)/e(z) = \sqrt{3}e^{-2z/3}$; the contribution divided by $\frac{2}{z}e(z)$ is at most $\frac{(\pi+2)z^2}{8e(z)} + \frac{\sqrt{3}z^2}{4}e^{-2z/3}$.
- (iii) $c > z/2$: $w = z/c < 2$, and by Lemma B.2(c), $(\varphi(c)/c)M(z/c) \leq f(c)$ with $f(y) := \frac{z^2}{4}y^{-2}[\kappa_1 \log(2y/z) + \kappa_3]$, which is positive and decreasing on $[z/2, \infty)$ (as $\kappa_1 < 2\kappa_3$). So $\sum_{c > z/2} f(c) \leq f(z/2) + \int_{z/2}^{\infty} f = \kappa_3 + \frac{z}{2}(\kappa_1 + \kappa_3)$, and divided by $\frac{2}{z}e(z)$ this is the last summand of ϱ .

Hence $|R_n| \leq \frac{2}{z}e(z)\varrho(z)$. For the term $c = 1$, Lemmas B.3 and B.5(c) give $T_1 \geq \frac{2}{z}e(z)\lambda_0(z) - \frac{\pi+1}{2}$ and $T_1 \leq \frac{\pi+1}{2} + \frac{z}{2} + \frac{2}{z}e(z)(1 - \frac{1}{z})$; this gives both inequalities. *Monotonicity.* λ_0 is increasing on $[1, \infty)$: it is a product of two non-negative increasing factors minus a decreasing term. Each of $z^k/e(z) \propto z^{k+1/2}e^{-z}$ ($k \leq 3$), $e^{-z/2}$ and $z^2e^{-2z/3}$ is decreasing on $[4\pi, \infty)$, so every subtracted term in λ_- decreases there, and λ_- increases. Finally $\lambda_-(4\pi) = 0.850105 > 0$, so $\mathcal{S}_n > 0$, and $a_n = n\mathcal{S}_n + \frac{1}{4}\delta_{n1} > 0$. $\square$

Proof of Theorem 6.2. Upper bound: $z(\lambda_+(z) - 1 + \frac{1}{z}) = (\frac{\pi+1}{2} + \frac{2}{z})\frac{z^2}{2e(z)} + z\varrho(z)$ is decreasing on $[4\pi, \infty)$, by the same monotonicity facts, and equals 0.34996 at $z = 4\pi$; so $\lambda_+(z) \leq 1 - 0.65/z$ and $a_n \leq \bar{a}_n(1 - 0.65/z_n) + \frac{1}{4}\delta_{n1} \leq \bar{a}_n$, because $0.65\bar{a}_1/(4\pi) = 265.6 > \frac{1}{4}$. Lower bound: $a_n \geq n\mathcal{S}_n \geq \bar{a}_n\lambda_-(z_n)$, and $z(\lambda_-(z) - 1 + \frac{2}{z}) = z(\lambda_0(z) - 1 + \frac{2}{z}) - z[\frac{(\pi+1)z}{4e(z)} + \varrho(z)]$, where $z(\lambda_0(z) - 1 + \frac{2}{z}) = 0.625 - (\frac{3}{32} + \frac{\pi^3}{16})z^{-1} + \frac{15}{32}z^{-2}$ is increasing (value 0.46630 at 4π) and the bracketed term is decreasing (value 0.34993 at 4π). So $\lambda_-(z) \geq 1 - 2/z$ on $[4\pi, \infty)$. $\square$

The bounds are sharp in the main term: the term $c = 1$ alone is $nT_1 = \frac{\sqrt{n}}{2\pi}I_1(z_n) - \frac{I_0(z_n)}{8\pi^2} + n[K_2(z_n) + \dot{J}_2(z_n)]$ by Lemma 4.1(c), and the certified values of Appendix A show $\sum_{c \geq 2} |\text{term}_c| \leq 3.6 \cdot 10^{-3} T_1$ for $n \leq 1500$, so Theorem 6.1 loses only a factor of about 8 in ϱ .

Remark 6.3 (The Weil bound is not needed). The Weil–Estermann bound $|S(m, n; c)| \leq d(c)(m, n, c)^{1/2}c^{1/2}$ [51, 21] (see also [30, Chapter 11]) would improve the constants, but the trivial bound already suffices for every n . The Weil bound enters the paper only through the tails of the certified coefficient enclosures (Appendix A), which are used by the window, by the numbers of the large- $|t|$ theorem and by the value of C .

6.2. The normalising constant. The function H_{raw} of Definition 5.8 has Γ -only transform $\hat{G}_{\text{raw}}$, whose coefficients are the a_n . The object of Theorem 4 is normalised by $H(0) = 1$, i.e. $H := H_{\text{raw}}/H_{\text{raw}}(0)$, provided $H_{\text{raw}}(0) \neq 0$. Its sign is decided by the physical side: we prove in Section 7 that $H_{\text{raw}} > 0$ on all of $\mathbb{R}$, so $H_{\text{raw}}(0) > 0$. The value is computer-assisted.

Proposition 6.4 (Computer-assisted). $H_{\text{raw}}(0) = 0.0011355097904 \pm 1.86 \cdot 10^{-9}$. Hence $C := 1/H_{\text{raw}}(0) \in [880.66027, 880.66316]$, and $\hat{G}_{\text{raw}}(0) \cdot C = C/(32\pi^2) = 2.788428 \pm 5 \cdot 10^{-6}$.

The certificate (Appendix A.4) evaluates the Mellin transform of $\hat{G}_{\text{raw}}/(2\pi\sqrt{x})$ at $s = \frac{1}{2}$ through a route that is independent of the physical-side formula of Section 7: the density (4.9) on $(0, 1]$, an explicit cusp- ∞ expansion of D on $[1, \infty)$, and closed-form Mellin kernels. The two routes share only the coefficient enclosures. Through the physical-side formula, $H_{\text{raw}}(0)$ is very

sensitive to a_1 (Remark 7.13); the route used here is far less so, which is why its radius is small. With Theorem 6.2, Proposition 6.4 gives the normalised bounds of Theorem 4: $C/(4\sqrt{2}\pi^2) \in [15.77370, 15.77376]$.

Remark 6.5 (Sign coherence three times). The positivity of C follows in three independent ways: from Proposition 6.4; from the certified value $\Pi(0) > 0$ of the physical-side function (Section 7), since $H_{\text{raw}}(0) = \Pi(0)/(2\pi^2 \cdot \frac{1}{16})$; and, given only that H_{raw} has constant sign on $\mathbb{R}$, from the exact identity $\int \Xi^2 H_{\text{raw}} = \frac{1}{32\pi^2} > 0$ of Proposition 5.11(b). The proof of the main theorems uses the second.

7. POSITIVITY ON THE PHYSICAL SIDE

The function H_{raw} is defined through a Mellin transform, and on the Fourier side $\widehat{G}_{\text{raw}}$ is of size $O(1)$ while $H_{\text{raw}}(t)$ grows roughly like $e^{\pi|t|/2-2\sqrt{2\pi}|t|}$; a direct Mellin inversion would have to resolve that growth out of a cancellation. Instead we apply the four-path functional of Section 5 to the Mellin transform of the kernel itself. The result is a formula for H_{raw} in which the growth is carried by a positive kernel, built from conical Legendre functions.

7.1. The conical kernel. $P_\nu(x) = F(\nu+1, -\nu; 1; \frac{1-x}{2})$, $-1 < x < 1$, is the Ferrers (on-the-cut) Legendre function of degree ν and order 0 [20, 14.3.1]. For complex z put $\mathbf{p}_z := P_{-1/2+iz}$, a conical function; for real t , $\mathbf{p}_t$ is real and even in t , since $P_\nu = P_{-\nu-1}$. Put

$$(7.1) \quad \mathbf{c}_t(\theta) := \frac{\sqrt{\sin \theta} [\mathbf{p}_t(\cos \theta) + \mathbf{p}_t(-\cos \theta)]}{2 \mathbf{p}_t(0)} \quad (0 < \theta < \pi).$$

Lemma 7.1 (Closed form of the kernel). *For $|\operatorname{Re} w| < \frac{1}{2}$ and $0 < \theta < \pi$, the function f_w of Lemma 5.1 is*

$$f_w(\theta) = \frac{\sqrt{\sin \theta} [P_{w-1/2}(\cos \theta) + P_{w-1/2}(-\cos \theta)]}{8 P_{w-1/2}(0)}, \quad P_{w-1/2}(0) = \frac{\sqrt{\pi}}{\Gamma(\frac{3}{4} + \frac{w}{2})\Gamma(\frac{3}{4} - \frac{w}{2})} \neq 0.$$

In particular $f_{it} = \frac{1}{4} \mathbf{c}_t$ for real t .

Proof. If y solves Legendre's equation $(1-x^2)y'' - 2xy' + \nu(\nu+1)y = 0$, then $\phi(\theta) := \sqrt{\sin \theta} y(\cos \theta)$ solves $\phi'' + [(\nu + \frac{1}{2})^2 + 1/(4\sin^2 \theta)]\phi = 0$ (Liouville normal form). With $\nu = w - \frac{1}{2}$ this is (5.3). Both $\sqrt{\sin \theta} P_{w-1/2}(\pm \cos \theta)$ solve it, so their sum is a solution symmetric about $\frac{\pi}{2}$, with value $2P_{w-1/2}(0)$ there. The value of $P_\nu(0)$ is [20, 14.5.1]; it is finite and non-zero for $|\operatorname{Re} w| < \frac{1}{2}$. Symmetric solutions form a line (proof of Lemma 5.1(iv)), and $f_w(\frac{\pi}{2}) = \frac{1}{4}$. $\square$

Theorem 7.2 (Positivity of the kernel). *For real t and $0 < \alpha < \pi$, Mehler–Dirichlet's formula [20, 14.12.1] gives*

$$(7.2) \quad \mathbf{p}_t(\cos \alpha) = \frac{\sqrt{2}}{\pi} \int_0^\alpha \frac{\cosh(t\phi)}{\sqrt{\cos \phi - \cos \alpha}} d\phi > 0,$$

and $\mathbf{p}_t(0) = \sqrt{\pi}/|\Gamma(\frac{3}{4} + \frac{it}{2})|^2 > 0$. Hence $\mathbf{c}_t(\theta) > 0$ for $0 < \theta < \pi$, $\mathbf{c}_t(\frac{\pi}{2}) = 1$ and $\mathbf{c}_{-t} = \mathbf{c}_t$. Moreover $\mathbf{p}_t(\cos \alpha)$ is even and non-decreasing in $|t|$, $\mathbf{p}_t(\cos \alpha) \leq \cosh(t\alpha)\mathbf{p}_0(\cos \alpha)$, and for complex z , $|\mathbf{p}_z(\cos \alpha)| \leq \mathbf{p}_{\operatorname{Re} z}(\cos \alpha)$.

Proof. (7.2) is [20, 14.12.1] with $\mu = 0$ and $\nu + \frac{1}{2} = it$; the integrand is positive. The other statements follow from $\cosh(t\phi) \leq \cosh(t\alpha)$ on $[0, \alpha]$ and $|\cosh(z\phi)| \leq \cosh(\phi \operatorname{Re} z)$. $\square$

7.2. The physical-side formula. Put, for $Y > 0$,

$$(7.3) \quad A(Y) := \sum_{n \geq 1} 2\pi n a_n \sqrt{Y} K_0(2\pi n Y), \quad B(Y) := \sum_{n \geq 1} (-1)^{n+1} 2\pi n a_n \sqrt{Y} K_0(2\pi n Y),$$

so that $f(iY) = -16\pi A(Y)$ and $f(\frac{1}{2} + iY) = 16\pi B(Y)$ by (5.1) and $\beta_n = -16\pi a_n$. Recall $\theta(Y) = 2 \operatorname{arccot}(2Y)$.

Theorem 7.3 (The physical-side formula). *For every real t ,*

$$(7.4) \quad H_{\text{raw}}(t) = \frac{\Pi(t)}{2\pi^2(t^2 + \frac{1}{4})^2}, \quad \Pi(t) := \int_1^\infty A(Y) \cos(t \log Y) dY + \frac{1}{2} \int_{1/2}^\infty B(Y) \mathbf{c}_t(\theta(Y)) dY,$$

with absolutely convergent integrals. Π is even and real, and $\Pi(t) = -R(it)/(4\pi)$. In particular $H_{\text{raw}}(t) > 0$ if and only if $\Pi(t) > 0$.

Proof. By Theorem 5.4(e) at $w = it$, $R(it) = \mathcal{L}[m_\tau(it)]$, the integrals converging absolutely: $|m_\tau(it)| \leq \frac{1}{4}\gamma_0(0)/|\gamma_0(it)|$ by (K4), and $|f| \leq C_0 e^{-2\pi Y}$ on the paths. By Lemma 5.1(i), $m_{iY}(it) + m_{i/Y}(it) = \frac{1}{4}(Y^{-it} + Y^{it}) = \frac{1}{2} \cos(t \log Y)$. On the arcs $|\tau_\pm(Y)| = 1$, $\arg \tau_+(Y) = \theta(Y)$, $\arg \tau_-(Y) = \pi - \theta(Y)$, so Lemma 5.1(iii) and Lemma 7.1 give $m_{\tau_\pm(Y)}(it) = \frac{1}{4} \mathbf{c}_t(\theta(Y))$. Inserting into (5.5),

$$R(it) = \frac{1}{4} \int_1^\infty f(iY) \cos(t \log Y) dY - \frac{1}{8} \int_{1/2}^\infty f(\frac{1}{2} + iY) \mathbf{c}_t(\theta(Y)) dY = -4\pi \Pi(t),$$

and Definition 5.8 gives (7.4). Evenness: $\cos(t \log Y)$ and $\mathbf{c}_t$ are even in t . $\square$

All Gamma factors have cancelled. The only cancellation left in (7.4) is near $t = 0$: there the two integrals are about $+5.585$ and -5.584 , and $\Pi(0) = 1.40 \cdot 10^{-3}$; as a function of the coefficients a_n the cancellation is much stronger (Remark 7.13). For large t the kernel $\mathbf{c}_t(\theta) \approx \frac{1}{2} e^{t(\pi/2 - \theta)}$ carries the growth, and the arc integral, whose integrand peaks at $Y \approx \sqrt{t/2\pi}$, dominates.

7.3. Large $|t|$. Fix $Y_z := 0.870$, $Y_s := 1$, $Y_3 := 4$, $\kappa := \frac{1}{2}$, $\theta_s := \theta(Y_s) = 2 \operatorname{arccot} 2 = 0.9272952 \dots$, so that $\frac{\pi}{2} - \theta_s = \arctan \frac{3}{4} = 0.6435011 \dots$; and $c_{\min} := \theta(Y_z) - \theta_s = 0.115941 \dots$, $t_{\text{mono}} := 1/(2c_{\min}) = 4.31252 \dots$

Theorem 7.4 (Large $|t|$). $\Pi(t) \geq 0.2554 e^{0.6435|t|} > 0$ for every real t with $|t| \geq 8$.

The proof is analytic except for the numerical inputs of Proposition 7.9: the sign of B on $[0.870, 4]$, the tail constant $\bar{\rho}(4)$, and three integrals evaluated at $t = 8$. A monotonicity argument carries the inequality from $t = 8$ to ∞ . We need four lemmas. Let $r(y) := |\Gamma(\frac{3}{4} + iy)|^2 e^{\pi y} / (2\pi \sqrt{y})$, so that by [20, 14.5.1]

$$(7.5) \quad 1/p_t(0) = r(t/2) \sqrt{2\pi t} e^{-\pi t/2} \quad (t > 0).$$

Lemma 7.5 (Stirling bracket). *For $y > 0$, $r_-(y) := \exp(-\frac{9}{32y^2} - \frac{1}{3y}) \leq r(y) \leq r_+(y) := \exp(\frac{9}{64y^2} + \frac{1}{3y})$; r_- is increasing and r_+ decreasing.*

Proof. Write $\log \Gamma(z) = (z - \frac{1}{2}) \log z - z + \frac{1}{2} \log 2\pi + R_1(z)$ with $|R_1(z)| \leq \sec^2(\frac{1}{2} \operatorname{ph} z) / (12|z|)$ [20, §5.11(ii)]. For $z = \frac{3}{4} + iy$, $\operatorname{ph} z \in (0, \frac{\pi}{2})$, so $|R_1| \leq 1/(6y)$; and $\operatorname{ph} z = \frac{\pi}{2} - \arctan \frac{3}{4y}$ gives $\log r(y) = \frac{1}{4} \log(1 + \frac{9}{16y^2}) + 2y \arctan \frac{3}{4y} - \frac{3}{2} + 2 \operatorname{Re} R_1(z)$. The first term lies in $[0, \frac{9}{64y^2}]$, the middle terms in $[-\frac{9}{32y^2}, 0]$ (as $x - x^3/3 \leq \arctan x \leq x$), and the last in $[-\frac{1}{3y}, \frac{1}{3y}]$. $\square$

Lemma 7.6 (Kernel bounds). *Let $t > 0$.*

(a) If $0 < \theta \leq \pi/3$ and $0 < \delta \leq \theta$, then

$$\mathbf{c}_t(\theta) \geq \frac{1}{2} \sqrt{\frac{\sin \theta}{\sin(\theta + \delta/2)}} e^{t(\pi/2 - \theta)} \operatorname{erf}(\sqrt{t\delta}) r_-(t/2).$$

(b) If $0 < \theta \leq \pi/2$, then

$$\mathbf{c}_t(\theta) \leq \frac{\sqrt{\sin \theta}}{2} r_+(\frac{t}{2}) \sqrt{2\pi t} \left[e^{-t(\frac{\pi}{2} - \theta)} \mathbf{p}_0(\cos \theta) + \frac{e^{t(\frac{\pi}{2} - \theta)} + e^{-t(\frac{3\pi}{2} - \theta)}}{2} \mathbf{p}_0(-\cos \theta) \right].$$

Proof. (a) Drop $\mathbf{p}_t(\cos \theta) > 0$ and use $\mathbf{p}_t(-\cos \theta) = \mathbf{p}_t(\cos \alpha)$, $\alpha = \pi - \theta$. In (7.2) keep only $\phi = \alpha - u$, $u \in [0, \delta]$ (note $\delta \leq \alpha$). There $\cos \phi - \cos \alpha = 2 \sin(\theta + \frac{u}{2}) \sin \frac{u}{2} \leq \sin(\theta + \frac{\delta}{2}) u$, since $\theta + \frac{\delta}{2} \leq \frac{\pi}{2}$, and $\cosh(t\phi) \geq \frac{1}{2} e^{t(\pi - \theta)} e^{-tu}$. With $\int_0^\delta e^{-tu} u^{-1/2} du = \sqrt{\pi/t} \operatorname{erf}(\sqrt{t\delta})$ this gives $\mathbf{p}_t(\cos \alpha) \geq e^{t(\pi - \theta)} \operatorname{erf}(\sqrt{t\delta}) / (\sqrt{2\pi t} \sqrt{\sin(\theta + \delta/2)})$. Combine with (7.5) and Lemma 7.5. (b) Use $\mathbf{p}_t(\cos \alpha) \leq \cosh(t\alpha) \mathbf{p}_0(\cos \alpha)$ (Theorem 7.2) for $\alpha = \theta$ and $\alpha = \pi - \theta$, $\cosh(t\theta) e^{-\pi t/2} \leq e^{-t(\pi/2 - \theta)}$, $\cosh(t(\pi - \theta)) e^{-\pi t/2} = \frac{1}{2} (e^{t(\pi/2 - \theta)} + e^{-t(3\pi/2 - \theta)})$, (7.5) and Lemma 7.5. $\square$

Lemma 7.7 (Signs of B). (a) $B > 0$ on $[Y_z, Y_3] = [0.870, 4]$.

(b) $B > 0$ on $[Y_3, \infty)$.

Proof. (a) is computer-assisted (Proposition 7.9(i)). (b) Since $|a_n| \leq \bar{a}_n$ (Theorem 6.2), $B(Y) \geq 2\pi a_1 \sqrt{Y} K_0(2\pi Y) (1 - \rho(Y))$ with $\rho(Y) := \sum_{n \geq 2} n \bar{a}_n K_0(2\pi n Y) / (a_1 K_0(2\pi Y))$. By $\sqrt{\pi/2x} e^{-x} (1 - \frac{1}{8x}) \leq K_0(x) \leq \sqrt{\pi/2x} e^{-x}$ for $x > 0$ [20, §10.40(ii)], $\rho(Y) \leq \bar{\rho}(Y) := \sum_{n \geq 2} \sqrt{n} (\bar{a}_n / a_1) e^{-2\pi(n-1)Y} / (1 - \frac{1}{16\pi Y})$, which is decreasing in Y ; and $\bar{\rho}(4) < 1$ (Proposition 7.9(ii)). $\square$

On $[\frac{1}{2}, Y_z]$ the function B changes sign; numerically its zeros there are near 0.5542 and at $\sqrt{3}/2$, the image of the elliptic point $e^{i\pi/3}$. We do not use any sign information on $[\frac{1}{2}, Y_z]$: only an upper bound for $B^- := \max(-B, 0)$ enters.

For $t > 0$ and $\theta = \theta(Y)$ put

$$\begin{aligned} p_+(t) &:= \frac{1}{4} \int_{Y_s}^{Y_3} B(Y) \sqrt{\frac{\sin \theta}{\sin(\theta + \kappa\theta/2)}} e^{t(\theta_s - \theta)} \operatorname{erf}(\sqrt{t\kappa\theta}) r_-(t/2) dY, \\ p_-(t) &:= \frac{1}{4} \int_{1/2}^{Y_z} B^-(Y) \sqrt{\sin \theta} r_+(t/2) \sqrt{2\pi t} \left[e^{-t(\pi - \theta - \theta_s)} \mathbf{p}_0(\cos \theta) \right. \\ &\quad \left. + \frac{e^{-t(\theta - \theta_s)} + e^{-t(2\pi - \theta - \theta_s)}}{2} \mathbf{p}_0(-\cos \theta) \right] dY. \end{aligned}$$

Lemma 7.8. (a) For $t > 0$, $e^{-t(\pi/2 - \theta_s)} \cdot \frac{1}{2} \int_{1/2}^\infty B(Y) \mathbf{c}_t(\theta(Y)) dY \geq p_+(t) - p_-(t)$.

(b) p_+ is non-decreasing on $(0, \infty)$, and p_- is non-increasing on $[t_{\text{mono}}, \infty)$.

(c) $|\int_1^\infty A(Y) \cos(t \log Y) dY| \leq I_{\text{ax}} := \int_1^\infty A(Y) dY$.

Proof. (a) Split $[\frac{1}{2}, \infty)$ at Y_z, Y_s, Y_3 . On $[\frac{1}{2}, Y_z]$, where $\theta \in [\theta(Y_z), \frac{\pi}{2}]$, bound the integrand below by $-B^- \mathbf{c}_t$ and use Lemma 7.6(b); multiplied by $e^{-t(\pi/2 - \theta_s)}$ this gives $-p_-(t)$. On $[Y_z, Y_s]$ and $[Y_3, \infty)$ the integrand is ≥ 0 , by Lemma 7.7 and Theorem 7.2. On $[Y_s, Y_3]$, $B \geq 0$ and $\theta \leq \theta_s < \pi/3$, and Lemma 7.6(a) with $\delta = \kappa\theta$ gives $p_+(t)$. (b) On $[Y_s, Y_3]$, $\theta \leq \theta_s$, so $e^{t(\theta_s - \theta)}$ is non-decreasing; erf and r_- are increasing; and $B \geq 0$. On $[\frac{1}{2}, Y_z]$ each exponent $c \in \{\pi - \theta - \theta_s, \theta - \theta_s, 2\pi - \theta - \theta_s\}$ is $\geq c_{\min}$, so $\sqrt{t} e^{-ct}$ is decreasing for $t \geq 1/(2c)$, hence on $[t_{\text{mono}}, \infty)$; and r_+ is decreasing. (c) $A > 0$ since every $a_n > 0$ (Theorem 6.1). $\square$

Proposition 7.9 (Computer-assisted). (i) $B > 0$ on $[0.870, 4]$: the certified cell minima are ≥ 0.5817 on $[0.870, 1]$ and $\geq 1.741 \cdot 10^{-7}$ on $[1, 4]$.

- (ii) $\bar{\rho}(4) \leq 4.1988957 \cdot 10^{-9}$.
- (iii) $p_+(8) \geq 2.2268109493$, $p_-(8) \leq 1.9389494542$, $I_{\text{ax}} \leq 5.5848933622$, and $I_{\text{ax}} e^{-8(\pi/2-\theta_s)} \leq 0.0324536121$.
- (iv) Consequently $p_+(8) - p_-(8) - I_{\text{ax}} e^{-8(\pi/2-\theta_s)} \geq 0.25540788 > 0$.

The integrals in (iii) are enclosed by Riemann sums of ball enclosures over 2000 cells for p_+ and 8000 cells for p_- , and I_{ax} by Gauss–Legendre quadrature with a rigorous error bound; all arithmetic is in ball arithmetic (Appendix A.3).

Proof of Theorem 7.4. For $t \geq 8 \geq t_{\text{mono}}$, Theorem 7.3 and Lemma 7.8 give

$$e^{-t(\pi/2-\theta_s)} \Pi(t) \geq p_+(t) - p_-(t) - I_{\text{ax}} e^{-t(\pi/2-\theta_s)} \geq p_+(8) - p_-(8) - I_{\text{ax}} e^{-8(\pi/2-\theta_s)},$$

which is ≥ 0.2554 by Proposition 7.9. Since Π is even, the same holds for $t \leq -8$. $\square$

7.4. The window.

Theorem 7.10 (Window; computer-assisted). $\Pi(t) > 0$ for every $t \in [0, 40]$. More precisely, $[0, 40]$ is covered by the 80 closed cells $[t_c - \frac{1}{4}, t_c + \frac{1}{4}]$, $t_c = \frac{1}{4}, \frac{3}{4}, \dots, 39\frac{3}{4}$, and on each cell a certified lower bound for Π is positive; on $[0, \frac{1}{2}]$ it is $1.1292 \cdot 10^{-3}$. Moreover $\Pi(0) \in [1.400011 \cdot 10^{-3}, 1.401747 \cdot 10^{-3}]$.

The certificate (Appendix A.2) works on each cell with the quadrature sum $Q(t)$ of the two integrals in (7.4) (Gauss–Legendre with 30 nodes on each of 30 panels of $[\frac{1}{2}, 13]$ for the arcs and 20 panels of $[1, 13]$ for the axis), which is a real-analytic function of t . It computes the Taylor polynomial of degree 19 of Q at t_c in power-series ball arithmetic, bounds the Taylor remainder by Cauchy’s estimate on the circle $|z - t_c| = 1$ with the majorant of Theorem 7.2, bounds the quadrature error through Bernstein ellipses, and bounds the tails in Y (beyond 13), in the conical series, and in n (via $|a_n| \leq \bar{a}_n$). Every error term is an explicit ball, and the certified lower bound on each cell is at least 0.295 times the value at its centre.

Corollary 7.11. $\Pi(t) > 0$ for every real t . Consequently $H_{\text{raw}}(t) > 0$ for every real t , $H_{\text{raw}}(0) = 8\Pi(0)/\pi^2 > 0$, and $H := H_{\text{raw}}/H_{\text{raw}}(0)$ satisfies $H(0) = 1$ and $H(t) > 0$ on $\mathbb{R}$; for $|t| \geq 8$, $H(t) \geq 0.2554 C e^{0.6435|t|} / (2\pi^2(t^2 + \frac{1}{4})^2)$ with $C = 1/H_{\text{raw}}(0)$.

Proof. Π is even; $[0, 40]$ is covered by Theorem 7.10 and $[8, \infty)$ by Theorem 7.4. Theorem 7.3 transfers the sign to H_{raw} . $\square$

Remark 7.12 (The certificate is sharp at small t). Positivity of Π near $t = 0$ depends on the Kloosterman terms with $c \geq 2$. If every a_n is replaced by its term $c = 1$, $n(\dot{J}_2 - \dot{I}_2)(z_n) + \frac{1}{4}\delta_{n1}$, the same certifier, with all error terms, proves that the resulting Π is negative on $[0, 1]$: its certified upper bounds are -0.25267 on $[-10^{-6}, 10^{-6}]$, -0.25060 on $[0, \frac{1}{2}]$ and -0.22702 on $[\frac{1}{2}, 1]$ (Appendix A.2). The cancellation at $t = 0$ is about 4000 : 1 between the axis and arc integrals, and the terms $c \geq 2$ change a_1 only by a relative $3.3 \cdot 10^{-3}$.

Remark 7.13 (Conditioning of $\Pi(0)$ in the coefficients). At $t = 0$, (7.4) is linear in the coefficients: $\Pi(0) = \sum_n a_n w_n$, where w_n is the value at $t = 0$ of the two integrals for the n -th terms of **A** and **B**. The terms $a_n w_n$ alternate in sign and are large, about $+51.7, -330, +719, -849, +680, -413, +203$ for $n = 1, \dots, 7$, while $\Pi(0) = 1.40 \cdot 10^{-3}$; so in coefficient space the cancellation is about $6 \cdot 10^5 : 1$. In particular $\partial\Pi(0)/\partial a_1 = w_1 = 0.011287\dots$, and a relative change of $-2.7 \cdot 10^{-5}$ in a_1 alone (about -0.124) would make $\Pi(0)$ vanish. The certificates take this into account: every a_n enters through a certified enclosure, a ball that includes the tail of its Kloosterman series for $n \leq 300$ (Appendix A.1) and the bounds $0 < a_n \leq \bar{a}_n$ beyond, and all radii are propagated. The enclosure of $\Pi(0)$ in Theorem 7.10 has radius $8.7 \cdot 10^{-7}$, of which $7.7 \cdot 10^{-7}$ is w_1 times the

radius $6.9 \cdot 10^{-5}$ of the ball for a_1 , and $\Pi(0)$ exceeds this radius by a factor of about 1600. A floating-point evaluation of $H_{\text{raw}}(0) = 8\Pi(0)/\pi^2$ through (7.4) needs a_1 to an absolute accuracy of about $2 \cdot 10^{-7}$ (relative $4 \cdot 10^{-11}$) to reach the precision of Proposition 6.4, so the c -series must be summed far: its partial sums for $\mathcal{S}_1$ up to $c = 10^4$ and up to $c = 3 \cdot 10^5$ differ by $3.2 \cdot 10^{-7}$. Proposition 6.4 is certified by the density route of Appendix A.4, which is far less sensitive to a_1 : with a ball for a_1 of radius $2.7 \cdot 10^{-5}$, the radius of $H_{\text{raw}}(0)$ is $1.86 \cdot 10^{-9}$. The figures in this remark, apart from the certified radii, are floating-point evaluations and are not used in any proof.

8. PROOFS OF THE MAIN RESULTS

The object. By Corollary 7.11, $H_{\text{raw}}(t) > 0$ for every real t ; in particular $H_{\text{raw}}(0) > 0$. Put

$$C := \frac{1}{H_{\text{raw}}(0)} > 0, \quad H := C H_{\text{raw}}, \quad F := \Xi^2 H.$$

By Proposition 5.9(c) and (5.8), the Γ -only transform of H in the sense of (3.1) is $\widehat{G}_H = C \widehat{G}_{\text{raw}} = -C\mathcal{G}/(16\pi^2)$, an analytic function of x on $\text{Re } x > 0$.

Proof of Theorem 4. (a) is Theorems 6.1 and 6.2, with Theorem 4.4 for the convergence of the series. (b) For $x > 1$, $\widehat{G}_H(x) = C \sin^2(\pi x) \sqrt{x} \sum_n a_n K_0(4\pi\sqrt{nx})$ by Theorem 5.4(c), and $\widehat{G}_H = C/(32\pi^2)$ at $x = 1$ by Theorem 5.4(d). The value of C is Proposition 6.4, and the bounds for $b_n = Ca_n$ follow from (a) and $C/(4\sqrt{2}\pi^2) \in [15.77370, 15.77376]$. (c) is Theorem 5.4(a),(d) and Theorem 7.3. $\square$

Proof of Theorem 1. H is entire, even and real on $\mathbb{R}$ (Proposition 5.9(a)), $H(0) = 1$ and $H > 0$ on $\mathbb{R}$ (Corollary 7.11), and (1.2) holds for every $\delta \in (0, 1)$ (Proposition 5.9(b)). In particular $H \in \mathcal{W}_\delta$ for every $\delta \in (0, 1)$. Fix $\delta \in (0, \frac{1}{2})$ and check the hypotheses of Proposition 3.3:

- (C2): $H > 0$ on $\mathbb{R}$.
- (C3): $\widehat{G}_H(\xi_k) = C \widehat{G}_{\text{raw}}(\xi_k) = 0$ for every integer $k \geq 2$ (Proposition 5.11(a)).
- (C4), indeed $\widehat{G}_H \geq 0$ on $[0, \infty)$: for $x > 1$, $\widehat{G}_H(x) = C \sin^2(\pi x) \sqrt{x} \sum_n a_n K_0(4\pi\sqrt{nx}) \geq 0$, since $C > 0$, every $a_n > 0$ (Theorem 6.1) and $K_0 > 0$; and $\widehat{G}_H = C/(32\pi^2) > 0$ at $x = 1$.

Proposition 3.3 gives $F \in \mathcal{C}_{\text{OPS}}$, $\widehat{F}(\xi_n) = 0$ for $n \geq 2$, $\int F = \widehat{F}(0) = \widehat{G}_H(0) > 0$, $\mathcal{A}(F) = 0$, and $\kappa^* \leq \kappa_{\text{OPS}}^* \leq 0$; the first inequality holds because $\mathcal{C}_{\text{OPS}} \subseteq \mathcal{C}$. For the derivatives: by (2.4) and (1.2) on the real line, $|F(t)| \leq C'(1 + |t|)^{6-9+\delta}$, so $tF \in L^1(\mathbb{R})$ for $\delta < 1$ and $\widehat{F}$ is C^1 ; as $\widehat{F} \geq 0$ on $\mathbb{R}$, each zero ξ_n is a minimum, and $\widehat{F}'(\xi_n) = 0$. Finally $q_{\min} = e^{-2\pi\kappa^*} \geq 1$, and likewise for the classical cone. $\square$

Proof of Theorem 2. By Theorem 1, $F \in \mathcal{C}$ and $\mathcal{A}(F) = 0$. For real t , $\Xi(t) = \gamma_\infty(t)\zeta(\frac{1}{2} + it)$ and $\gamma_\infty(t) \neq 0$, so the real zeros of Ξ are exactly Z_ζ ; and H has no real zeros. Hence $Z(\widehat{F}) = Z_\zeta \subseteq Z_\zeta \cup \{0\}$, and Corollary I.3.9(a), which rests on Theorem I.3.6, gives $\mathcal{K} \subseteq \{p_\zeta\}$. $\square$

No condition on the zeros of $\widehat{F}$ or of $\widehat{G}_H$ is needed here, and strict positivity of H on $\mathbb{R}$ is used only to place $Z(F)$ inside $Z_\zeta \cup \{0\}$.

Proof of Corollary 3. By Theorem 2 and Theorem I.2.9(c), (i) $\Leftrightarrow$ (ii) $\Leftrightarrow$ (iii). By Theorem 1, $\kappa^* \leq 0$, so (iii) $\Leftrightarrow$ (iv). If they hold, then $\mathcal{K} = \{p_\zeta\}$ by Corollary I.3.9(a); $0 = \kappa^* \leq \kappa_{\text{OPS}}^* \leq 0$; and $F/\int F$, which lies in $\mathcal{C}_{\text{OPS}} \subseteq \mathcal{C}$ and has $\mathcal{A}(F/\int F) = 0$, is a minimiser for both. If RH fails, then $\kappa^* \neq 0$ and $\kappa^* \leq 0$, so $\kappa^* < 0$, and $\mathcal{K} = \emptyset$ by Theorem I.2.7(a). $\square$

Remark 8.1 (Where each input enters). Theorem 1 uses Part I only through Definitions 1.2.1 and 1.2.4, Lemma 1.2.3 (in Corollary 1.4.4) and the bound (2.4). Theorem 2 uses in addition Lemma 1.2.13 and Theorem 1.3.6 (whose proof in Part I uses Lemma 1.2.6, Theorem 1.2.9(b) and the membership of the Bondarenko–Radchenko–Seip functions in $\mathcal{T}$, item (P5) of Section 2), and Corollary 3 uses Theorems 1.2.7(a) and 1.2.9; Section 2 lists which of these proofs are reproduced. Within this paper, the positivity of the a_n enters through (C4) and through the positivity of A in Lemma 7.8(c); the upper bound $a_n \leq \bar{a}_n$ of Theorem 6.2 enters Lemma 7.7(b) and the tail estimates of the certificates. The positivity of H enters through (C2) and, for Theorem 2, through the location of the real zeros of F . The certified enclosures of the a_n enter through the window, the numbers of Proposition 7.9 and the value of C .

9. REMARKS AND OPEN QUESTIONS

9.1. Other archimedean data. The reduction of Section 3 uses only that the Dirichlet coefficients of ζ^2 are non-negative and supported on the integers, a set closed under multiplication. The Dedekind zeta functions of $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\sqrt{-3})$, with the Gamma factors $\Gamma_{\mathbb{R}}^2$ and $\Gamma_{\mathbb{C}}$, share this property, so an analogue of the decoupling holds for their squares; we do not write it out. An analogue of the Γ -only integer-critical function for those data would then give an exact magic function, hence slack ≤ 0 . But Part I, Proposition 5.11, certifies strictly positive slack for both fields, at the gap ξ_2 and at their own gaps ($\kappa^* \geq 1.64 \cdot 10^{-5}$ and $\kappa^* \geq 1.50 \cdot 10^{-4}$). So, granted the decoupling, no such analogue exists for them, and the existence statement of this paper is specific to the data of ζ . In the construction this specificity is visible in the fit between the Gamma factor $\Gamma_{\mathbb{R}}$, the conductor 1 and the lattice $\mathbb{Z}$: the kernel $K_0(4\pi\sqrt{nx})$ comes from Whittaker functions of eigenvalue $\frac{1}{4}$ for $\mathrm{SL}_2(\mathbb{Z})$, and $\sin^2(\pi x)$ from a second difference with step 1.

9.2. Consequences for the questions of Part I. With Theorems 1 and 2, the uniqueness questions left open in Part I, §6.3 (non-integer zero weights, extra prime-side mass together with extra zeros, infinitely many extra zeros without a summability condition) are settled, since every admissible pair is p_{ζ} . Part I asked whether (U) forces an exact magic function without spurious zeros; the function F of Theorem 1 is one, with $Z(F) = Z_{\zeta}$. The existence statement (E) without RH, the last question of Part I, is equivalent to RH by Corollary 3. Part I, Corollary 3.12, characterised (S) by atomicity of the admissible zero measures; under RH there is now only one admissible pair.

9.3. What remains open.

- (1) *RH*, equivalently (E), equivalently $\kappa^* = 0$ (Corollary 3).
- (2) *Uniqueness of the integer-critical function.* We do not know whether H is the only even function in $\mathcal{W}_{\delta}$, with $H(0) = 1$, whose Γ -only transform vanishes at all integers ≥ 2 and is non-negative beyond the gap.
- (3) *The prime-power family of Part I.* Whether the exact Hermite family $\Xi^2 P_J(t^2)$ of Part I converges, and to what, is open (Part I, Conjecture 6.1). The transform $\hat{F}$ of our function vanishes doubly at ξ_n for all integers $n \geq 2$, not only at the prime powers, and F is not obtained as a limit. Canonical polynomial interpolants at the integer nodes approach it numerically and then diverge.
- (4) *A conceptual proof of positivity.* The positivity of H for $|t| < 8$ rests on a certified computation, as did the original proofs of the corresponding statements for the magic functions of sphere packing [50, 14, 15]. Remark 7.12 shows that it depends on the arithmetic of the Kloosterman terms: it fails if only the term $c = 1$ is kept. (This concerns the positivity of H ; the positivity of the a_n holds for every n by Theorem 6.1.)

- (5) *Other Gamma factors and lattices.* Which archimedean data admit an exact magic function, and through which automorphic objects?

AI-USE STATEMENT

This paper was produced by AI agents under the author’s direction. The construction, the proofs, the certificate scripts, the Lean spine and the text were produced mainly by Anthropic’s Claude, with supplementary assistance from OpenAI’s Astra. Separate (AI) referees, also Claude agents, checked each component and re-ran every certificate, in part with independently written code; their requested corrections are incorporated. The author initiated and directed the work, designed its verification process and reviewed the results, but did not check every step of the proofs. **No human mathematician has yet checked the proofs line by line.** The author takes full responsibility for the content of this paper, including any errors, and welcomes independent verification and corrections. Section 1.4 states precisely what is claimed and what is computer-assisted.

APPENDIX A. THE CERTIFICATES

This appendix describes the four computer-assisted ingredients listed in Section 1.4. Each has a standalone script in the ancillary files, with its exact inputs pinned by SHA-256, the command that was run, and the log of a fresh run; the file `README.md` of the ancillary files lists the expected decisive output lines and the runtimes. Paths below are relative to the top directory of the ancillary files. In the scripts and logs the function Π of Theorem 7.3 is called `R`, `A` is `Phi`, `B` is `g`, `ct` is `fhat_t`, and the large- $|t|$ theorem and the window are called Theorem L and Certificate W.

Conventions. All bounds are computed in ball arithmetic (Arb [31], through python-flint, version 0.9.0, built on FLINT [47]): every quantity is a midpoint–radius interval with outward rounding that contains the exact value, and every truncation, tail and quadrature error is bounded explicitly. Binary64 numbers are used only to choose parameters (cell centres, partitions, ellipse parameters), which are then used as the exact binary numbers they are. The modified Bessel function K_0 for real arguments is taken from the validated library of Part I (`lib/besselk.py`, shipped byte-identical); Arb’s own `bessel.k` gives correct but very wide enclosures at moderate arguments and is not used for certified quantities. For complex arguments only the majorant $|K_\nu(w)| \leq K_\nu(\operatorname{Re} w)$ ($\operatorname{Re} w > 0$, $\nu = 0, 1$) is used, which follows from $K_\nu(w) = \int_0^\infty e^{-w \cosh s} \cosh(\nu s) ds$.

A.1. The coefficient enclosures. *Statement.* Certified balls for $a_n = n\mathcal{S}_n + \frac{1}{4}\delta_{n1}$: for $n \leq 1500$ from the terms $c \leq 10^4$ ($n \leq 60$) or $c \leq \max(600, \lceil 4\pi\sqrt{n} \rceil + 1)$ ($60 < n \leq 1500$); for $n \leq 6$ from $c \leq 3 \cdot 10^5$; for $n \leq 2$ from $c \leq 6 \cdot 10^5$; in each case plus a rigorous bound for the remaining terms. The positivity certificates use $n \leq 300$ and the bound $|a_n| \leq \bar{a}_n$ beyond; with $c \leq 3 \cdot 10^5$,

$$a_1 = 4581.370553408 \pm 6.9 \cdot 10^{-5}, \quad a_2 = 1025727.234918 \pm 2.7 \cdot 10^{-4},$$

$a_3 = 62590223.689992 \pm 5.9 \cdot 10^{-4}$, and every a_n , $n \leq 1500$, is certified positive, in agreement with Theorem 6.1.

Method (`coefficients/cert_an.py`, `coefficients/cert_small_n.py`, `coefficients/coefficientlib.py`, `coefficients/kloost_fast.py`). For each prime power $q \leq 10^4$ the whole table $j \mapsto S(1, j; q)$ is obtained from one discrete Fourier transform in Arb, and composite moduli by twisted multiplicativity, $S(1, n; qr) = S(1, n\bar{r}^2; q)S(1, n\bar{q}^2; r)$ for $(q, r) = 1$ (an elementary consequence of the Chinese remainder theorem). For prime powers $10^4 < q \leq 6 \cdot 10^5$ the sums are computed with exact integer residues and a cosine table with a proved error at most 10^{-14} , and the rounding of the summation is bounded by Higham’s bound [28]; the proved

error of each such sum, times its weight in $\mathcal{S}_n$, is below 10^{-16} . The order derivatives $\dot{J}_2, \dot{I}_2$ are evaluated from the series of Lemma B.1 with a geometric tail bound for arguments ≤ 8 , and from the closed forms of Lemma 4.1(c), with the validated K_0, K_1 , beyond. The terms $c > X$ are bounded by the Weil–Estermann bound $|S(\pm 1, n; c)| \leq d(c)c^{1/2}$ [51, 21] (see also [30, Chapter 11]), the small-argument bound of Lemma B.2, and partial summation with $\sum_{c \leq u} d(c) \leq u(1 + \log u)$. The stored balls are rounded outward. The three coefficient files read by the positivity certificates are in `coefficients/data/`; a fresh run of the scripts reproduces their certified numbers (`coefficients/compare_outputs.py`).

A.2. The window. *Statement.* Theorem 7.10; script `positivity/run_window.py` (with `positivity/cert_window.py` and `positivity/poslib.py`).

Conical series. For $Y \geq \frac{1}{2}$ put $z_1 := \sin^2 \frac{\theta(Y)}{2} = 1/(4Y^2 + 1) \leq \frac{1}{2}$, $s := t^2$ and $c_k(s) := \prod_{j < k} ((\frac{1}{2} + j)^2 + s)/k!^2$. Then $\sqrt{\sin \theta} = 2\sqrt{Yz_1}$,

$$\mathbf{p}_t(\cos \theta) = \sum_{k \geq 0} c_k(s) z_1^k,$$

$$\mathbf{p}_t(-\cos \theta) = \frac{\cosh \pi t}{\pi} \sum_{k \geq 0} c_k(s) [2\psi(k+1) - 2\operatorname{Re} \psi(k + \frac{1}{2} + it) - \log z_1] z_1^k$$

by [20, 14.3.1, 15.8.10], and $1/\mathbf{p}_t(0) = \Gamma(\frac{3}{4} + \frac{it}{2})\Gamma(\frac{3}{4} - \frac{it}{2})/\sqrt{\pi}$. All terms of the first series are positive and increase with s , and $c_{k+1}(s)/c_k(s) \leq 1 + s/(k+1)^2$. Both series converge geometrically: the ratio of consecutive terms tends to $z_1 \leq \frac{1}{2}$. They are truncated at $k_{\max} = \lfloor 300 + 8T \rfloor$, $T := |t_c| + 1$, and the tails are bounded with the following lemma, valid uniformly for complex t with $|t| \leq T$.

Lemma A.1. *Let $T \geq 0$, $k_{\max} \geq 299 + 8T$, $k \geq k_{\max} + 1$, and $|z| \leq T$. Then $b_k(z) := 2\psi(k+1) - \psi(k + \frac{1}{2} + iz) - \psi(k + \frac{1}{2} - iz)$ satisfies $|b_k(z)| \leq 4(\frac{1}{2} + T)/(k_{\max} + \frac{3}{2} - T) \leq \frac{4}{7}$.*

Proof. On the segments from $k + \frac{1}{2} \pm iz$ to $k + 1$ the real part is at least $k + \frac{1}{2} - T \geq 1$, and $|\psi'(w)| \leq \psi'(\operatorname{Re} w) \leq 2/\operatorname{Re} w$ for $\operatorname{Re} w \geq 1$. Hence $|\psi(k+1) - \psi(k + \frac{1}{2} \pm iz)| \leq 2(\frac{1}{2} + T)/(k + \frac{1}{2} - T)$; finally $7(\frac{1}{2} + T) \leq k_{\max} + \frac{3}{2} - T$. $\square$

Quadrature. On each panel $[a, b]$ the integrals in (7.4) are replaced by the Gauss–Legendre rule with $m = 30$ nodes, whose nodes and weights are computed as Arb balls; there are 30 panels on $[\frac{1}{2}, 13]$ for the arcs and 20 on $[1, 13]$ for the axis. The error is bounded with the following form of [49, Theorem 19.3].

Lemma A.2. *Let f be analytic in the open Bernstein ellipse E_ρ of $[-1, 1]$, $\rho > 1$, with $|f| \leq M$ there, and let $m \geq 2$. The m -point Gauss–Legendre rule G_m satisfies $|\int_{-1}^1 f - G_m(f)| \leq \frac{64}{15} M \rho^{-2m}/(1 - \rho^{-2})$.*

Proof. The Chebyshev coefficients satisfy $|a_k| \leq 2M\rho^{-k}$. The rule is exact for polynomials of degree $\leq 2m - 1$, and both the rule and the integral vanish on the odd T_k . For even $k \geq 4$, $|\int T_k| = 2/(k^2 - 1)$ and $|G_m(T_k)| \leq \sum w_j = 2$, so the error on T_k is at most $\frac{32}{15}$; sum over even $k \geq 2m$. (This is [49, Theorem 19.3] with $n + 1 = m$ nodes.) $\square$

On each panel, M is bounded on an outward Arb box containing the image of E_ρ , uniformly for t in the cell, using $|K_0(z)| \leq K_0(\operatorname{Re} z)$, the absolute conical series with $|z_1| \leq 1/(4(x_{\text{lo}}^2 - y_{\text{hi}}^2) + 1)$, $|\cos(t \log Y)| \leq \cosh(t \arctan(y_{\text{hi}}/x_{\text{lo}}))$, and the coefficient tail below. The analyticity of the integrand on the box ($\operatorname{Re} Y > 0$, $\operatorname{Re}(4Y^2 + 1) > 1.3$, $|\arg Y| + |\arg(4Y^2 + 1)| < \pi/2$) is checked in Arb. Beyond $Y = 13$: $\int_{13}^\infty \mathbf{A} \leq \sup_{[13, 14]} \mathbf{A}/(1 - \sqrt{14/13} e^{-2\pi})$, which follows from $\mathbf{A}(Y + 1) \leq \sqrt{(Y + 1)/Y} e^{-2\pi} \mathbf{A}(Y)$ (as $(\log K_0)' \leq -1$ and $a_n > 0$); $|\mathbf{B}| \leq \mathbf{A}$; and $\mathbf{c}_t$ is bounded

from its series at $z_1 = 1/677$. The K_0 -sums use the coefficient balls for $n \leq 300$ and $|a_n| \leq \bar{a}_n$ with $K_0(x) \leq \sqrt{\pi/2x} e^{-x}$ and a geometric series beyond.

Continuum coverage. On the cell $[t_c - \frac{1}{4}, t_c + \frac{1}{4}]$, the quadrature sum Q , computed with the truncated conical series, is an analytic function of t on $|\operatorname{Im} t| < \frac{3}{2}$; the truncation does not cancel the poles of $1/\mathfrak{p}_t(0)$ at $t = \pm i(\frac{3}{2} + 2m)$. Its Taylor coefficients at t_c up to degree 19 are computed by power-series arithmetic in Arb. The Taylor remainder is at most $M_Q(d/r_t)^{20}/(1 - d/r_t)$ with $d = \frac{1}{4}$, $r_t = 1$, and $M_Q \geq \sup_{|z-t_c|=1} |Q(z)|$; M_Q uses $|\mathfrak{p}_z(\cos \alpha)| \leq \mathfrak{p}_T(\cos \alpha)$ (Theorem 7.2), $|\cos(z \log Y)| \leq \cosh(\log Y)$, and an Arb bound of $1/|\mathfrak{p}_z(0)|$ on 96 boxes covering the circle. The Taylor polynomial is evaluated on 8 sub-balls covering the cell, and all error terms are subtracted. Everything is computed at 192 bits.

Results (`positivity/logs/window_0_40.log`). All 80 cells are certified; the lower bounds on $[0, \frac{1}{2}]$ and $[39\frac{1}{2}, 40]$ are $1.12926 \cdot 10^{-3}$ and $9.034 \cdot 10^{16}$. Relative to the printed quadrature value $\Pi(t_c)$ at the cell centre, the certified lower bound is at least $0.2952 \Pi(t_c)$ on every cell (the smallest ratio, 0.29528, is on $[\frac{1}{2}, 1]$), the quadrature error is at most $1.35 \cdot 10^{-7} \Pi(t_c)$ (largest ratio $1.3420 \cdot 10^{-7}$, on $[39\frac{1}{2}, 40]$) and the Taylor remainder at most $1.33 \cdot 10^{-7} \Pi(t_c)$; these ratios are informational. The run takes 30 seconds on 8 cores. The enclosure $\Pi(0) \in [1.400011 \cdot 10^{-3}, 1.401747 \cdot 10^{-3}]$ is computed in the same way on a tiny cell around 0 (`positivity/check_r0.py`).

Negative control (`positivity/negctl.py`; not used in any proof). With a_n replaced by its term $c = 1$, the same certifier proves the upper bounds of Remark 7.12.

Independent cross-check (`coefficients/cert_Ht.py`, `coefficients/compare_window.py`; not used in any proof). The density route of Appendix A.4, run at $s = \frac{1}{2} + it$, gives certified point values of $H_{\text{raw}}(t)$ that do not use the conical functions or the four-path formula. At all 16 cell centres in $[0, 8]$ and at $t = 10.25, 15.25, 20.25, 30.25, 39.75$, these enclosures of $\Pi(t) = 2\pi^2(t^2 + \frac{1}{4})^2 H_{\text{raw}}(t)$ overlap those of the window certificate (21 of 21). The two routes share only the coefficient enclosures.

A.3. The large- $|t|$ numbers. *Statement.* Proposition 7.9; scripts `positivity/cert_large_t.py` (items (i), (iii), (iv)) and `positivity/cert_gtail.py` (item (ii)).

Method. $p_+(8)$ is bounded below by a Riemann sum over 2000 geometric cells of $[1, 4]$, with the integrand enclosed on each cell and the sum rounded down; $p_-(8)$ is bounded above by a Riemann sum over 8000 cells of $[\frac{1}{2}, 0.870]$, with B^- bounded by the negative of the lower end of the ball of B , rounded up; $\mathfrak{p}_0(\pm \cos \theta)$ come from the conical series at $t = 0$. The cell minima of B in (i) are the minima of the exact lower ends over 300 cells of $[0.870, 1]$ and the 2000 cells of $[1, 4]$. I_{ax} is computed by Gauss–Legendre quadrature with Lemma A.2 (quadrature error about $2.5 \cdot 10^{-17}$, tail beyond $Y = 13$ about $5 \cdot 10^{-32}$, both included in the bound). For (ii), the terms $2 \leq n \leq 399$ of $\bar{\rho}(4)$ are summed in Arb with a_1 replaced by the lower end of its ball, which gives an upper bound for $\bar{\rho}(4)$, and the terms $n \geq 400$ are at most twice the term $n = 400$, since consecutive ratios are at most $(1 + \frac{1}{n})e^{2\pi/\sqrt{n}-8\pi} < 10^{-10}$. The lower bound in (iv) is formed in Arb from the exact endpoints of the bounds in (iii); the run takes 138 seconds on one core.

A.4. The normalising constant. *Statement.* Proposition 6.4; script `coefficients/cert_H0.py`.

The formula. Put $g_{\text{raw}} := \hat{G}_{\text{raw}}/(2\pi\sqrt{x})$. By (5.9), $H_{\text{raw}}(0) = \int_0^\infty g_{\text{raw}}(x)x^{-1/2} dx / \gamma_\infty(0)^2$ with $\gamma_\infty(0)^2 = \pi^{-1/2}\Gamma(\frac{1}{4})^2/64$. Let $d := -D/(16\pi^2) = v^{-5/2} \sum_n 2na_n K_0(2\pi n/v)$, $d_{\text{grow}} := -D_{\text{grow}}/(16\pi^2)$ and $d_{\text{rest}} := d - d_{\text{grow}}$. By Theorem 5.4(d) and the integrals in its proof, $\hat{G}_{\text{raw}}(x) = \sin^2(\pi x)\sqrt{x} [P_L(x) + \int_0^\infty d_{\text{rest}}(v)\sqrt{v} K_0(2\pi vx) dv]$ with $P_L := -R_L/(16\pi^2)$, that is,

$P_L(x) = \frac{1}{32\pi^4(x^2-1)} + \frac{1}{8\pi^4(x^2-1)^2}$. Fubini's theorem gives

$$2\pi \int_0^\infty g_{\text{raw}}(x)x^{-1/2} dx = \int_0^\infty \sin^2(\pi x)x^{-1/2} P_L(x) dx + \int_0^\infty d_{\text{rest}}(v)\sqrt{v} \mathcal{Q}(v) dv,$$

$$\mathcal{Q}(v) = \frac{\Gamma(\frac{1}{4})^2 [1 - {}_2F_1(\frac{1}{4}, \frac{1}{4}; \frac{1}{2}; -v^{-2})]}{2^{5/2}(2\pi v)^{1/2}},$$

where $\mathcal{Q}(v) = \int_0^\infty \sin^2(\pi x)x^{-1/2} K_0(2\pi vx) dx$ is evaluated by [25, 6.699.12]. On $(0, 1]$, d_{rest} is computed from the series for d and the closed form of d_{grow} ; on $[1, 8]$ from the following expansion at the cusp ∞ , with the coefficients $A_n(1)$ and $A'_n(1) = 4\mathcal{S}_n$ of Theorem 4.4.

Lemma A.3. *For $v > 0$, with $y_n := 2\pi nv$,*

$$D(v) = D_{\text{grow}}(v) + 4\pi v^{3/2} [K_1(y_1) - y_1 K_0(y_1)]$$

$$+ \sum_{n \geq 1} 4\pi n v^{3/2} \left[A'_n(1) (K_1(y_n) - y_n K_0(y_n)) + A_n(1) \left(\frac{K_0(y_n)}{y_n} - 2K_1(y_n) \right) \right].$$

Proof. Write $\dot{P} = \sum_n p_n(v) \sin 2\pi nu$ and $P_1 = \sum_n r_n(v) \sin 2\pi nu$ as in Proposition 4.9(c). From (4.8), $D = -E_1(iv) - \partial_u E_0(iv) = \sum_n 2\pi n [3vp_n + 2v^2 p'_n - 2vr_n]$. Insert p_n, r_n and use $K'_0 = -K_1$, $(yK_1)' = -yK_0$, $I'_0 = I_1$; the terms $-\frac{I_0(2\pi v)}{2\pi\sqrt{v}}$ of p_1 and $\sqrt{v}I_1(2\pi v)$ of r_1 give D_{grow} . $\square$

The two expressions for d_{rest} give overlapping enclosures at $v = 1$. The integrals are computed with Arb's rigorous integration routine, with K_ν of complex argument evaluated at exact mid-points plus a derivative bound for the radius; the tails, $n > 60$ at the cusp 0 and $n > 30$ at the cusp ∞ , use $|a_n| \leq \bar{a}_n$, $|A'_n(1)| \leq 4\bar{a}_n/n$ and Lemma B.6. The ends are bounded analytically: on $[0, 10^{-8}]$, $0 \leq d \leq 10^{-6}\sqrt{v}/(8\pi^2)$ and $|\mathcal{Q}(v)| \leq 2^{-3/2}\Gamma(\frac{1}{4})^2(2\pi v)^{-1/2}$ give a contribution at most $3.2 \cdot 10^{-14}$; on $[8, \infty)$ a termwise majorant from Lemma A.3, $K_0 \leq K_1 \leq K_{3/2}$ and $|\mathcal{Q}(v)| \leq \sqrt{2}\pi^2\Gamma(\frac{5}{4})^2(2\pi v)^{-5/2}$ gives at most $3.5 \cdot 10^{-20}$; and the part $x > 2000$ of the first integral is bracketed with the second mean value theorem.

Results (coefficients/logs/cert_H0.log, 160 bits). With the enclosures of a_1, a_2 for $c \leq 6 \cdot 10^5$: the first integral is $0.0034380158943 \pm 2.9 \cdot 10^{-13}$; the second is $-0.0152642540638 \pm 7.7 \cdot 10^{-10}$ on $(0, 1]$ and $0.0126529953780 \pm 5.8 \cdot 10^{-10}$ on $[1, \infty)$; so $2\pi \int g_{\text{raw}} x^{-1/2} = 0.00082675720847 \pm 1.4 \cdot 10^{-9}$ and $H_{\text{raw}}(0) = 0.0011355097904 \pm 1.86 \cdot 10^{-9}$. The radius is dominated by the bound for the terms $c > 6 \cdot 10^5$ of $\mathcal{S}_1$. A second run that splits at $v = 1.25$ instead of $v = 1$ gives an overlapping enclosure. The run takes about 90 seconds.

A.5. The constants of Section 6. The constants $\lambda_0(4\pi)$, $\varrho(4\pi)$, $\lambda_\pm(4\pi)$, the two monotone envelopes in the proof of Theorem 6.2 and $\varrho_\alpha(4\pi)$ are elementary expressions evaluated at one point; `coefficients/analytic_constants.py` evaluates them in Arb, for example $\lambda_-(4\pi) = 0.850104915628 \pm 2 \cdot 10^{-13}$.

A.6. The Lean formalisation. The ancillary directory `lean/` contains a Lean 4 project, built on Mathlib and on the corresponding project for Part I, that formalises the logical spine of the proof. It states Theorems 1 and 2, Corollary 3 and the reduction (Lemma 3.2(a) and Proposition 3.3), and derives them from Mathlib, from six axioms of Part I's ledger, and from exactly two axioms for this paper. The six axioms of Part I are the explicit formula (Lemma I.2.3) and the bound (2.4), used for Theorem 1; the zero-side support theorem (Theorem I.3.6) and Theorem I.2.9(b), added for Theorem 2; the duality theorem (Theorem I.2.7(a)), added for Corollary 3; and the lower bound for κ^* of Part I, Proposition 2.8, used only to show that κ^* and κ_{OPS}^* are finite. The Part I axiom for Theorem I.3.6 states only its first conclusion, $\nu = \nu_\zeta$; the remaining conclusions, RH and $\mu = \mu_\zeta$, are derived in Lean from Theorem I.2.9(b), as in

Part I. The two axioms for this paper are: first, one axiom stating parts (b) and (c) of the Voronoi decoupling (Lemma 3.2); second, one joint existence axiom for the object: an $H \in \mathcal{W}_\delta$, for some $\delta < \frac{1}{2}$, with (C3), $\widehat{G}_H \geq 0$ on $[0, \infty)$ and $H > 0$ on $\mathbb{R}$, which is what Theorem 4 and Corollary 7.11 provide. The project builds with no unproved step (**sorry**) and passes its audit script; its `README.md`, `LEDGER.md` and `FAITHFUL.md` describe the axioms and list every difference between the Lean statements and the paper's. In particular the Lean form of the first part of Theorem 1 asserts less about H than the paper does: membership in $\mathcal{W}_\delta$ for one $\delta < \frac{1}{2}$, that is, the exponent 8 of (C1) instead of $9 - \delta$ for every $\delta < 1$, and no entireness. Moreover the differentiability of $\widehat{F}$ is not formalised: the Lean form of $\widehat{F}'(\xi_n) = 0$ says that the derivative of $\widehat{F}$ at ξ_n , if it exists (`HasDerivAt`), is 0. The Lean kernel therefore checks the assembly of the proof, namely that the main statements follow from these inputs. It does not check the inputs: Sections 4–7, the appendices and the certificates are not formalised.

APPENDIX B. BOUNDS FOR BESSEL FUNCTIONS AND THEIR ORDER DERIVATIVES

We collect the bounds for $\dot{I}_2 = \partial_\mu I_\mu|_{\mu=2}$ and $\dot{J}_2 = \partial_\mu J_\mu|_{\mu=2}$ used in Sections 4 and 6. Throughout $w > 0$, $\mathbf{e}(w) = e^w/\sqrt{2\pi w}$, $\psi(3) = \frac{3}{2} - \gamma_E = 0.92278\dots$, $\kappa_1 = 1 + e/3$ and $\kappa_2 = 2e/9$.

Lemma B.1 (Ascending series). *With $t_k(w) := (w/2)^{2k+2}/(k!(k+2)!)$,*

$$\dot{I}_2(w) = \sum_{k \geq 0} t_k(w) [\log \frac{w}{2} - \psi(k+3)], \quad \dot{J}_2(w) = \sum_{k \geq 0} (-1)^k t_k(w) [\log \frac{w}{2} - \psi(k+3)].$$

Proof. Differentiate the series $I_\mu(w) = \sum_k (w/2)^{2k+\mu}/(k!\Gamma(k+\mu+1))$ and $J_\mu(w) = \sum_k (-1)^k (w/2)^{2k+\mu}/(k!\Gamma(k+\mu+1))$ [20, 10.25.2, 10.2.2] termwise in μ (they converge locally uniformly in μ), and put $\mu = 2$. $\square$

Lemma B.2 (Small argument). *Let $0 < w \leq 2$, $v := w^2/4$ and $L := \log(2/w) \geq 0$. Then*

- (a) $\dot{I}_2(w) < 0$ and $|\dot{I}_2(w)| \geq \frac{w^2}{8}(L + \psi(3))$;
- (b) $|\dot{I}_2(w)|, |\dot{J}_2(w)| \leq B(w) := \frac{w^2}{8}[(L + \psi(3))(1 + \frac{v}{3}e^v) + \frac{v}{9}(1 + v)e^v]$;
- (c) $B(w) \leq \frac{w^2}{8}[\kappa_1(L + \psi(3)) + \kappa_2]$.

Proof. Every term of the series for $\dot{I}_2$ is negative, since $\log \frac{w}{2} - \psi(k+3) = -(L + \psi(k+3))$; the term $k = 0$ gives (a). Both series are bounded in absolute value by $\sum_k t_k(L + \psi(k+3))$, with $t_k = \frac{v}{2} \cdot \frac{2v^k}{k!(k+2)!}$. For $k \geq 1$, $2/(k!(k+2)!) \leq 1/(3(k-1)!)$, so $\sum_k 2v^k/(k!(k+2)!) \leq 1 + \frac{v}{3}e^v$; and $\psi(k+3) - \psi(3) \leq k/3$, so $\sum_k \frac{2v^k}{k!(k+2)!}(\psi(k+3) - \psi(3)) \leq \frac{1}{9} \sum_{k \geq 1} \frac{kv^k}{(k-1)!} = \frac{v}{9}(1 + v)e^v$. This gives (b), and $v \leq 1$ gives (c). $\square$

Lemma B.3 (Uniform bounds). *For every $w > 0$:*

- (a) $|\dot{J}_2(w)| \leq \frac{\pi+1}{2}$;
- (b) $|\dot{I}_2(w)| \leq e^w + \frac{1}{2}$;
- (c) *if $w \leq 2$, then $|\dot{I}_2(w)|, |\dot{J}_2(w)| \leq \frac{w^2}{8}e^{w^2/4}[\log \frac{2}{w} + \psi(3) + \frac{w^2}{12}]$.*

Proof. (a) By Schlöfli's integral [20, 10.9.6],

$$J_\mu(w) = \frac{1}{\pi} \int_0^\pi \cos(\mu\theta - w \sin \theta) d\theta - \frac{\sin \mu\pi}{\pi} \int_0^\infty e^{-w \sinh t - \mu t} dt.$$

Differentiating at $\mu = 2$ (the integrands are C^1 in μ and dominated uniformly near $\mu = 2$), $\dot{J}_2(w) = -\frac{1}{\pi} \int_0^\pi \theta \sin(2\theta - w \sin \theta) d\theta - \int_0^\infty e^{-w \sinh t - 2t} dt$, which is at most $\frac{\pi}{2} + \frac{1}{2}$ in absolute value. (b) Likewise from

$$I_\mu(w) = \frac{1}{\pi} \int_0^\pi e^{w \cos \theta} \cos \mu\theta d\theta - \frac{\sin \mu\pi}{\pi} \int_0^\infty e^{-w \cosh t - \mu t} dt$$

[20, 10.32.4]: $|\dot{I}_2(w)| \leq \frac{e^w}{\pi} \int_0^\pi \theta |\sin 2\theta| d\theta + \frac{1}{2} = e^w + \frac{1}{2}$. (c) In the bound $\sum_k t_k(L + \psi(k+3))$ use $(k+2)! \geq 2k!$, $\psi(k+3) \leq \psi(3) + \frac{k}{3}$, $\sum_k v^k/k!^2 \leq e^v$ and $\sum_k kv^k/k!^2 \leq ve^v$. $\square$

Lemma B.4 (Inequalities for I_0, I_1, K_2). *For $w > 0$: (a) $I_1(w) \leq e(w)$; (b) $I_1(w) \geq e(w)(1 - \frac{3}{8w} - \frac{15}{32w^2})$; (c) $0 < I_0(w) - I_1(w) \leq \frac{\pi^3}{16} e(w)/w$; (d) $K_2(w) \leq 2/w^2$.*

Proof. (a), (b): by [20, 10.32.2], $I_1(w) = \frac{w}{\pi} \int_{-1}^1 \sqrt{1-t^2} e^{wt} dt$, and $t = 1 - s/w$ gives $I_1(w) = \frac{\sqrt{2}e^w}{\pi\sqrt{w}} \int_0^{2w} \sqrt{s} \sqrt{1-s/2w} e^{-s} ds$. For (a) use $\sqrt{1-x} \leq 1$ and $\int_0^\infty \sqrt{s} e^{-s} ds = \frac{\sqrt{\pi}}{2}$. For (b), with $x = s/2w$, $\sqrt{1-x} \geq (1-x)(1+\frac{x}{2}) = 1 - \frac{x}{2} - \frac{x^2}{2}$ on $[0, 1]$ (equivalent to $1 - \frac{3}{4}x^2 - \frac{1}{4}x^3 \leq 1$), and the right side is ≤ 0 for $x \geq 1$; so extending the integral to $[0, \infty)$ only decreases it, and $\int_0^\infty \sqrt{s}(1 - \frac{s}{4w} - \frac{s^2}{8w^2}) e^{-s} ds = \frac{\sqrt{\pi}}{2}(1 - \frac{3}{8w} - \frac{15}{32w^2})$. (c) By [20, 10.32.3], $I_0 - I_1 = \frac{1}{\pi} \int_0^\pi e^{w \cos \theta} (1 - \cos \theta) d\theta > 0$; with $1 - \cos \theta \leq \theta^2/2$ and $\cos \theta \leq 1 - 2\theta^2/\pi^2$ on $[0, \pi]$, $I_0 - I_1 \leq \frac{e^w}{2\pi} \int_0^\pi \theta^2 e^{-2w\theta^2/\pi^2} d\theta = \frac{\pi^3}{16} e(w)/w$. (d) $(w^2 K_2(w))' = -w^2 K_1(w) < 0$ and $w^2 K_2(w) \rightarrow 2$ as $w \rightarrow 0^+$ [20, 10.29.4, 10.30.2]. $\square$

Lemma B.5 (The order derivative $\dot{I}_2$). (a) $-\dot{I}_2(w) = K_2(w) + \frac{2}{w} I_1(w) - \frac{2}{w^2} I_0(w)$.

(b) $-\dot{I}_2(w) > 0$ for every $w > 0$.

(c) For $w \geq 1$: $\frac{2}{w} e(w) \lambda_0(w) \leq -\dot{I}_2(w) \leq \frac{2}{w^2} + \frac{2}{w}(1 - \frac{1}{w})e(w)$, with λ_0 as in Section 6.

(d) For $w \geq 2$: $|\dot{I}_2(w)| \leq \frac{1}{2} + \frac{2}{w} e(w)$.

Proof. (a) is Lemma 4.1(c). (b) For $w \leq 2$ use Lemma B.2(a). For $w \geq 2$, the series give $wI_1(w) - I_0(w) = -1 + \sum_{j \geq 1} (2j-1)v^j/j!^2 \geq v - 1 \geq 0$ with $v = w^2/4 \geq 1$, and $K_2 > 0$. (c) Write $-\dot{I}_2 = K_2 + \frac{2}{w}(1 - \frac{1}{w})I_1 - \frac{2}{w^2}(I_0 - I_1)$. For the upper bound use $K_2 \leq 2/w^2$, $I_1 \leq e$ and $I_0 - I_1 > 0$; for the lower bound drop $K_2 > 0$ and use Lemma B.4(b),(c), noting that $1 - \frac{3}{8w} - \frac{15}{32w^2} > 0$ for $w \geq 1$. (d) follows from (b), (c) and $2/w^2 \leq \frac{1}{2}$. $\square$

Lemma B.6 (The coefficients at $\nu = 1$). *For every $n \geq 1$, $1.610e(z_n) \leq -A_n(1) \leq 2.325e(z_n)$, where $z_n = 4\pi\sqrt{n}$.*

Proof. As in the proof of Theorem 6.1, with J_2 and I_2 in place of $\dot{J}_2$ and $\dot{I}_2$. The term $c = 1$ of $-A_n(1)/2$ is $I_2(z) - J_2(z)$, and $|J_2| \leq 1$ [20, 10.14.1], $0 < I_2 = I_0 - \frac{2}{w}I_1 \leq I_0 \leq e(w)(1 + \frac{\pi^3}{16w})$, and $I_2 \geq (1 - \frac{2}{w})e(w)(1 - \frac{3}{8w} - \frac{15}{32w^2})$ for $w \geq 2$, by Lemma B.4. For $c = 2$ and for $3 \leq c \leq z/2$ we have $|J_2| + I_2 \leq 1 + (1 + \frac{\pi^3}{32})e(w)$ with $w = z/c \geq 2$, and for $c > z/2$, $|J_2(w)| + I_2(w) \leq \kappa_1 w^2/4$ (from the series, as in Lemma B.2) together with $\sum_{c > z/2} (z/c)^2 \leq 4 + 2z$. The resulting bound $|A_n(1) - 2(J_2 - I_2)(z)| \leq 2e(z)\varrho_\alpha(z)$, with $\varrho_\alpha(z) := [\frac{1}{2}(1 + (1 + \frac{\pi^3}{32})e(\frac{z}{2})) + \frac{z}{2}(1 + (1 + \frac{\pi^3}{32})e(\frac{z}{3})) + \kappa_1(1 + \frac{z}{2})]/e(z)$ decreasing on $[4\pi, \infty)$, gives the claim after evaluating the monotone envelopes at $z = 4\pi$ ($\varrho_\alpha(4\pi) = 0.0082$). $\square$

Lemma B.6 and $|A'_n(1)| = 4\mathcal{S}_n \leq 4\bar{a}_n/n$ (Theorem 6.1 and the proof of Theorem 6.2) are used only as tail bounds in the certificate for C (Appendix A.4).

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Email address: njohns@gmail.com